STRUCTURAL SYSTEMS 1 *ARC 2713*

CONCRETE BUILDING

Lecture 12

1. CONCRETE a brief history of ideas and interconnected relationships
2. CONCRETE prestressed and precast
3. PRECAST PROTOTYPES
4. CONCRETE BUILDING
5. SPAN & LOAD TABLES

…mass on Earth in 2016

Biomass: 1.1 Trillion Tons ***Anthromass: 1.2 Trillion Tons***

...mass on Earth in 2016

CONCRETE:
992 billion tons on earth
(33 billion tons each year)

Biomass: 1.1 Trillion Tons

Anthromass: 1.2 Trillion Tons

CONCRETE

…an ancient material, the second most consumed in the world, after water

CONCRETE

…a material of the Earth and from the earth

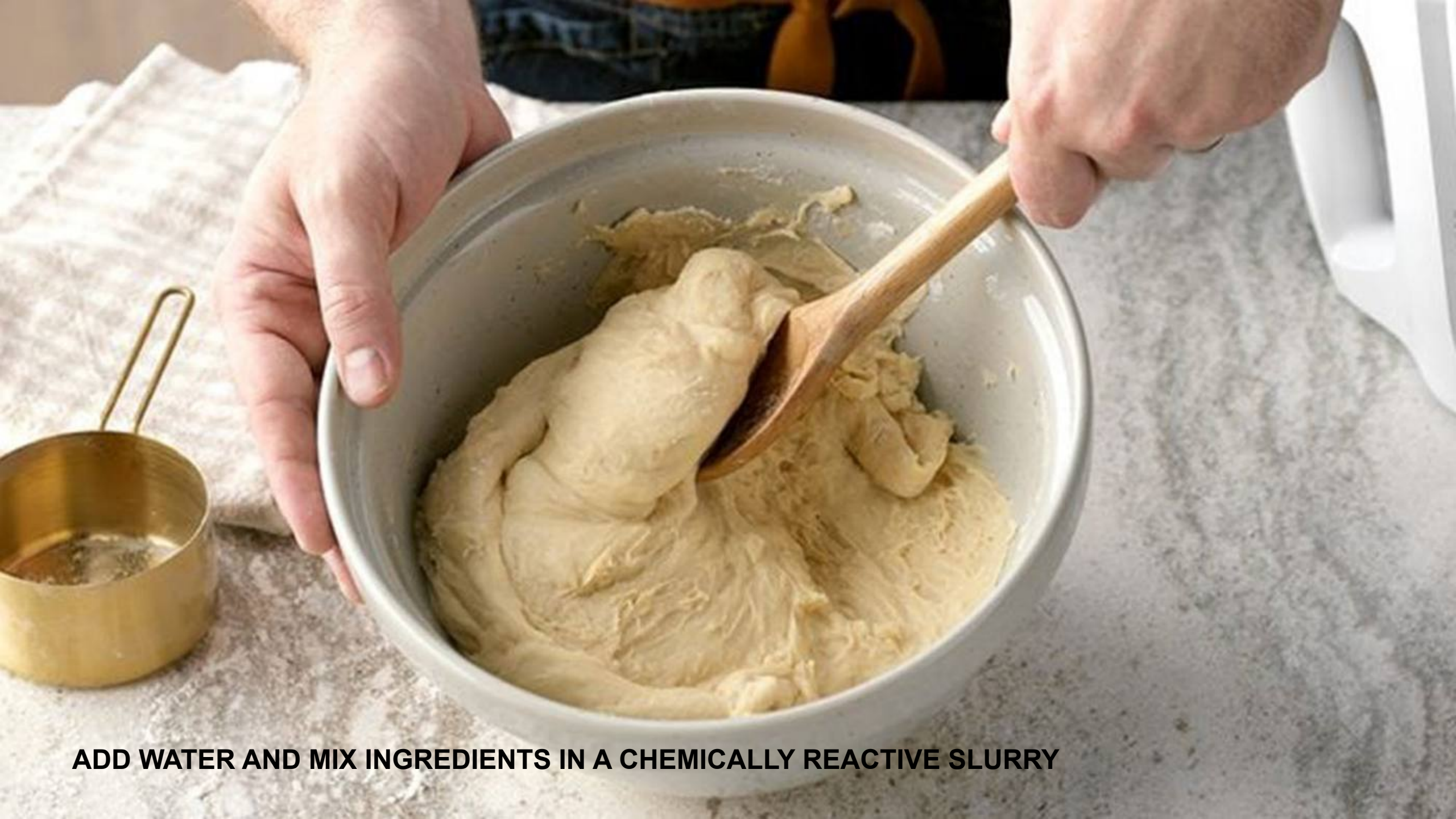
CONCRETE

…a material of readily available local materials and simple ingredients

CONCRETE

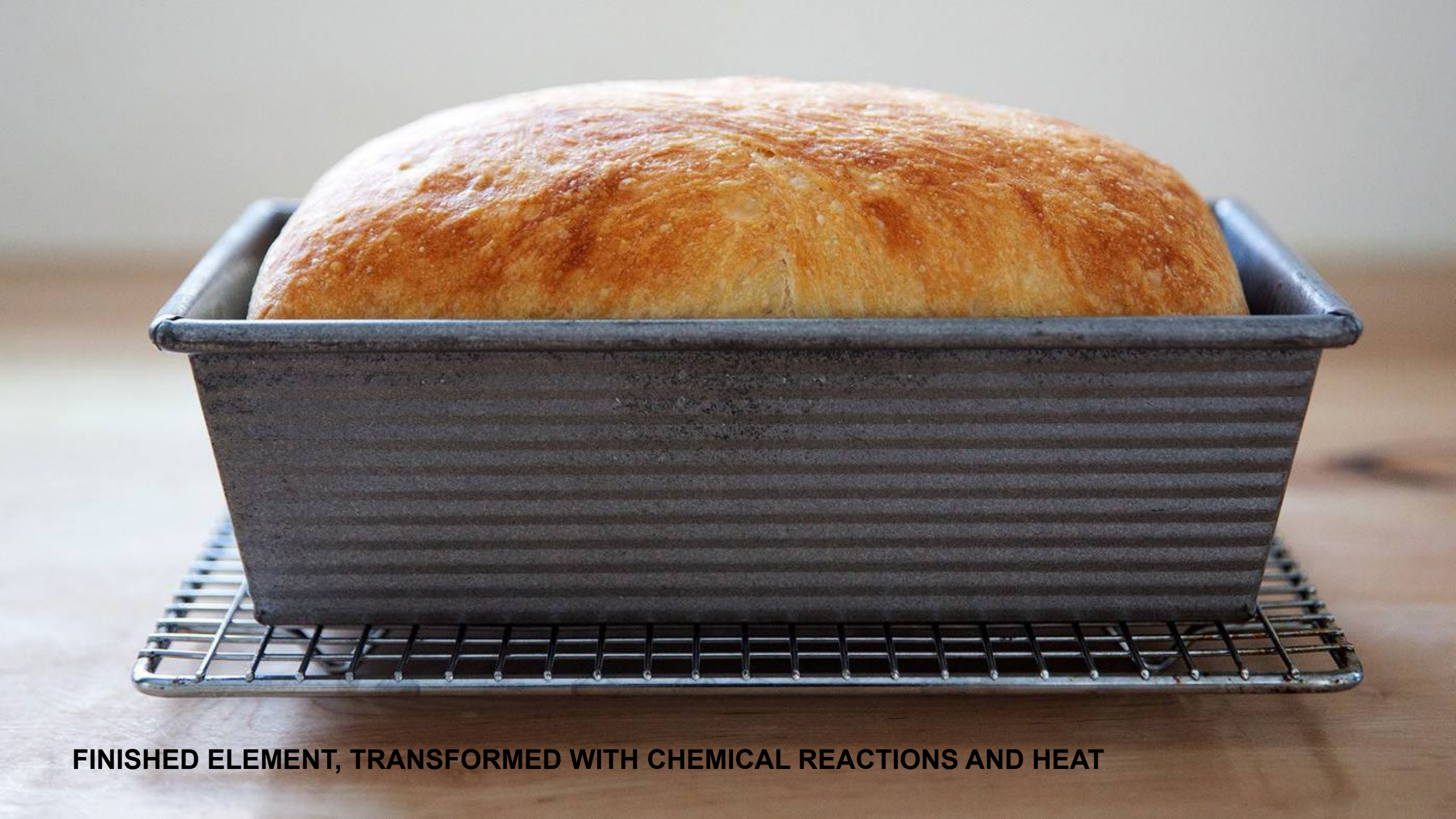
…a material that magically (chemically) transforms from dry solids, to liquid, to a hard solid

Water
Flour
Salt
Yeast
Oil
SIMPLE INGREDIENTS



ADD WATER AND MIX INGREDIENTS IN A CHEMICALLY REACTIVE SLURRY

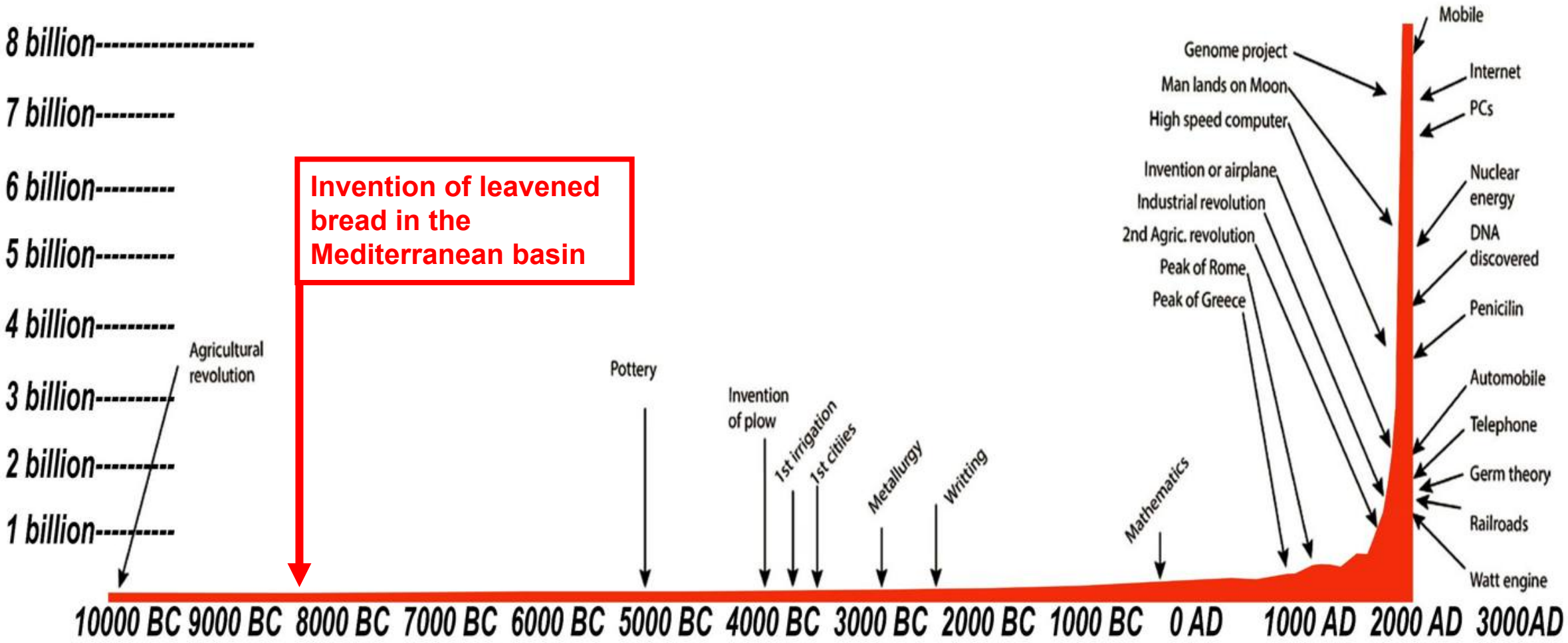
SLURRY IN REUSABLE FORMWORK, OF A SPECIFIC SIZE



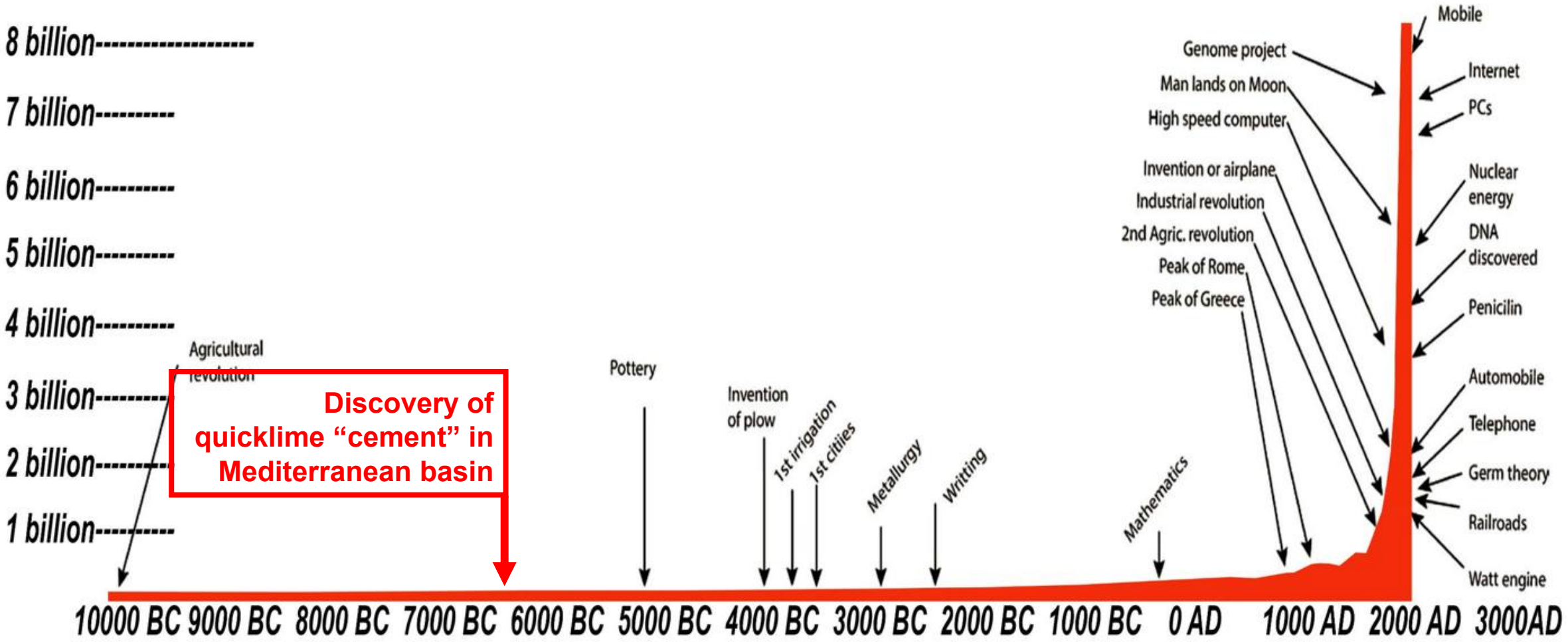
FINISHED ELEMENT, TRANSFORMED WITH CHEMICAL REACTIONS AND HEAT

CONCRETE

…a brief history of the evolution of concrete



HISTORY OF TECHNOLOGY and HUMAN POPULATION



DISCOVERY AND USE OF LIME

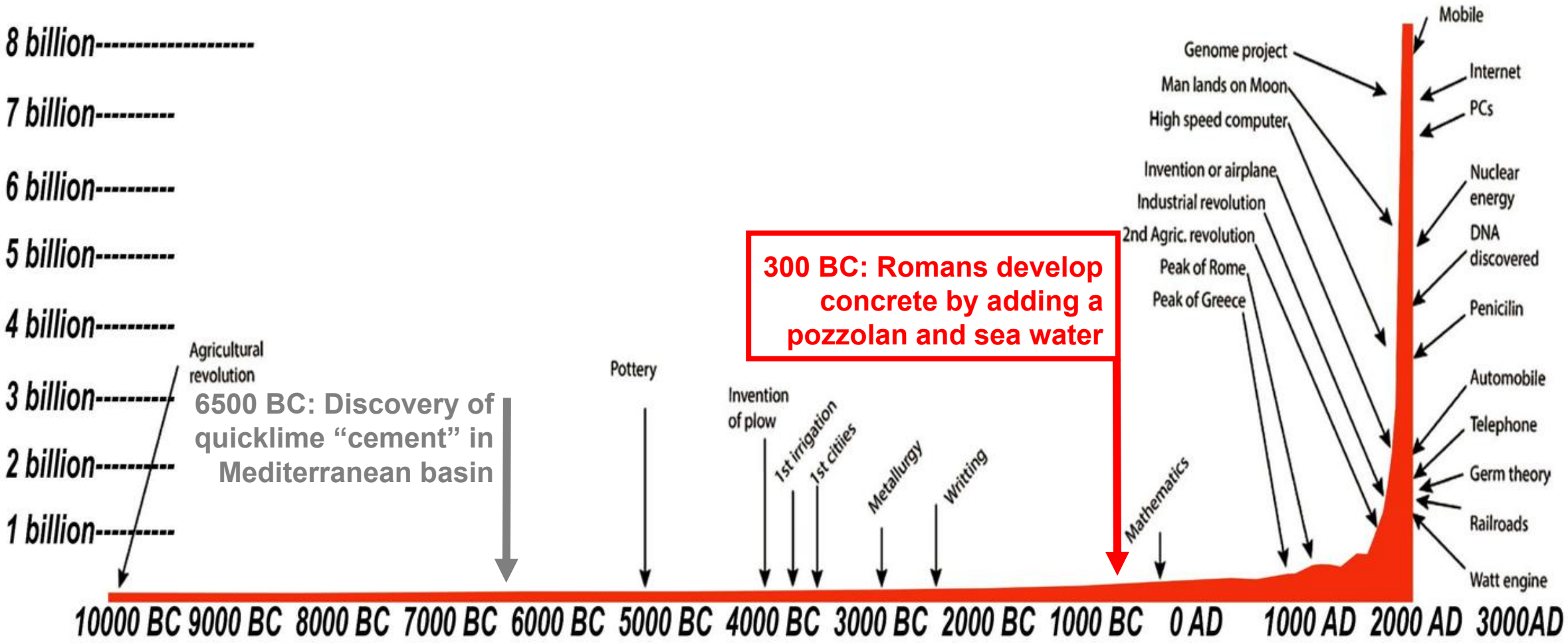
calcium carbonate + heat = calcium oxide & carbon dioxide
HEATED SHELLS CREATES “QUICKLIME”

calcium carbonate + heat = calcium oxide & carbon dioxide
HEATED LIMESTONE CREATES “QUICKLIME”

SLAKING LIME WITH WATER (exothermic reaction gives off heat)

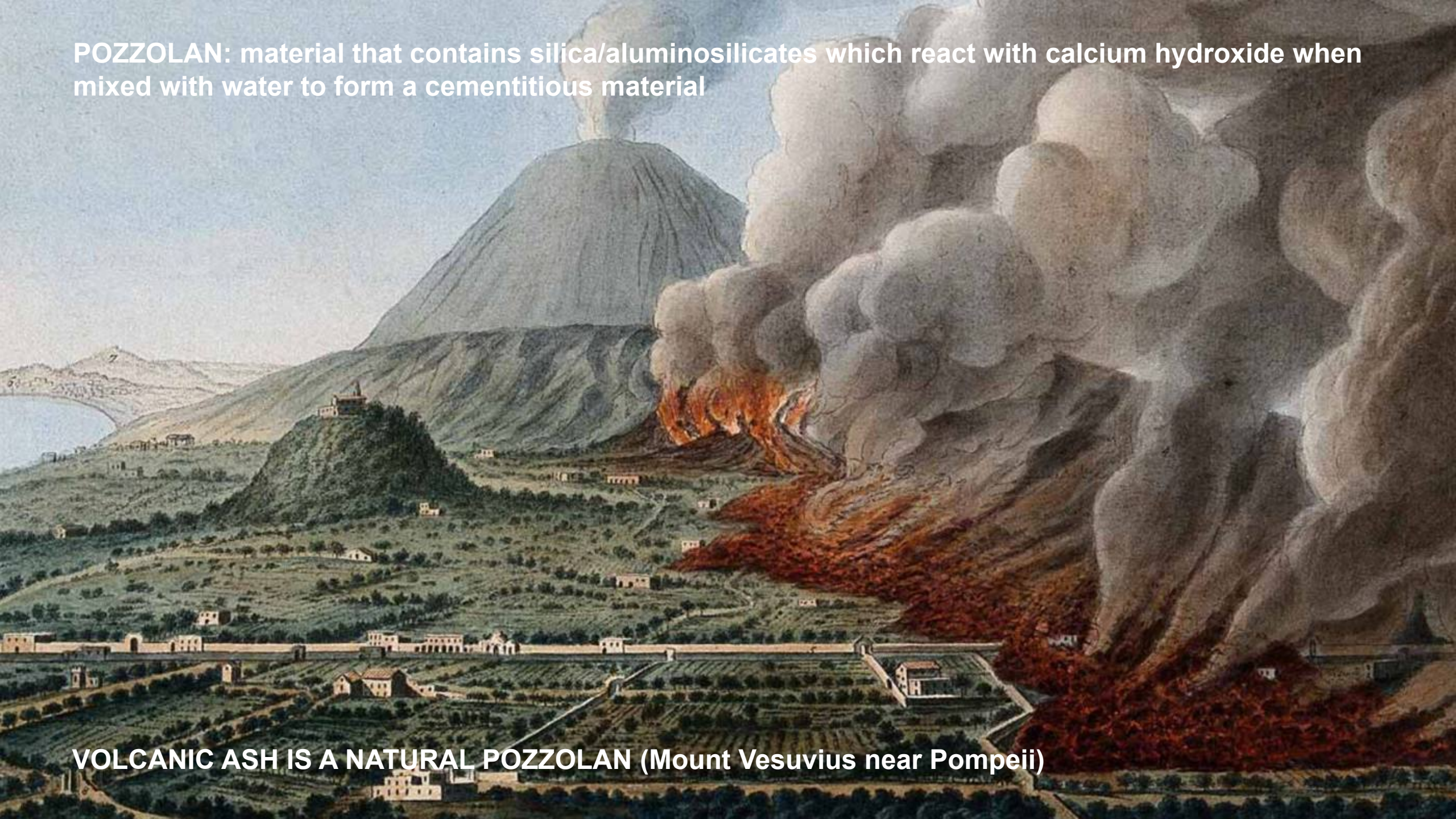
LIME, SAND, AND WATER bind together to make mortar and plaster (exothermic chemical reaction)

MORTAR USED TO BIND STONES TO FORM STRUCTURE



DEVELOPMENT OF CONCRETE

POZZOLAN: material that contains silica/aluminosilicates which react with calcium hydroxide when mixed with water to form a cementitious material



VOLCANIC ASH IS A NATURAL POZZOLAN (Mount Vesuvius near Pompeii)

Volcanic Pumice
Lime
Seashells
Volcanic Ash
Aggregate
SIMPLE INGREDIENTS OF ROMAN CONCRETE (+ water)

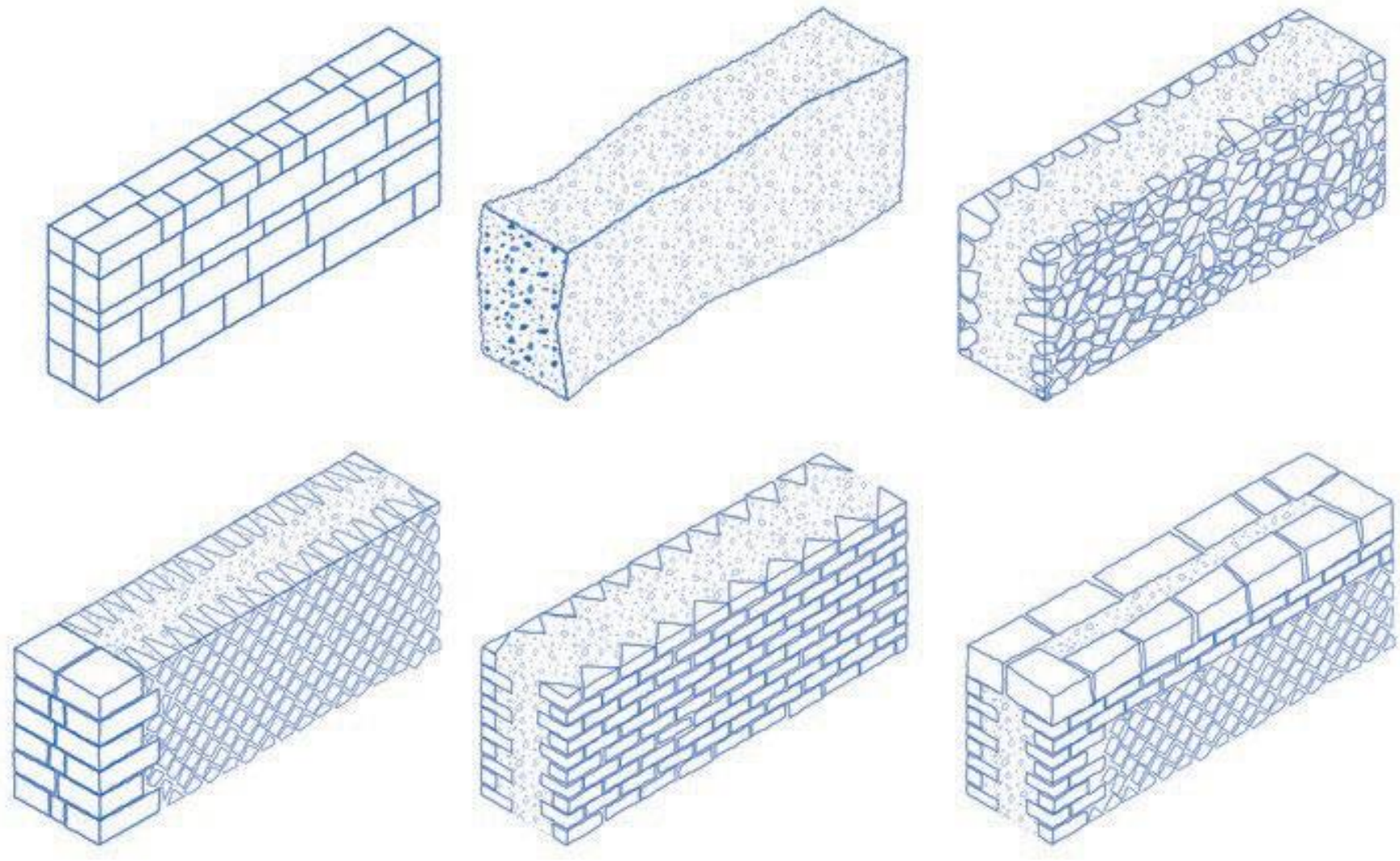
CONCRETE historical Roman concrete

When quicklime and pozzolan react with water, a complex chemical process takes place that results in the formation of a cementitious material. The calcium oxide in quicklime reacts with water to form calcium hydroxide, which then reacts with the silica and alumina in the pozzolan to produce calcium silicate hydrates (C-S-H) and calcium aluminate hydrates (C-A-H), which are the primary binding agents in cement.

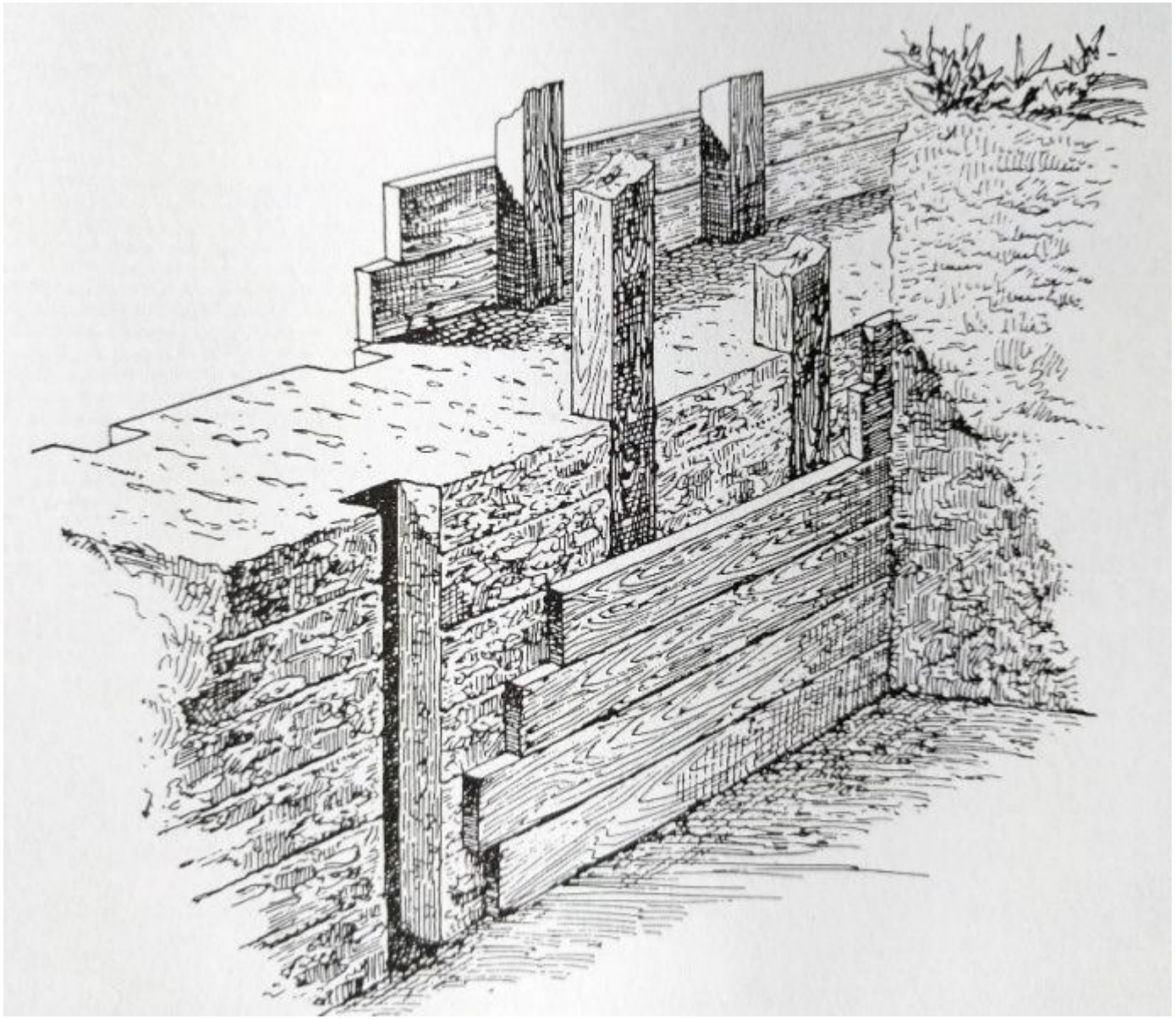
This reaction, known as the pozzolanic reaction, also produces significant heat and releases hydrogen gas. The C-S-H and C-A-H bind together to form a strong, durable material that hardens over time.

The chemical reaction between quicklime and pozzolan is highly complex and depends on a variety of factors, including the type and composition of the pozzolan, the temperature and humidity, and the presence of other minerals or additives to form a highly durable binding material.

Roman concrete contain small nodules of quicklime called “lime clasts” which have self-healing properties. When water enters a crack, the clasts dissolves and turns into calcium carbonate, filling the crack.



ROMAN WALLS USING CONCRETE AS INFILL AND STONE OR BRICK AS “FORMWORK”



ROMAN CONCRETE WALL WITH WOOD FORMWORK

CONCRETE DOME OF PANTHEON Rome, Italy Constructed in 126

LARGEST UNREINFORCED CONCRETE DOME IN THE WORLD (142 foot diameter, 142 feet high)

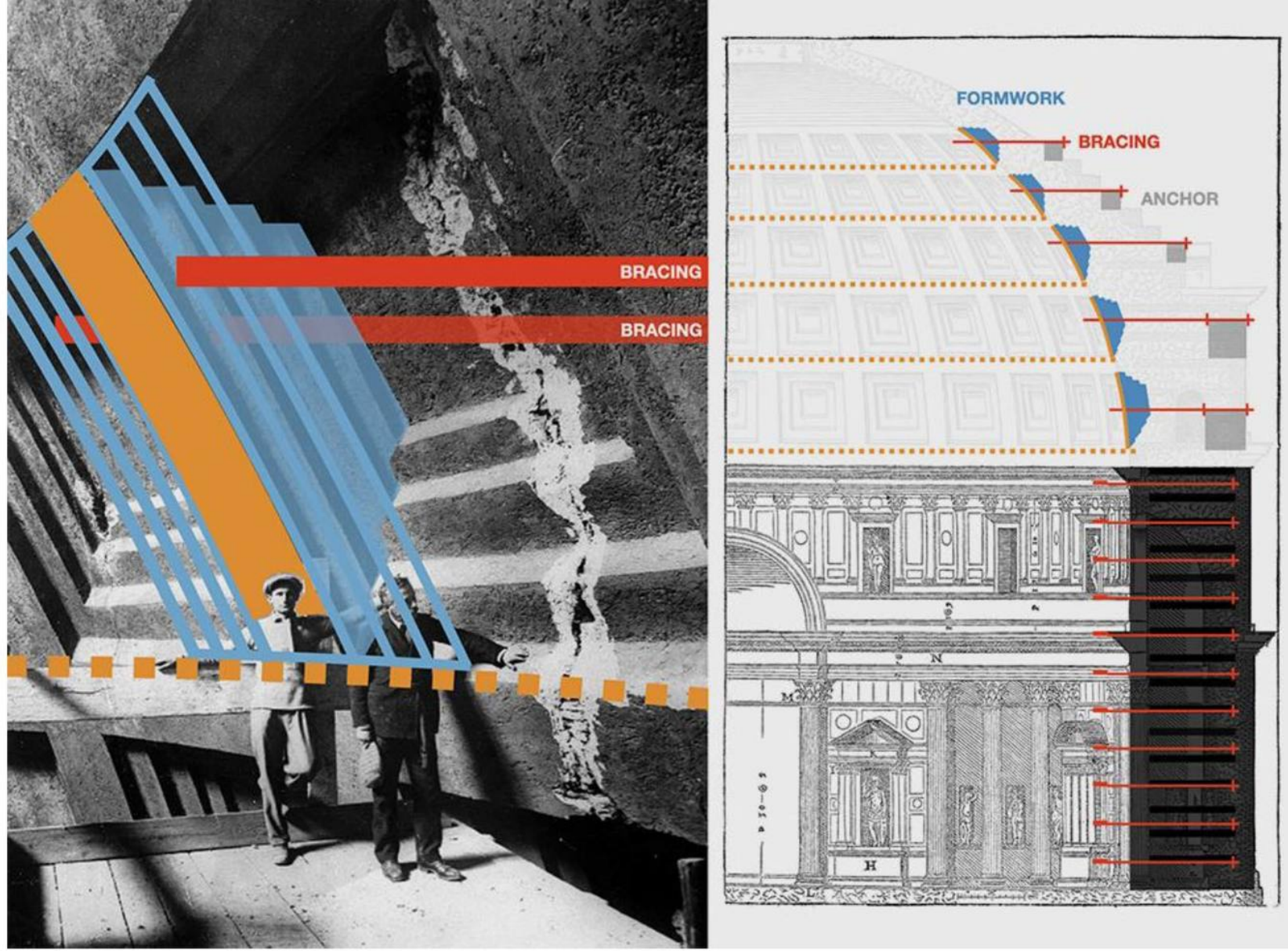


Fig. 2. The person with the hat is stretching out his arms indicating that this is how each coffer was cast using formwork placed in the space between each coffer. Each mold was stepped and then braced backwards using the outer wall as counterweight/anchor.

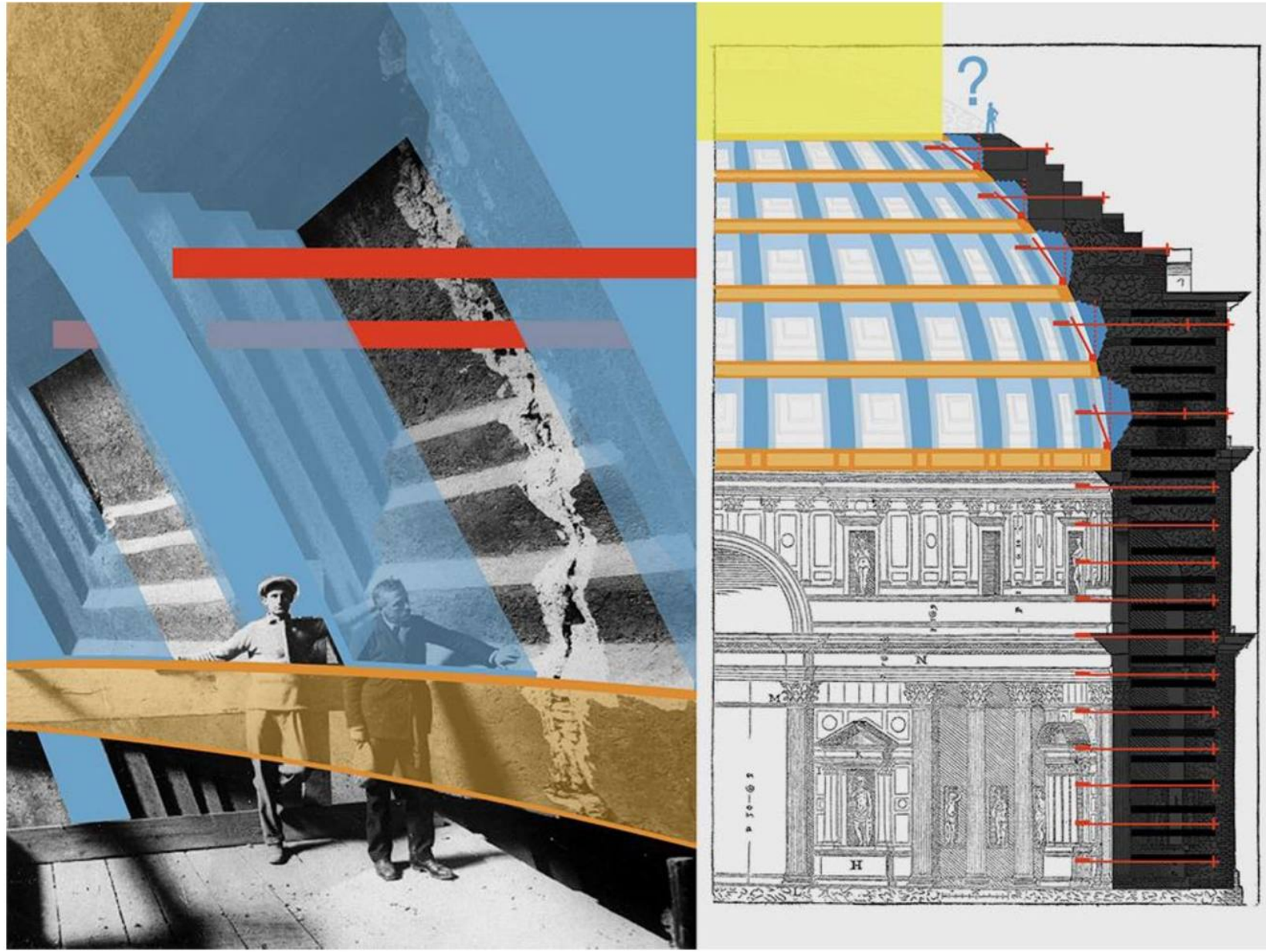


Fig. 5. If using the putlog system craftsmen can cast the entire coffer area of the dome of the Pantheon without any standing or flying centering system. Notice that the vertical edges are cast perfectly but horizontal surfaces are not, indicating that the mold was open.

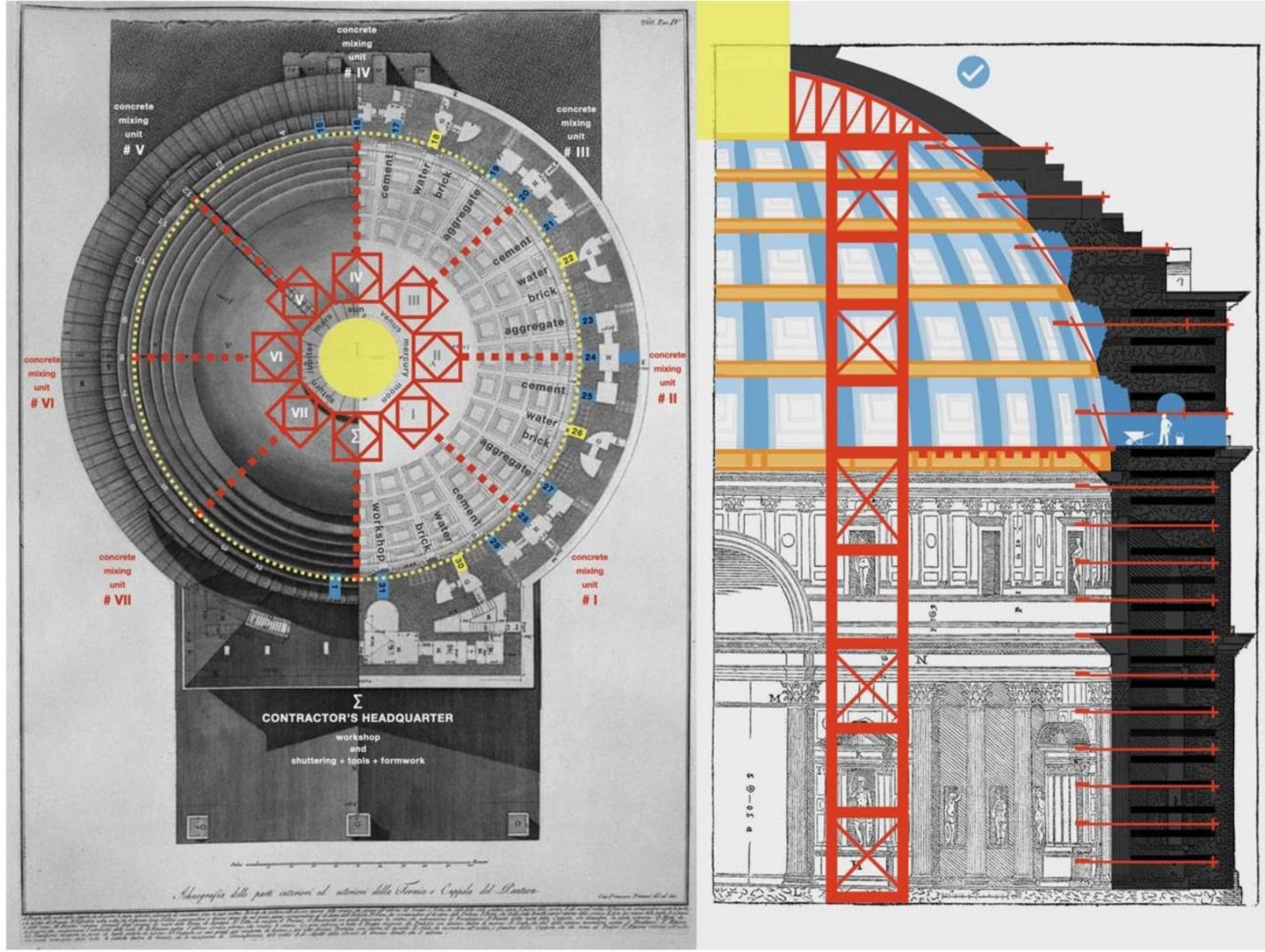
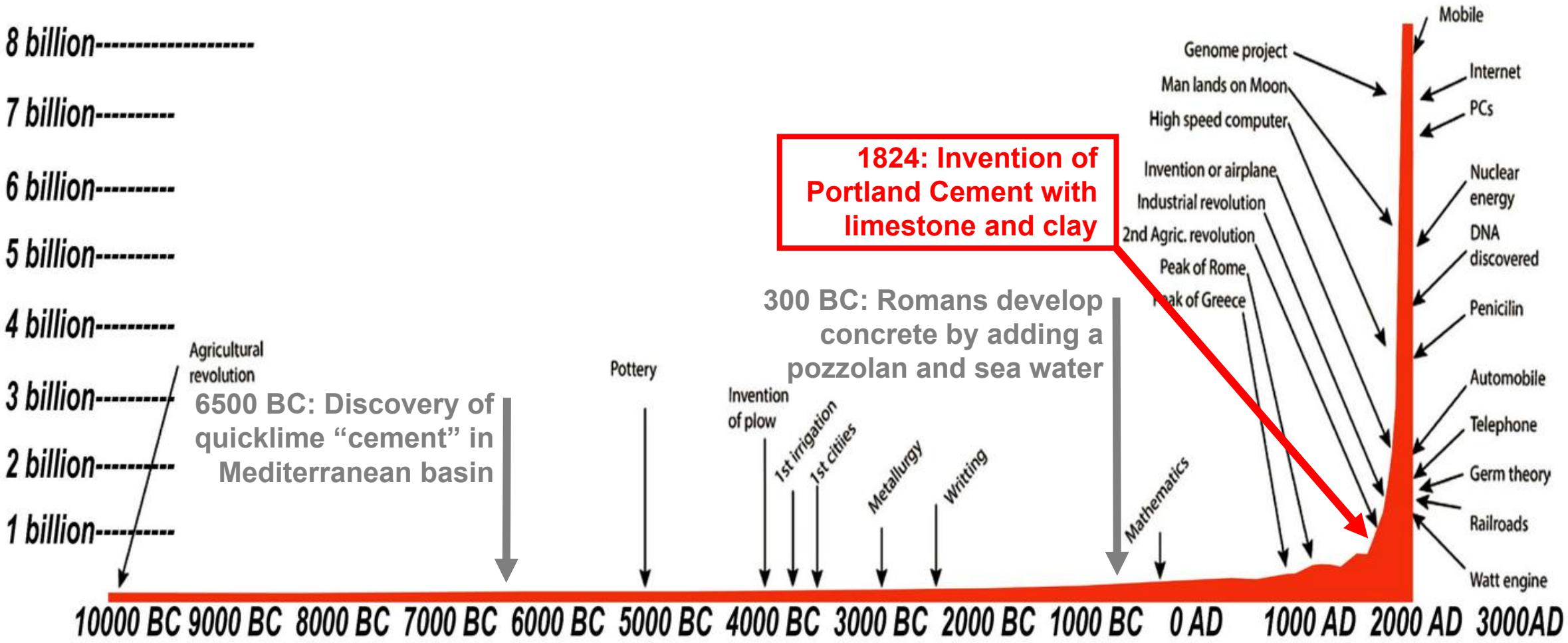


Fig. 6. To complete the casting of the dome a different approach was needed to execute the top with no coffers. By interlocking 8 or perhaps only 4 individual towers a top mold could be installed and each void in the springing of the dome could be utilized for mixing concrete. Notice that rooms in the springing of the dome and that stairs top roof line up with the towers. It is a factory for casting.



DEVELOPMENT OF CONCRETE

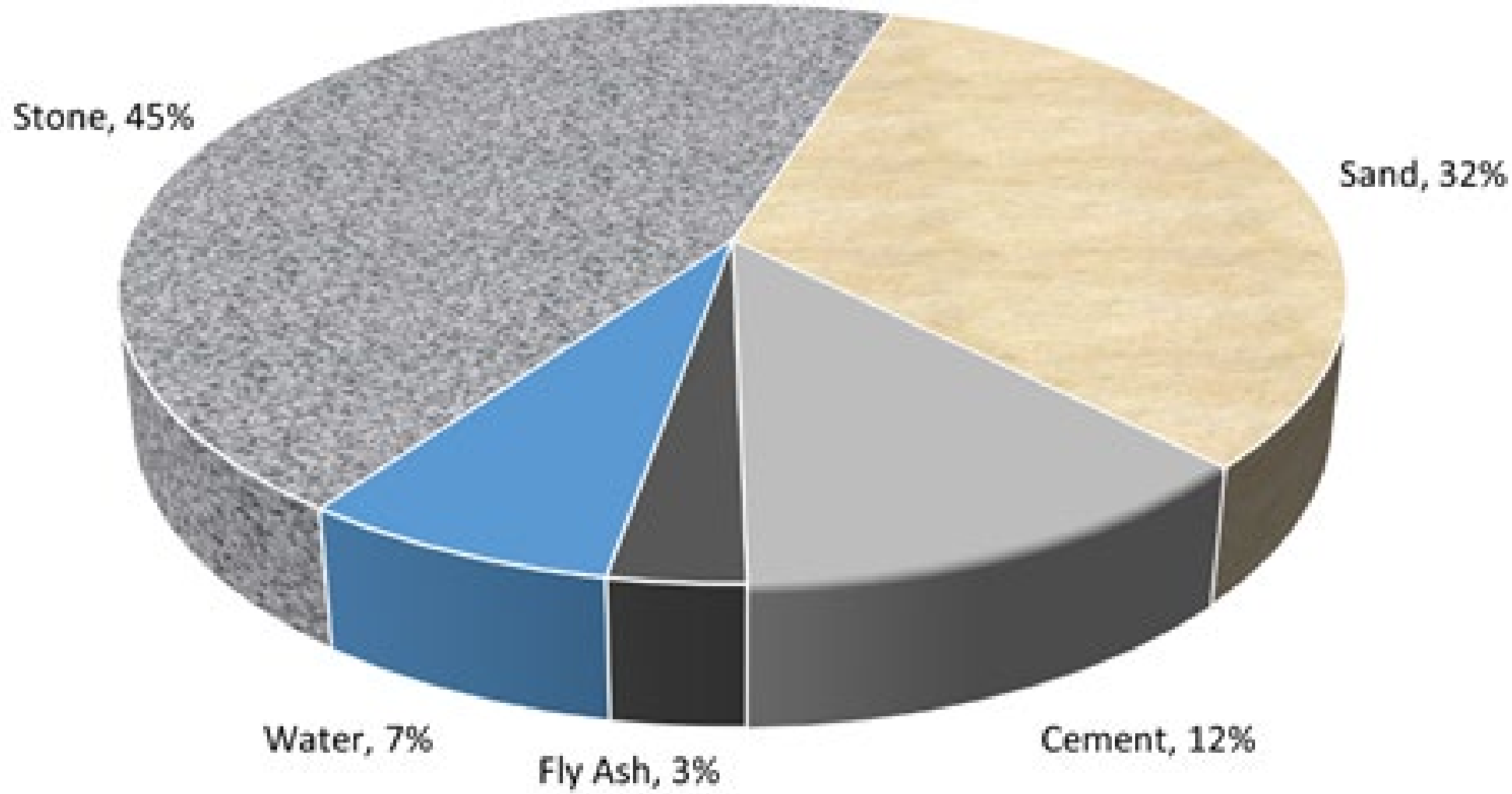
CONCRETE contemporary Portland cement

Portland cement-based concrete uses Portland cement as a binding material. Portland cement is made from a mixture of limestone, clay, and other materials that are heated to high temperatures in a kiln. This process produces a material called clinker, which is ground up and mixed with gypsum to form Portland cement.

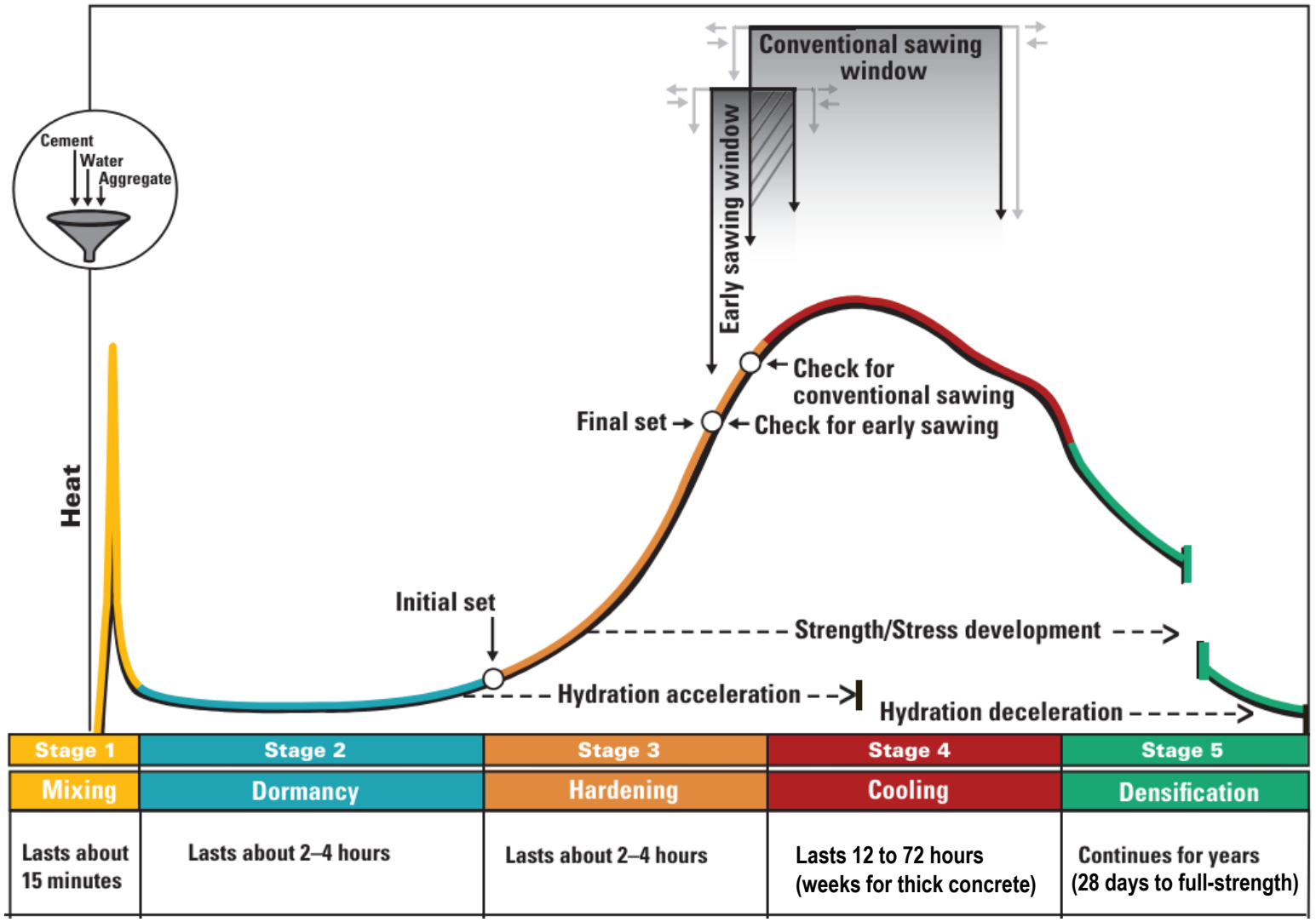
While Portland cement-based concrete is strong and durable, it is simply not as long-lasting as ancient concrete. This is because Portland cement-based concrete is more susceptible to chemical and physical degradation over time, especially when exposed to environmental factors such as salts and freeze-thaw cycles.

However, Portland cement-based concrete is made from readily available materials such as limestone, clay, and other minerals, making it more accessible and less expensive than pozzolana-based concrete, which required the importation of materials like volcanic ash from specific locations. Additives today also allow modern concrete to be formulated to resist specific environmental factors like freeze-thaw cycles or chemical exposure.

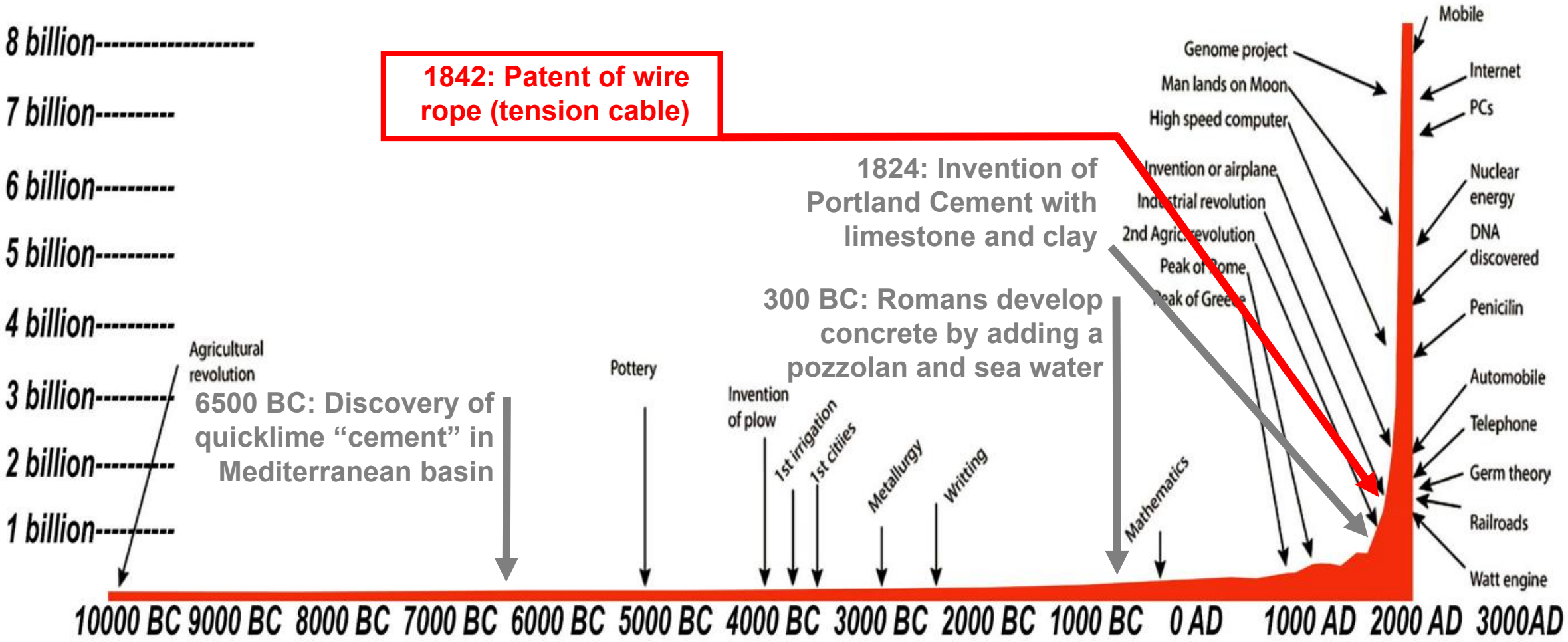
SIMPLE INGREDIENTS



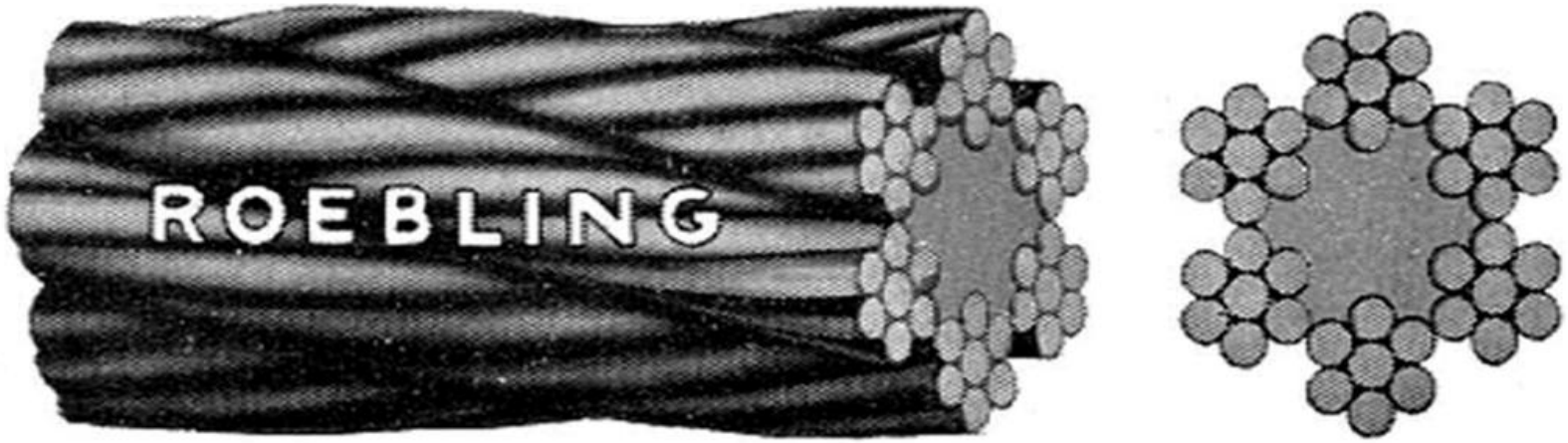
CONCRETE INGREDIENTS BY WEIGHT



CONCRETE CURING PROCESS in FIVE STAGES



DEVELOPMENT OF CONCRETE



Standard Coarse Laid Rope

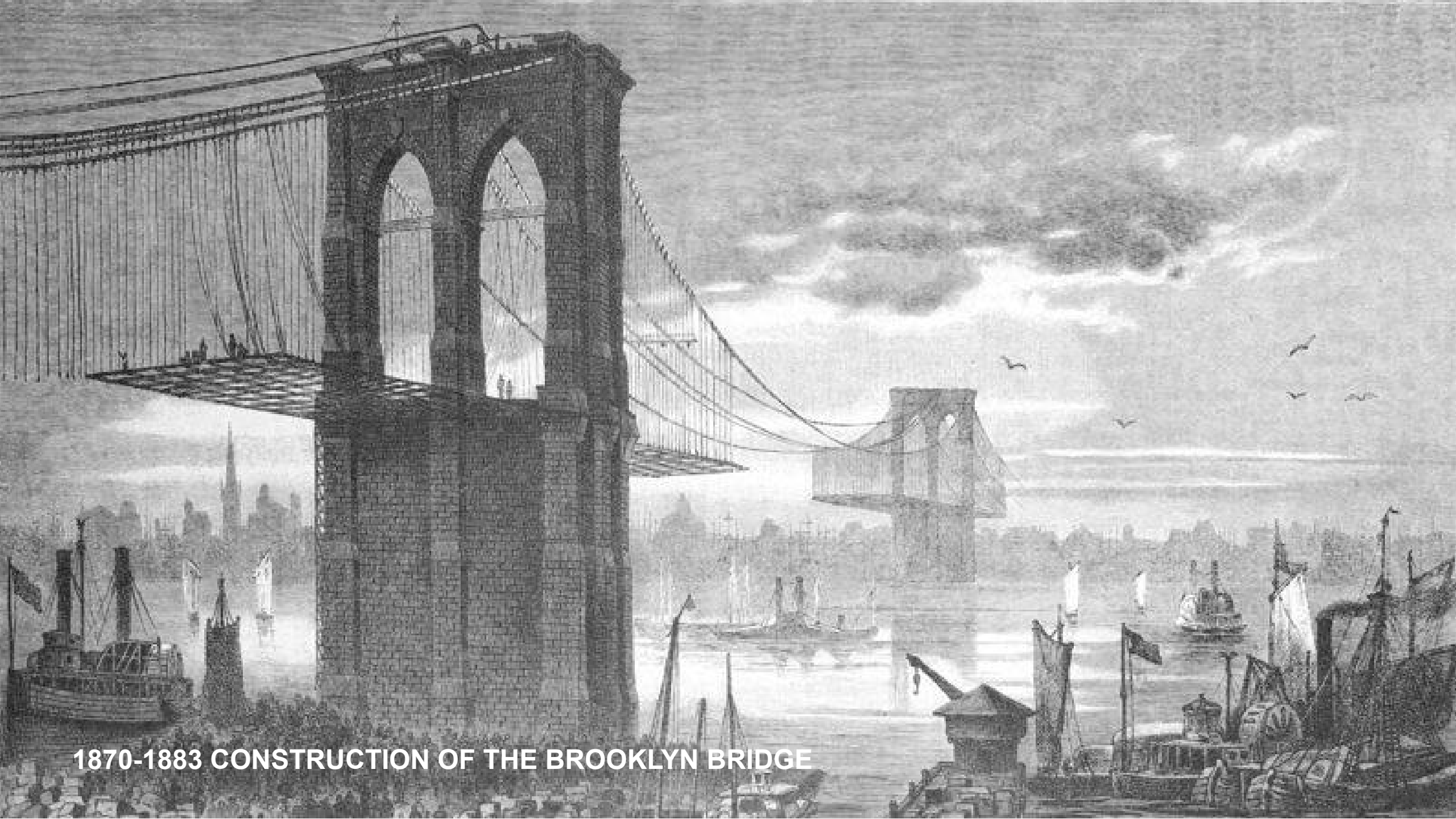
For Haulages and Transmissions

Composed of 6 Strands and a Hemp Center,

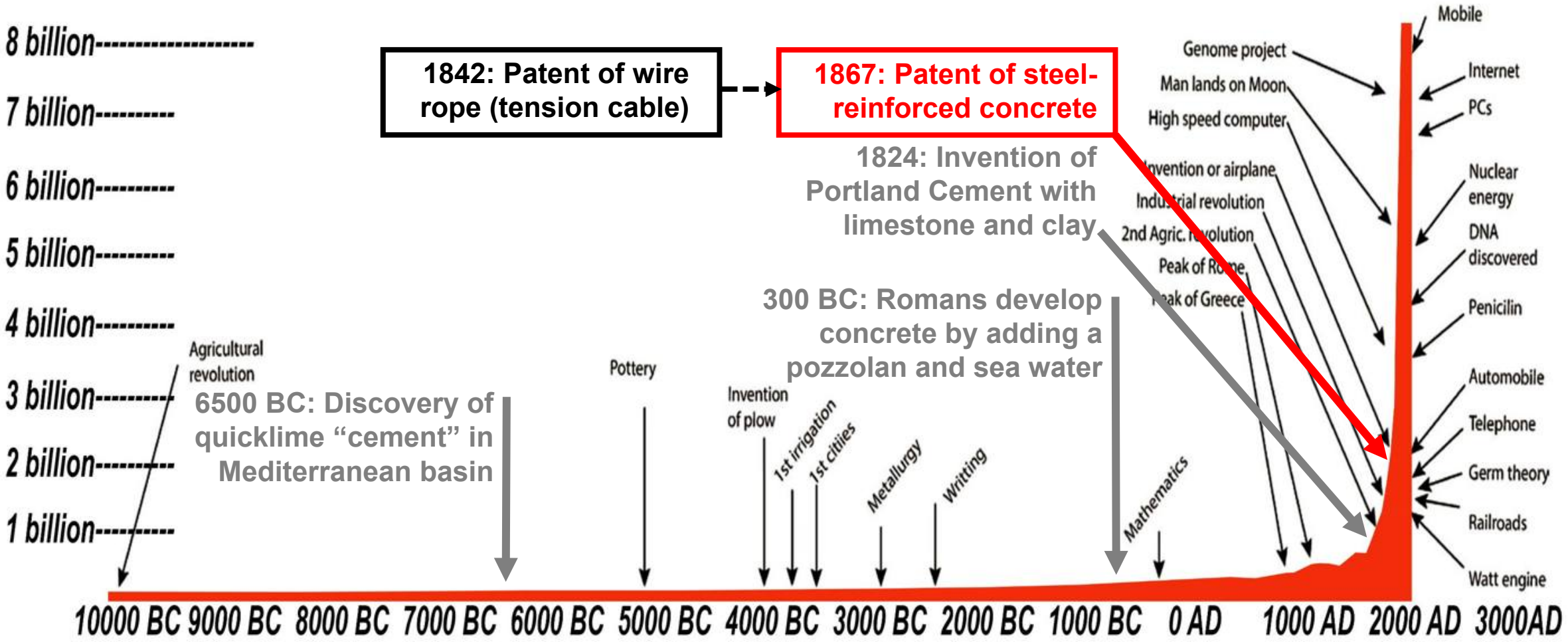
7 Wires to the Strand

"Blue Center" Steel

1842 PATENT FOR WIRE ROPE (John Roebling)



1870-1883 CONSTRUCTION OF THE BROOKLYN BRIDGE



DEVELOPMENT OF CONCRETE

Premier feuillet.

REPUBLIQUE FRANÇAISE.

Liberté, Égalité, Fraternité.

PREFECTURE DE PÓLICE.

1re DIVISION. — 2e BUREAU. — 2e SECTION.

Jardinier Fleuriste

N° 1739.

Paris, ce 10 juin 1851

Le Sr Monier,

Joseph,

natif de St Quentin département Aisne.

âgé de 28 ans, taille 1 mètre 70 cent.;

cheveux et sourcils chat; front bombé;

yeux chat; nez fort; bouche moyen;

menton rond, visage ovale; teint ordre;

Signes particuliers

déclarant demeurer à Paris

rue Lobineau, N° 2

1867 Joseph Monier (French gardener) FIRST PATENT OF STEEL-REINFORCED CONCRETE

1892 Francois Hennebique FIRST PATENT OF STEEL-REINFORCED CONCRETE BUILDING SYSTEM

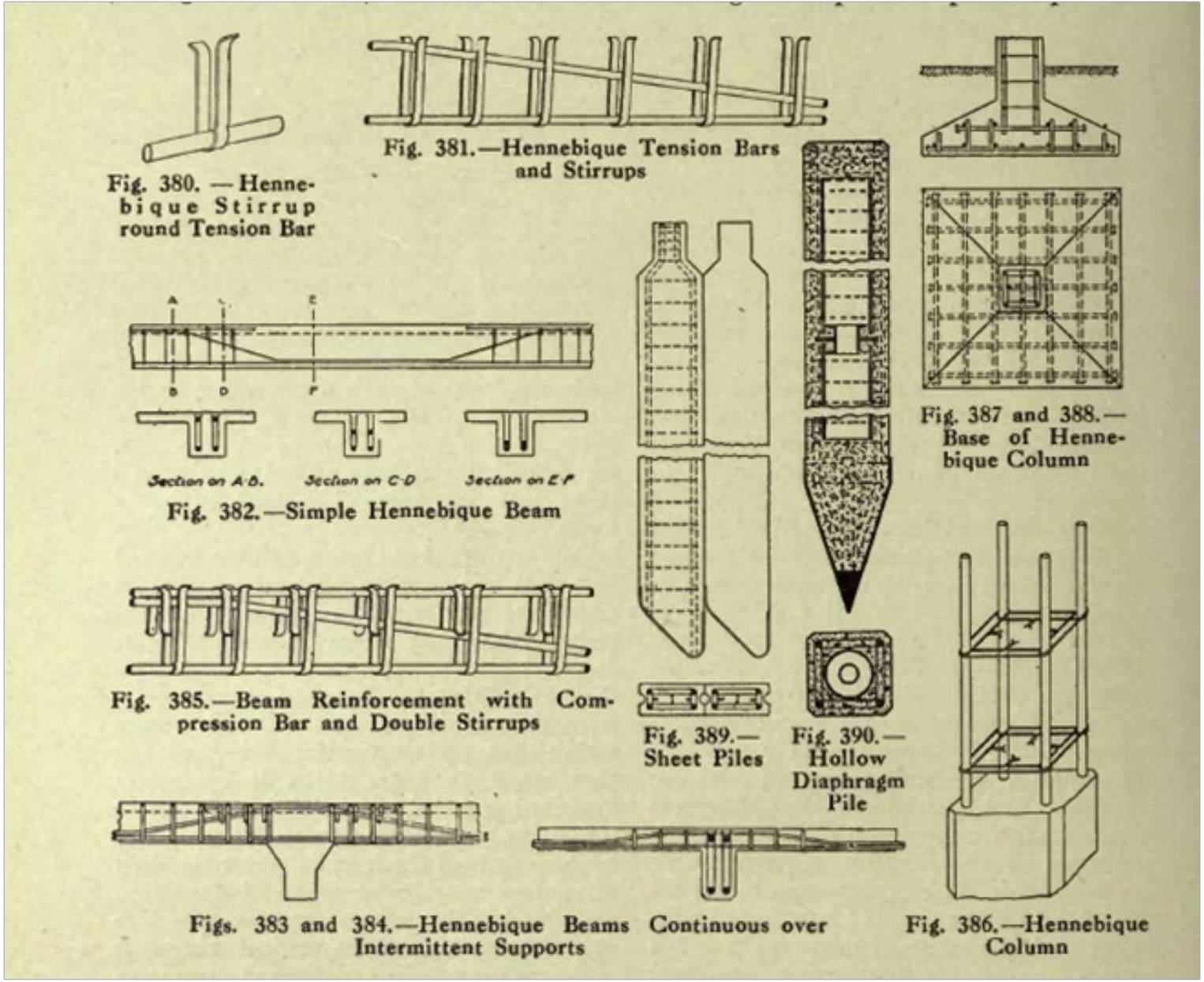


Fig. 380. — Hennebique Stirrup round Tension Bar

Fig. 381.—Hennebique Tension Bars and Stirrups

Fig. 382.—Simple Hennebique Beam

Fig. 385.—Beam Reinforcement with Compression Bar and Double Stirrups

Fig. 389.—Sheet Piles

Fig. 390.—Hollow Diaphragm Pile

Fig. 387 and 388.—Base of Hennebique Column

Figs. 383 and 384.—Hennebique Beams Continuous over Intermittent Supports

Fig. 386.—Hennebique Column

1892 Francois Hennebique FIRST PATENT OF STEEL-REINFORCED CONCRETE BUILDING SYSTEM

1894 CONSTRUCTION OF EGLISE SAINT-JEAN DE MONTMART, PARIS (Reinforced Concrete)

1894 CONSTRUCTION OF EGLISE SAINT-JEAN DE MONTMART, PARIS (Reinforced Concrete)

PRINCIPLES

OF

REINFORCED CONCRETE CONSTRUCTION

BY

F. E. TURNEAURE
Dean of the College of Engineering, University of Wisconsin

AND

E. R. MAURER
Professor of Mechanics, University of Wisconsin

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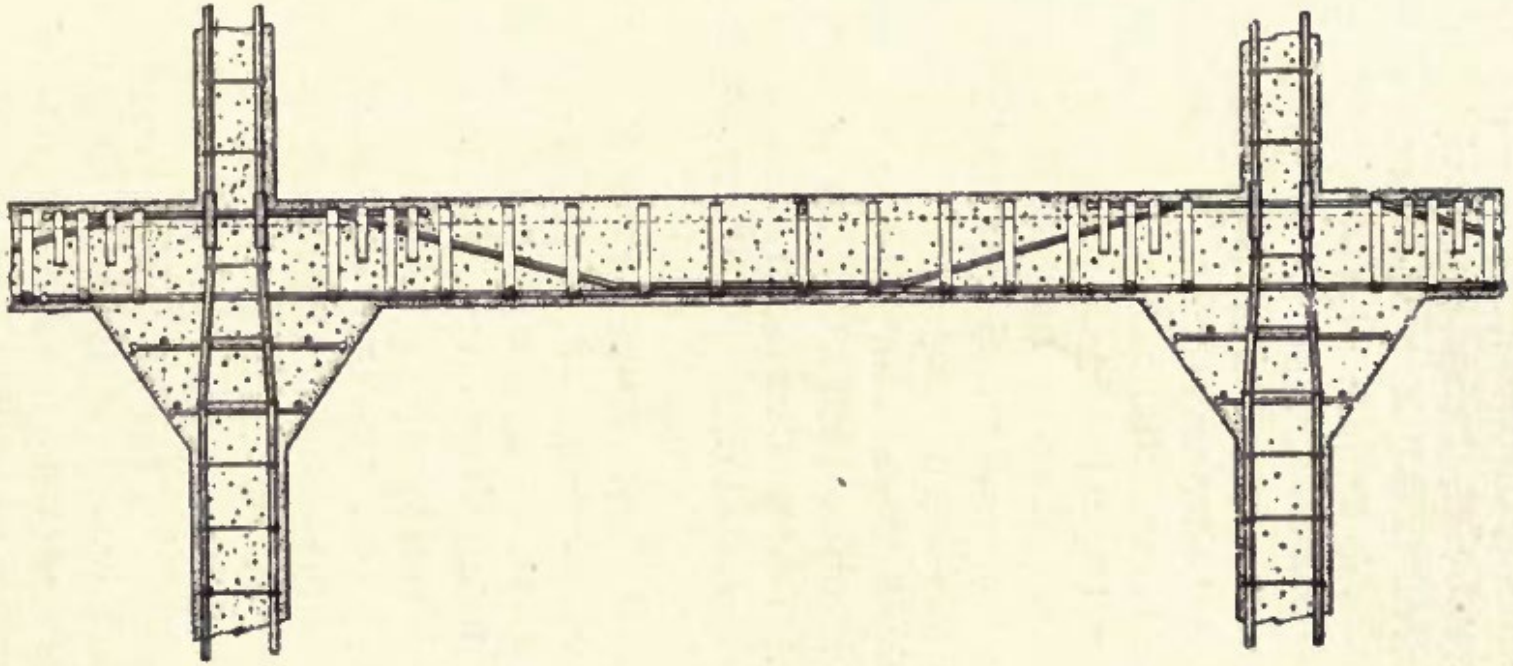


FIG. 41. TYPICAL ARRANGEMENT OF REINFORCEMENT IN HENNEBIQUE BEAMS AND COLUMNS

1907 EARLY PRINCIPLES AND CONFIGURATIONS OF REINFORCED CONCRETE in the U.S.

COMMON FORMULAS.

Rectangular beam:

	Reference to Page		Reference to Page
Depth of steel, $d = C\sqrt{\frac{M}{b}}$	418	Steel area, $A_s = pbd$	418
Tension in steel, $f_s = \frac{M}{A_s j d}$	420	Compression in concrete, $f_c = \frac{2M}{bd^2jk}$	420
Unit shear, $v = \frac{V}{bjd}$	447	Unit bond, $u = \frac{V}{jd\,\Sigma o}$	457
Ratio span to depth not requiring stirrups, $\frac{l}{d} > \frac{\alpha}{1.74\,C^2 v}$	456		

T-Beam:

Horizontal steel, $A_s = \frac{M}{f_s\left(d - \frac{t}{2}\right)}$ (approx.)	426	Economical depth, $d - \frac{t}{2} = \sqrt{\frac{r\,M}{f_s b}}$	425
Area required by shear, $b'\left(d - \frac{t}{2}\right) > \frac{V}{120}$	424	Length of haunch, $x = \frac{l}{5}\,\frac{M_b - M_r}{M_b}$ (approx.)	430
Depth of neutral axis, $kd = \frac{2\,nd\,A_s + bt^2}{2\,n\,A_s + 2\,bt}$	755	$s = \frac{3\,kd - 2\,t}{2\,kd - t}\,\frac{t}{3}$	755
Moment arm, $jd = d - s$	755		
Tension in steel, $f_s = \frac{M}{Ajd}$	756	Compression in concrete, $f_c = \frac{M\,kd}{bt\,(kd - \frac{1}{2}t)jd}$	756

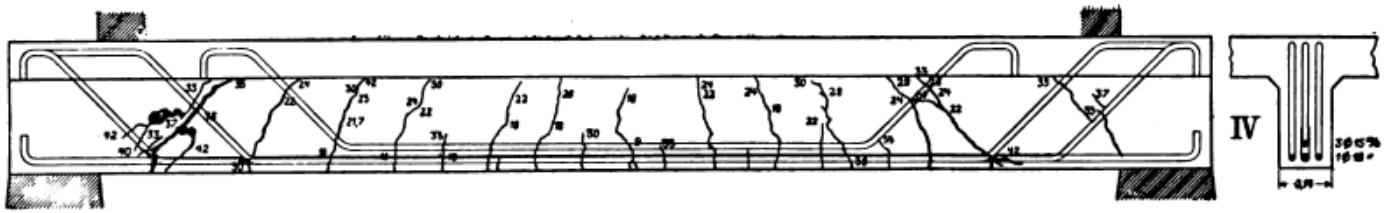


FIG. 147.—Beam IV, breaking load 42 t. (46.2 tons).

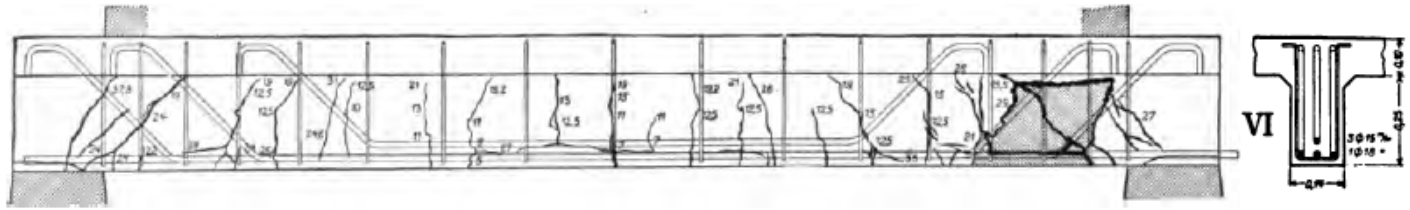


FIG. 148.—Beam VI, breaking load 37.8 t. (41.5 tons).

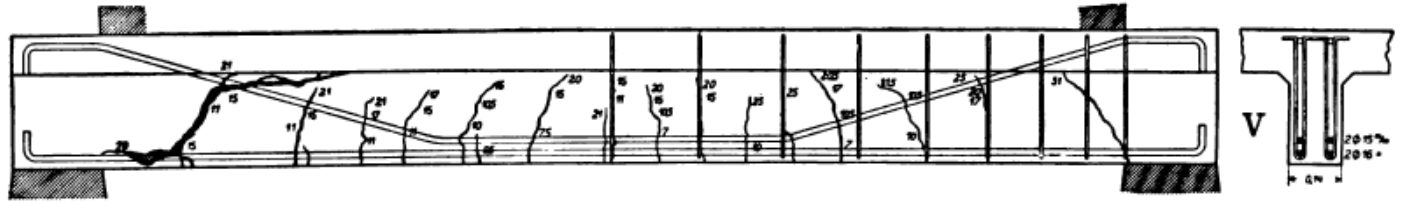
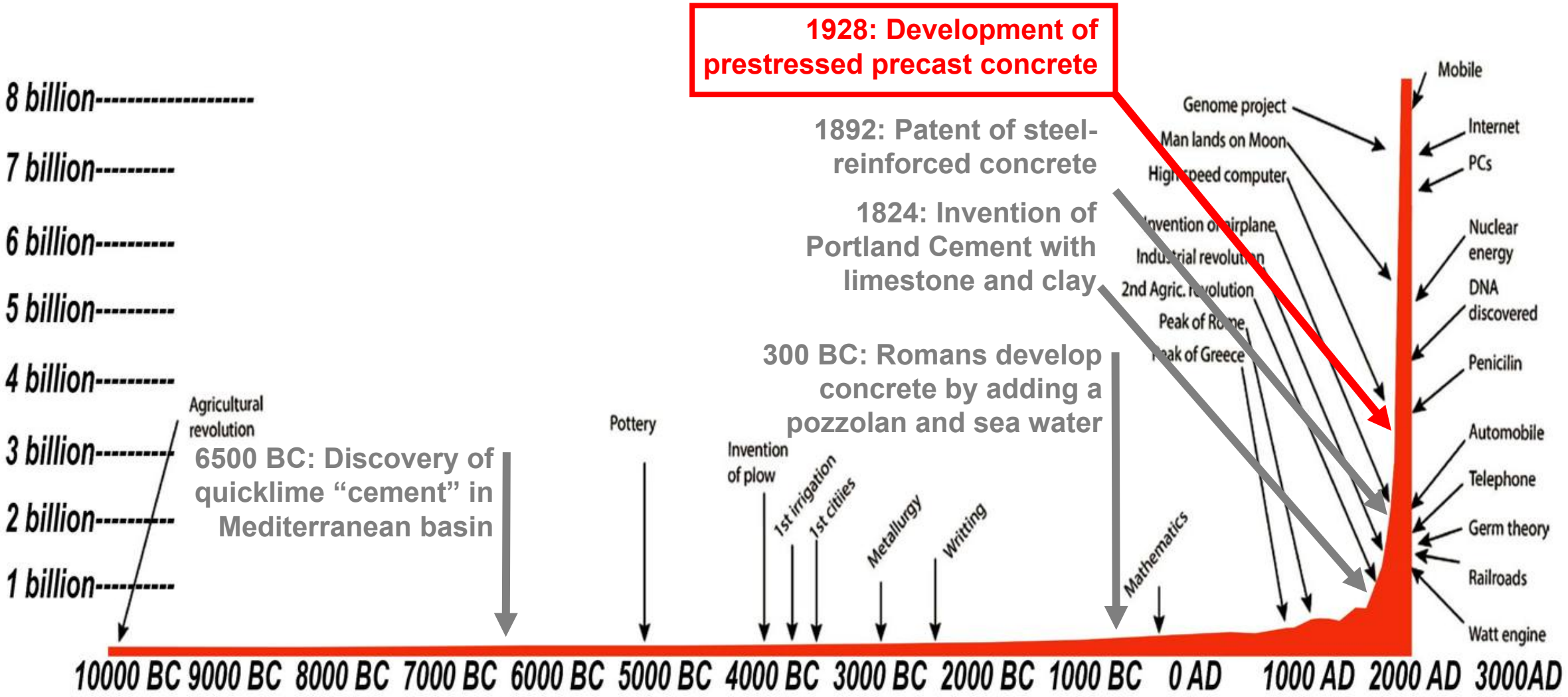
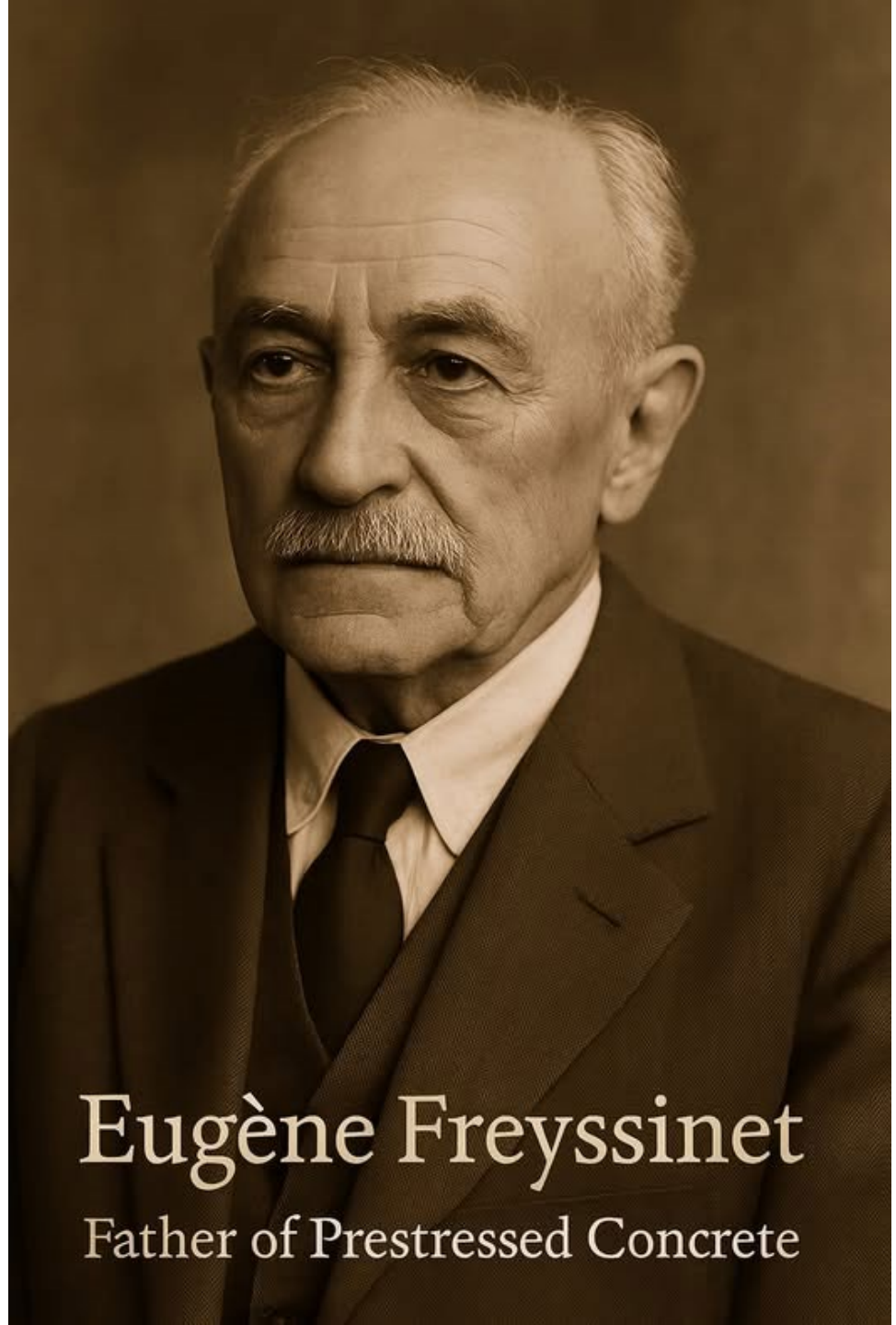


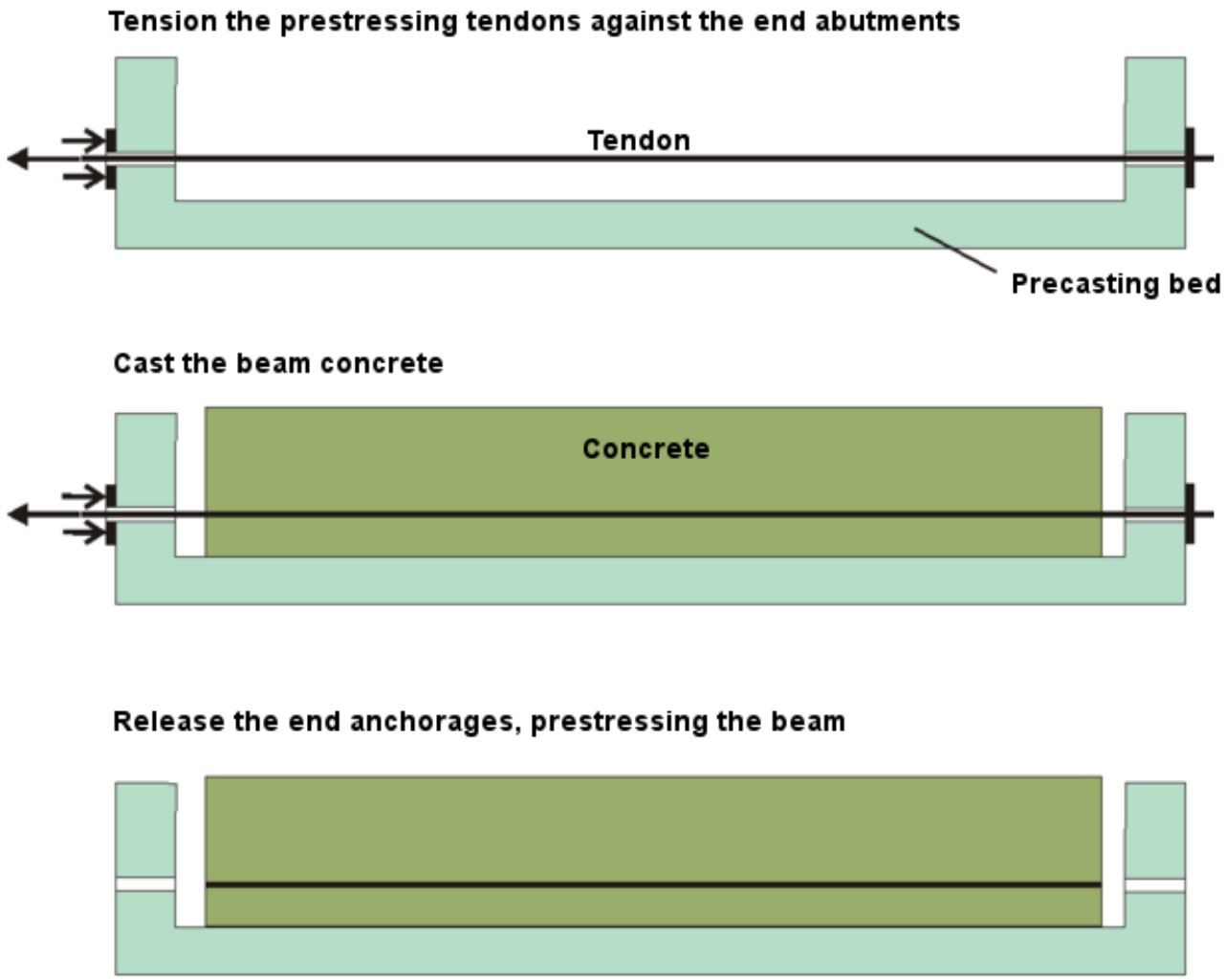
FIG. 149.—Beam V, breaking load 31 t. (34.1 tons).

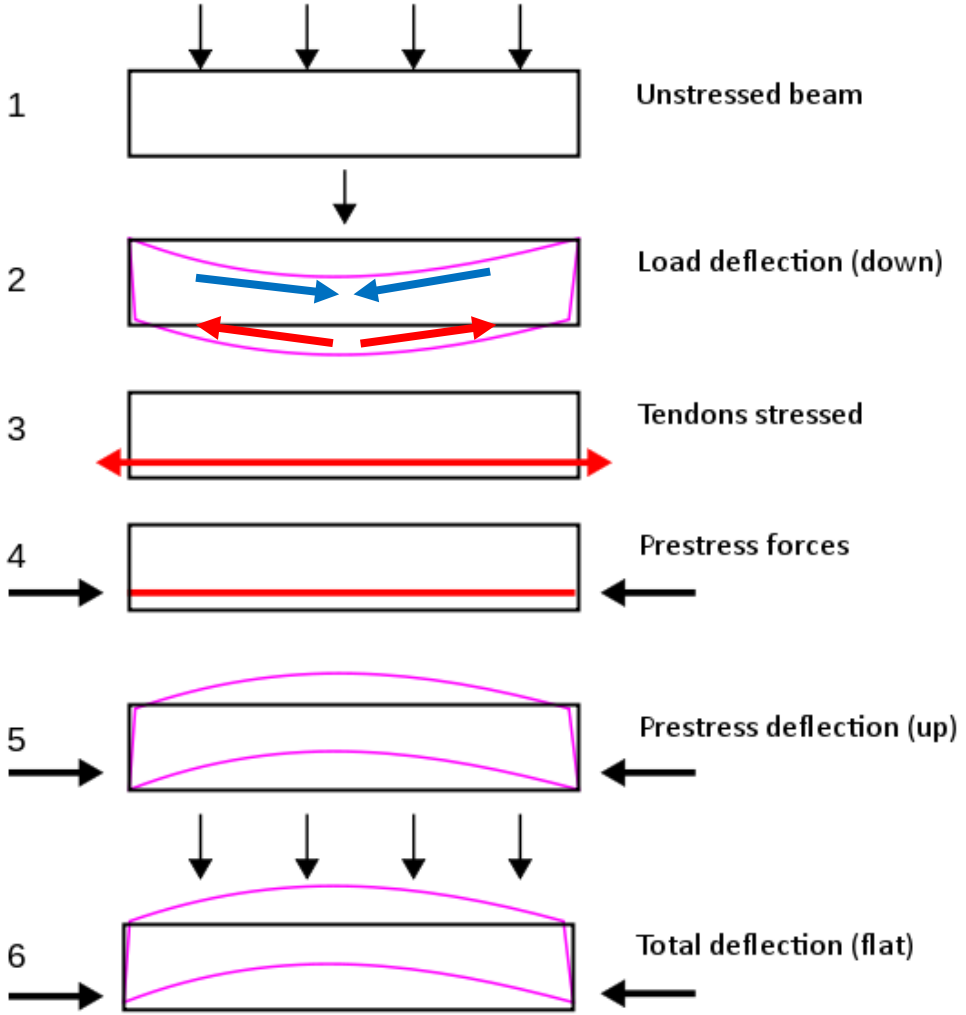
1907 EARLY CALCULATIONS AND TESTING OF COMPRESSION, TENSION & BENDING FORCES



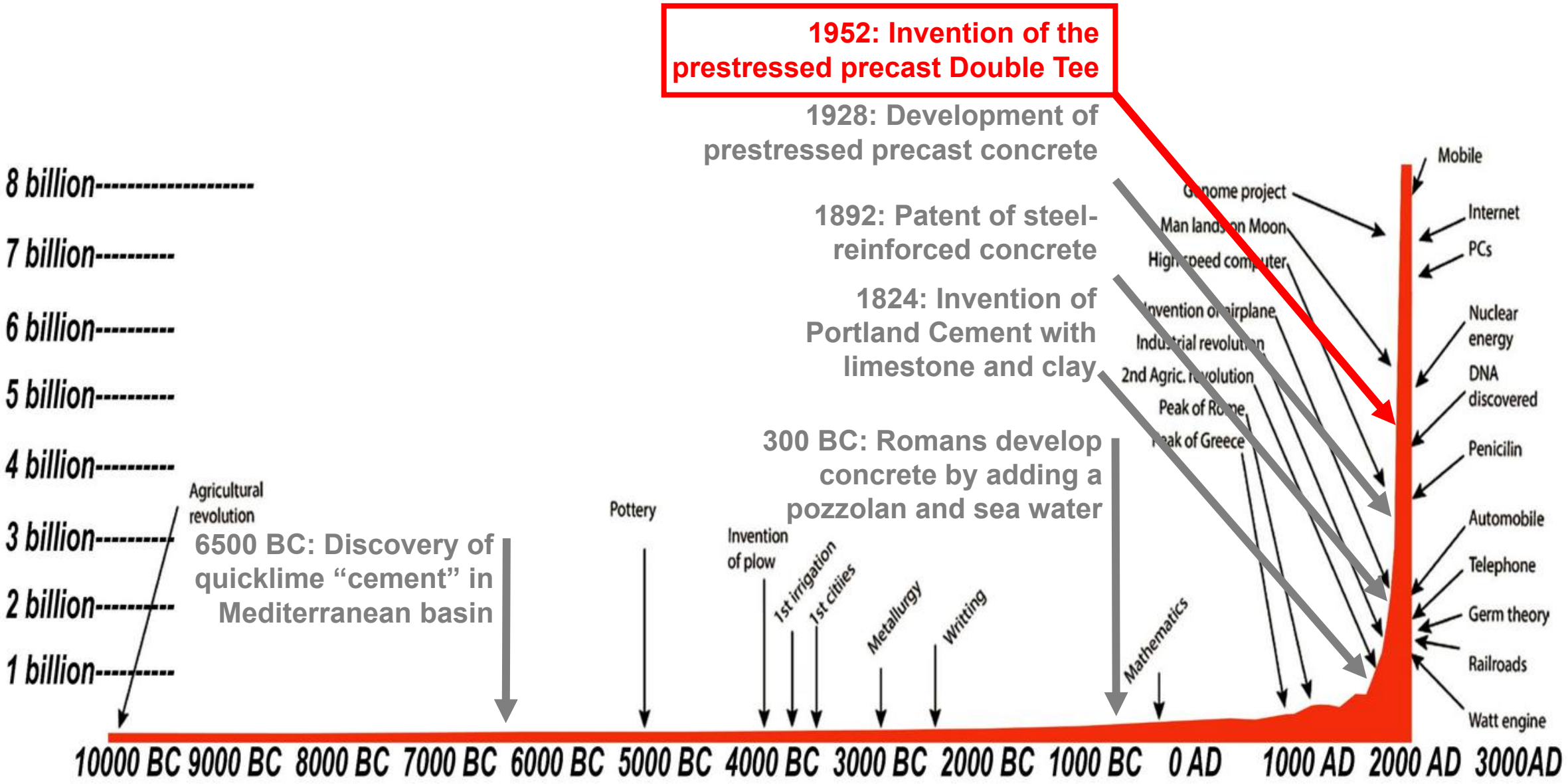
DEVELOPMENT OF CONCRETE


Eugène Freyssinet
Father of Prestressed Concrete

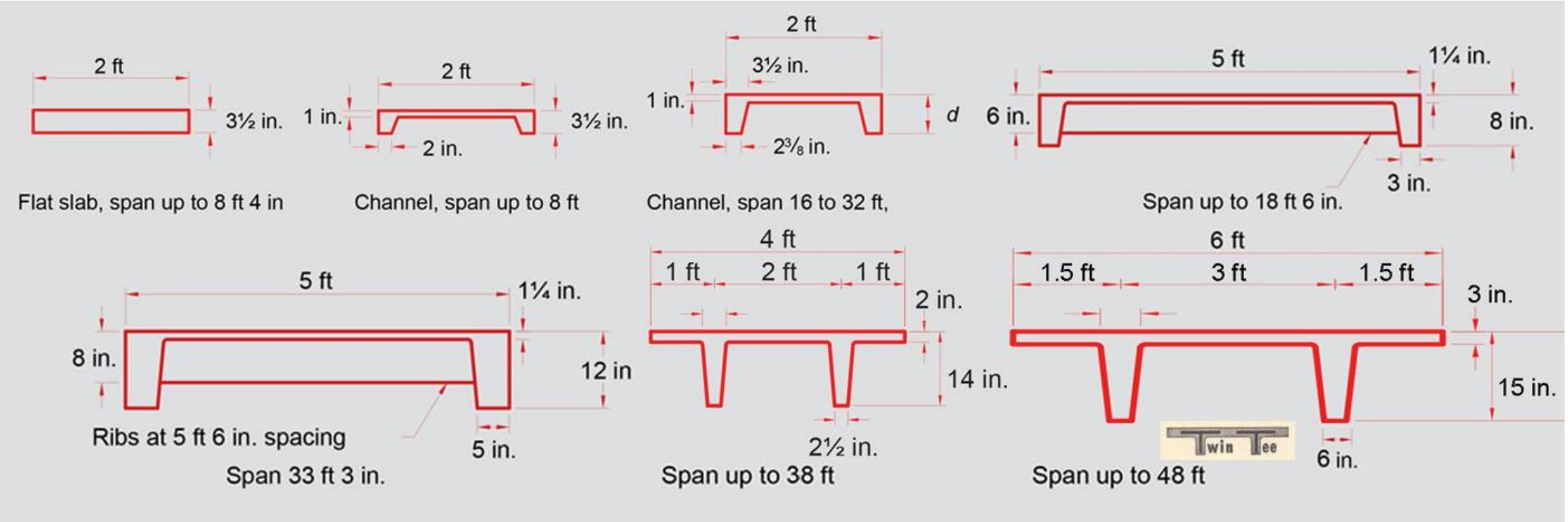




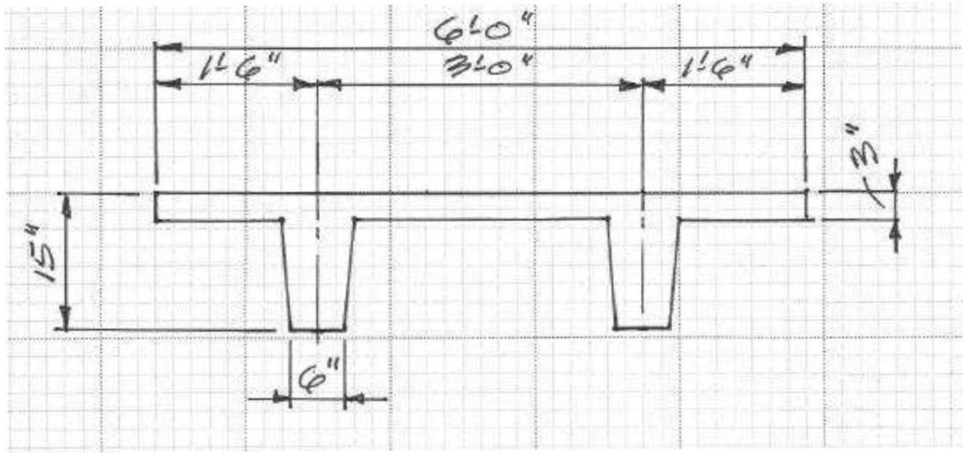
PRESTRESSED STEEL TENDONS COUNTERACT TENSION FORCES AND DEFLECTION



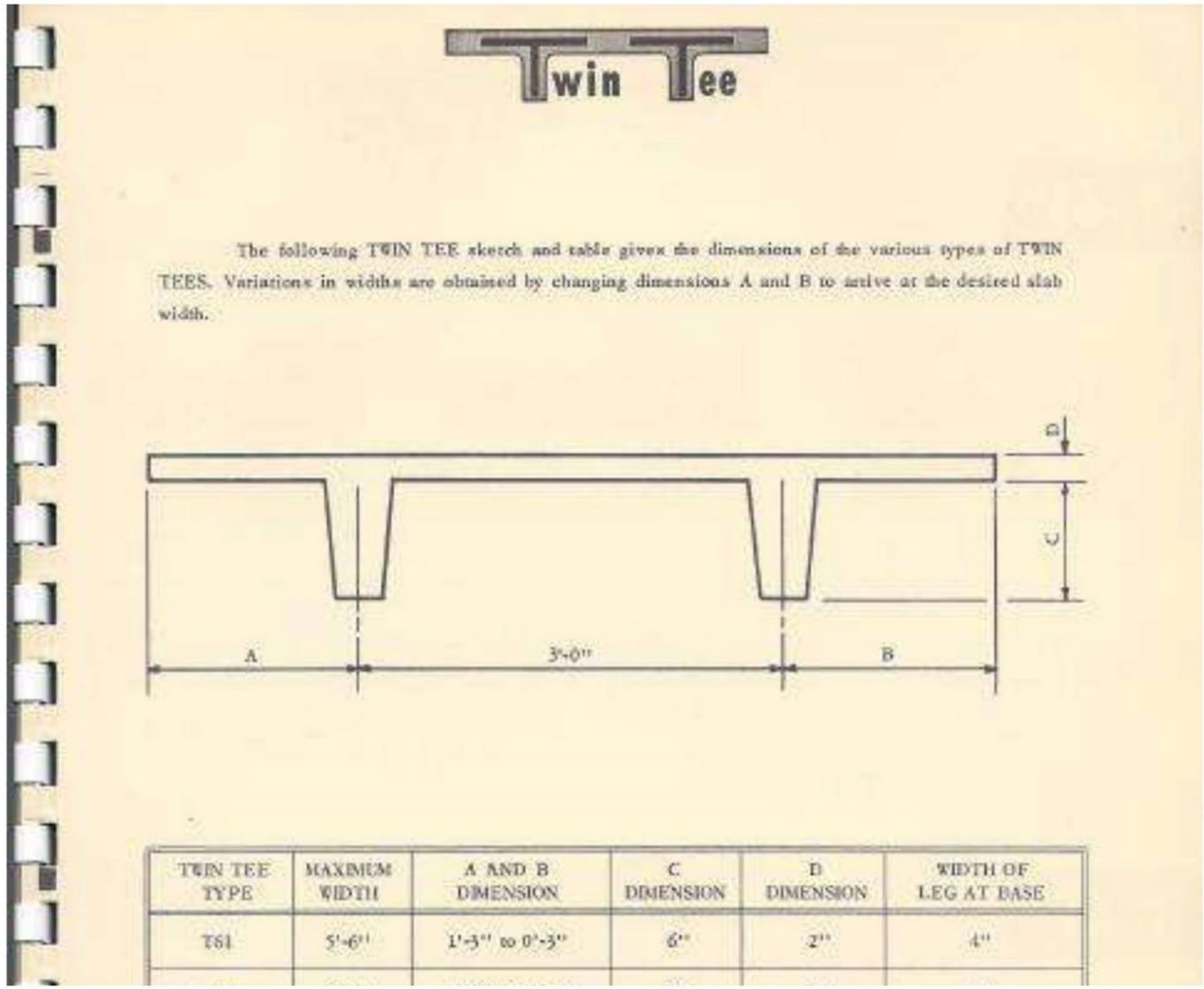
DEVELOPMENT OF CONCRETE



EVOLUTION OF THE DOUBLE TEE



First prestressed double tee (Twin Tee)
Prestressed Concrete of Colorado
Jack and Leonard Perlmutter, owners
Nat Sachter, architect
George Hanson, structural engineer

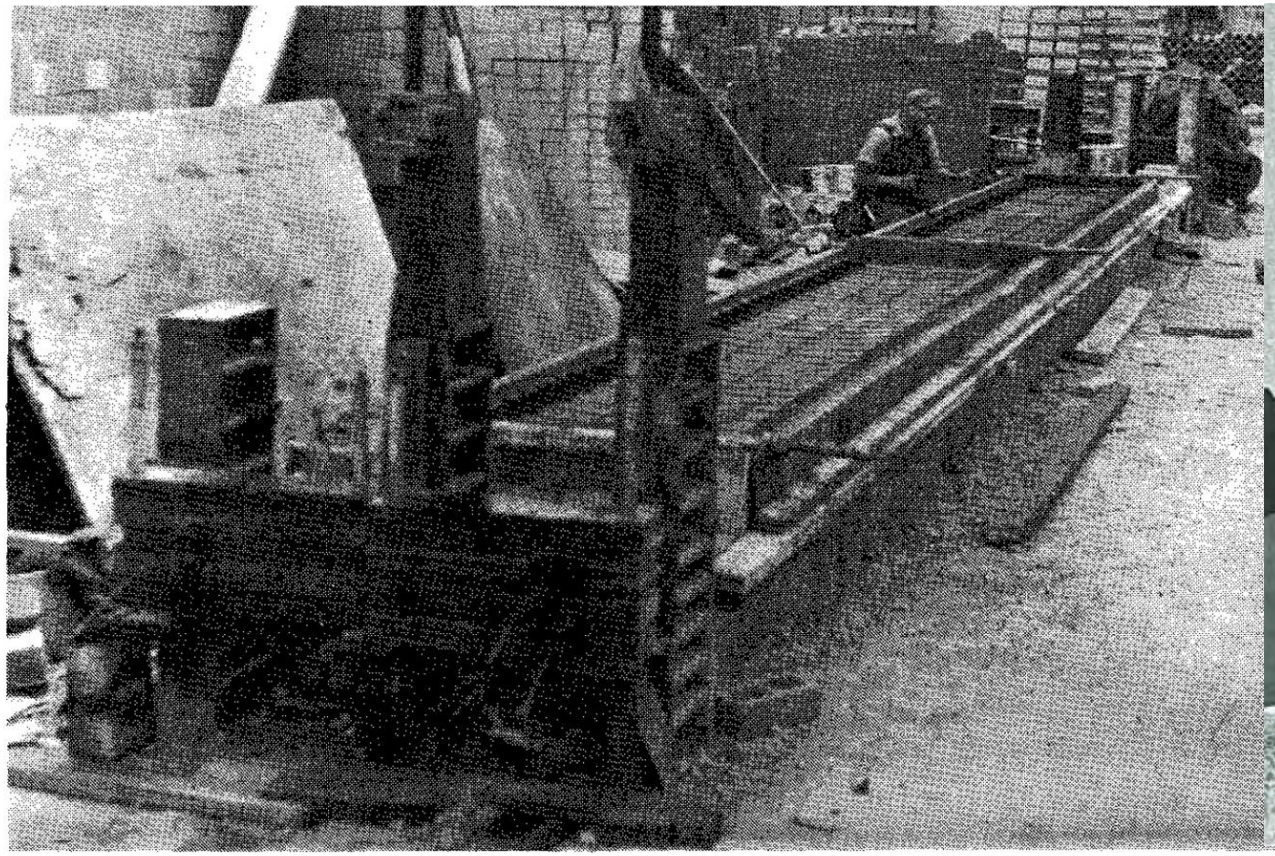


Twin Tee

The following TWIN TEE sketch and table gives the dimensions of the various types of TWIN TEES. Variations in widths are obtained by changing dimensions A and B to arrive at the desired slab width.

TWIN TEE TYPE	MAXIMUM WIDTH	A AND B DIMENSION	C DIMENSION	D DIMENSION	WIDTH OF LEG AT BASE
T61	5'-6"	1'-3" to 0'-3"	6"	2"	4"

1952 PATENT OF THE “TWIN TEE” BY AN ARCHITECT AND AN ENGINEER IN COLORADO



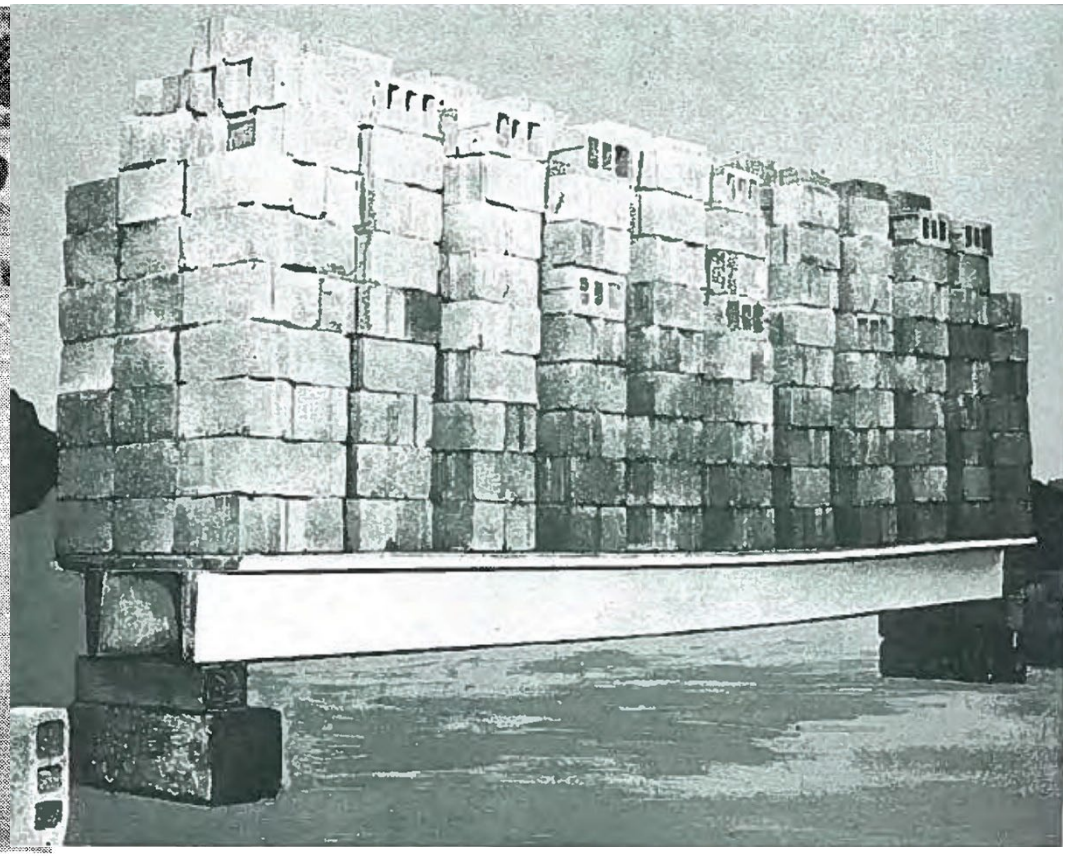


Fig. 2. Forms and reinforcement are placed in Colorado's first pretensioning bed prior to casting channel slab.

1952 EARLY CASTING BED AND LOAD TESTING

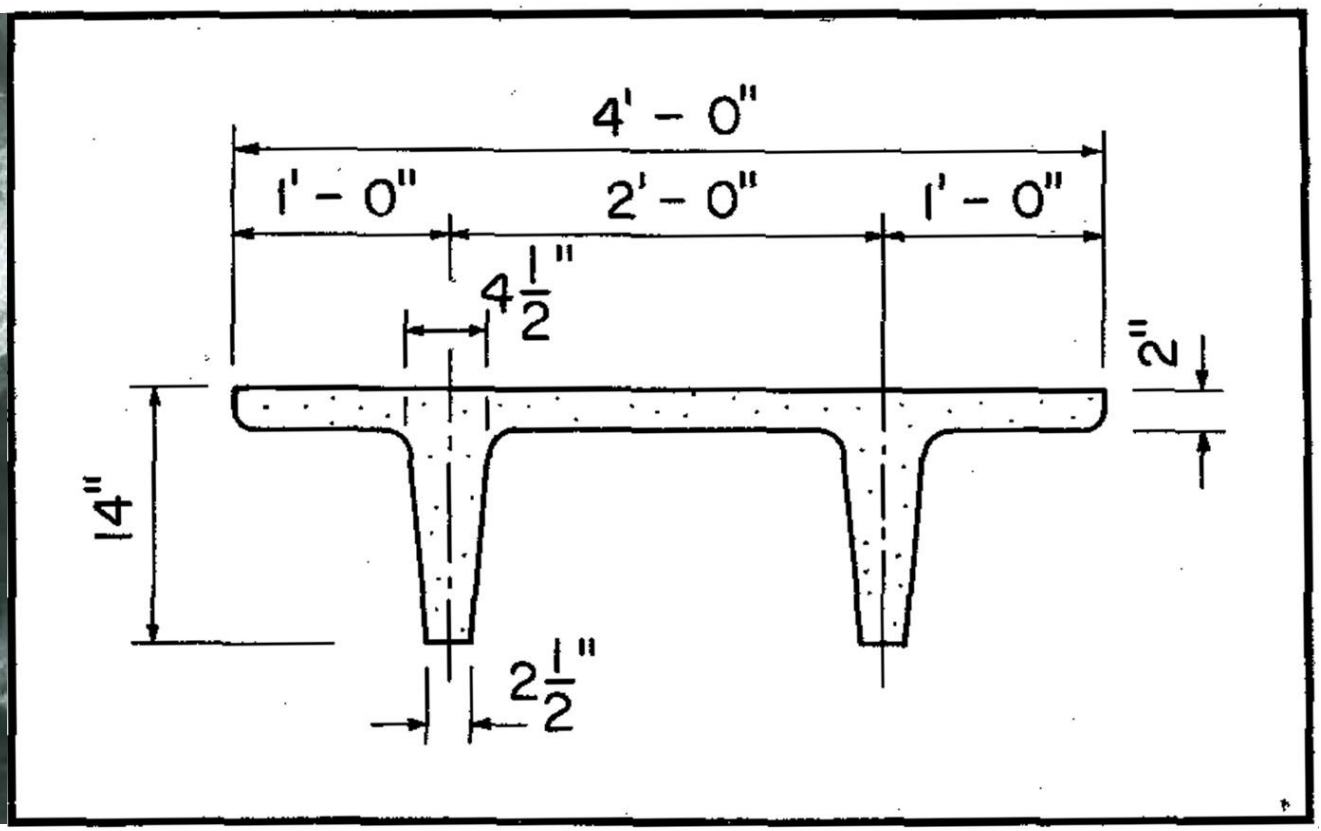
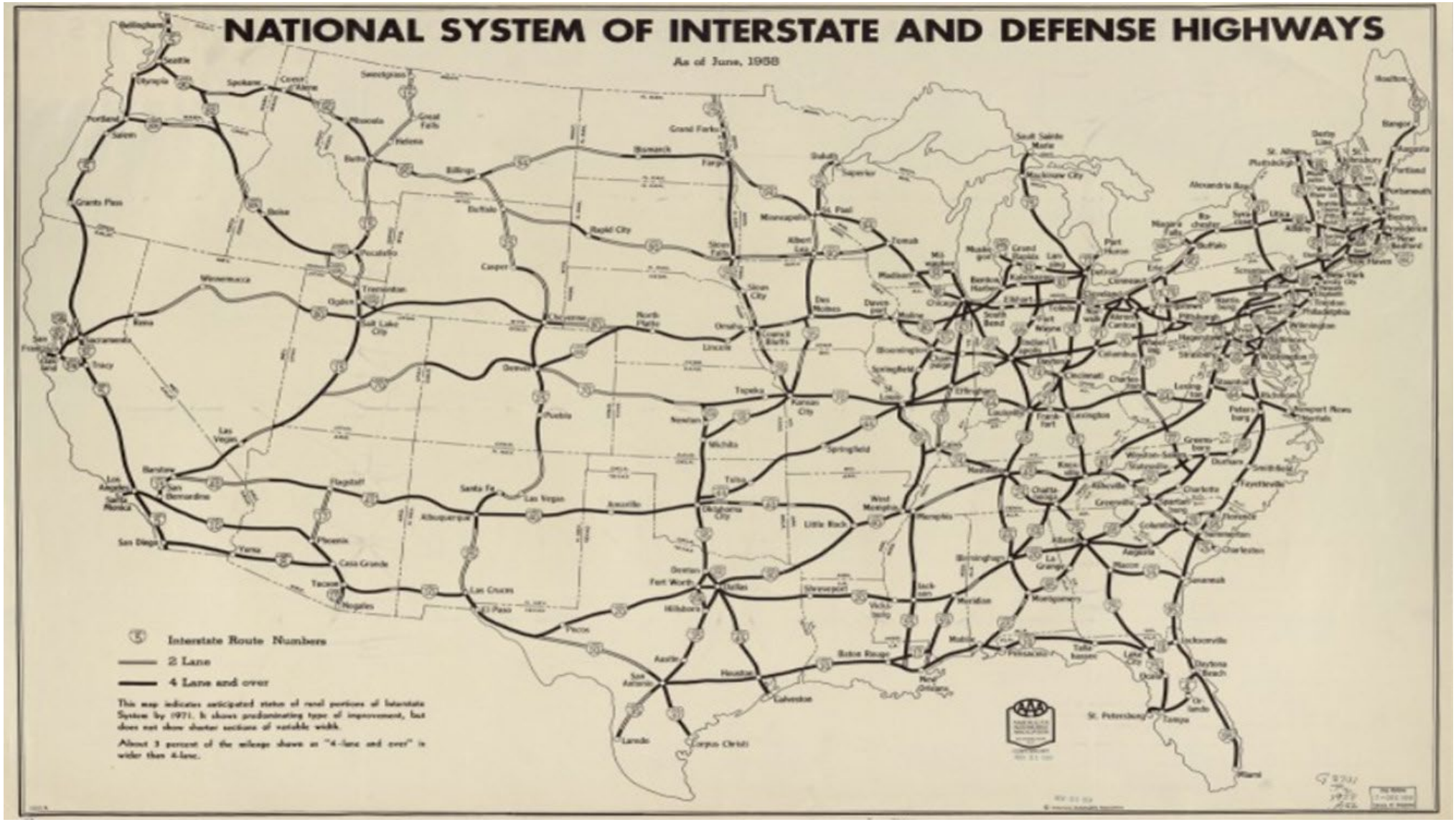


Fig 10. Dimensions of original double-tee.

1952 FIRST USED FOR A COLD-STORAGE WAREHOUSE IN COLORADO FOR BEATRICE FOODS



OFFSITE PRECAST COINCIDES WITH THE DEVELOPMENT OF THE HIGHWAY SYSTEM 1956-1992

OFFSITE PRECAST AND TRANSPORTATION GO TOGETHER

CONCRETE prestressed precast

…how do you make a double tee?

REUSABLE FORMWORK OF A SPECIFIC SIZE, PREPARED FOR REINFORCING STEEL & CONCRETE

PREPARING STEEL REINFORCING STIRRUPS

PRESTRESSING STEEL TENDONS (size and location determined by span and load)

TRUCK PREPARING TO DELIVER CONCRETE SLURRY

FINISHED ELEMENT, TRANSFORMED BY CHEMICAL HEAT, CRANE-LIFTED AFTER 24 HOURS

FINISHED ELEMENT, CURES FOR 14 to 28 DAYS

CONCRETE prestressed precast

…how do you make a hollow-core plank?

300-FOOT LONG CASTING BEDS WITH ZERO-SLUMP CONCRETE

HOLLOW-CORE PLANK CUT TO LENGTH

HOLLOW-CORE THICKNESS DISPLAY

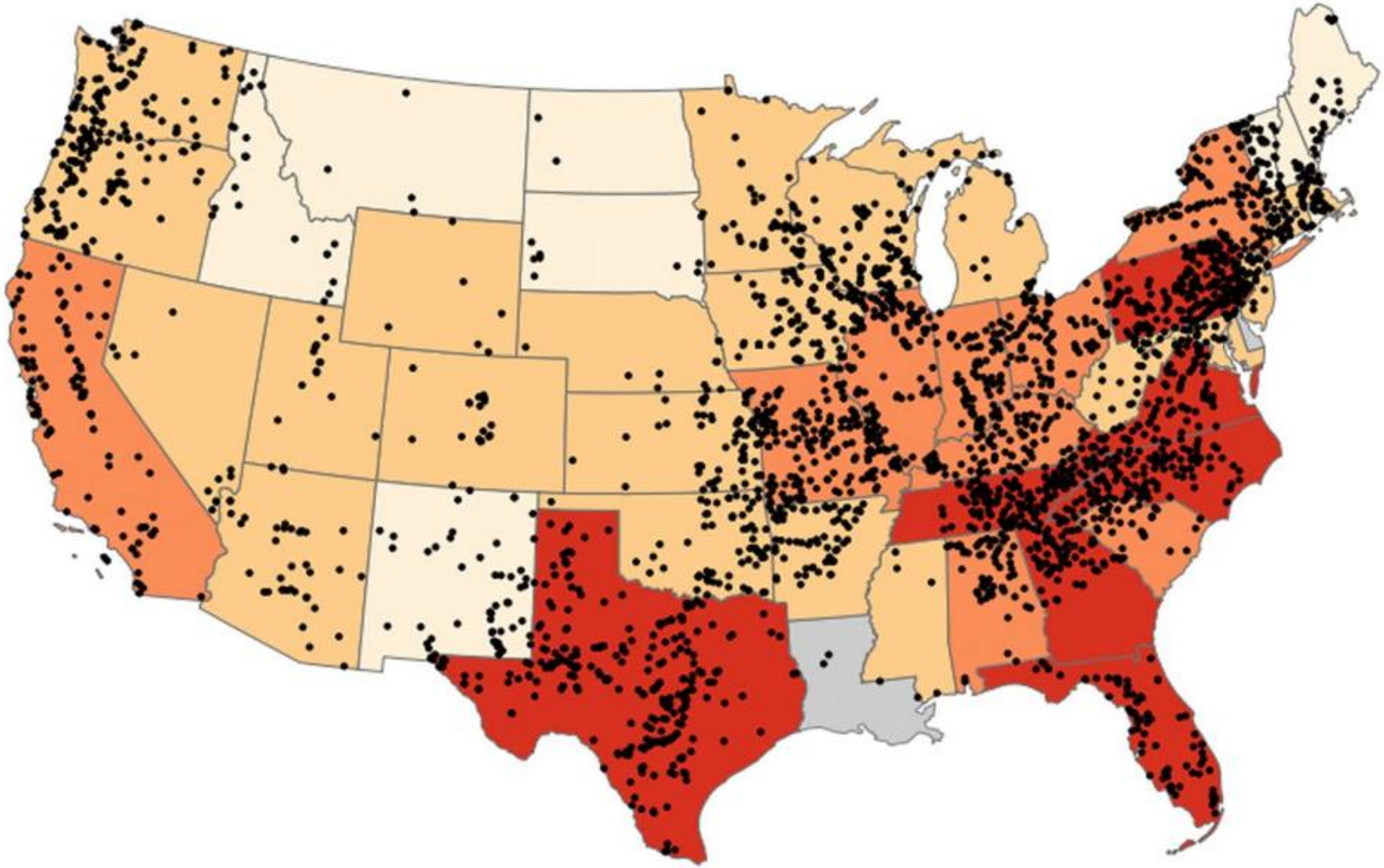
INSULATED BEARING WALL PANELS

CASTING BED FOR PRECAST STAIRS

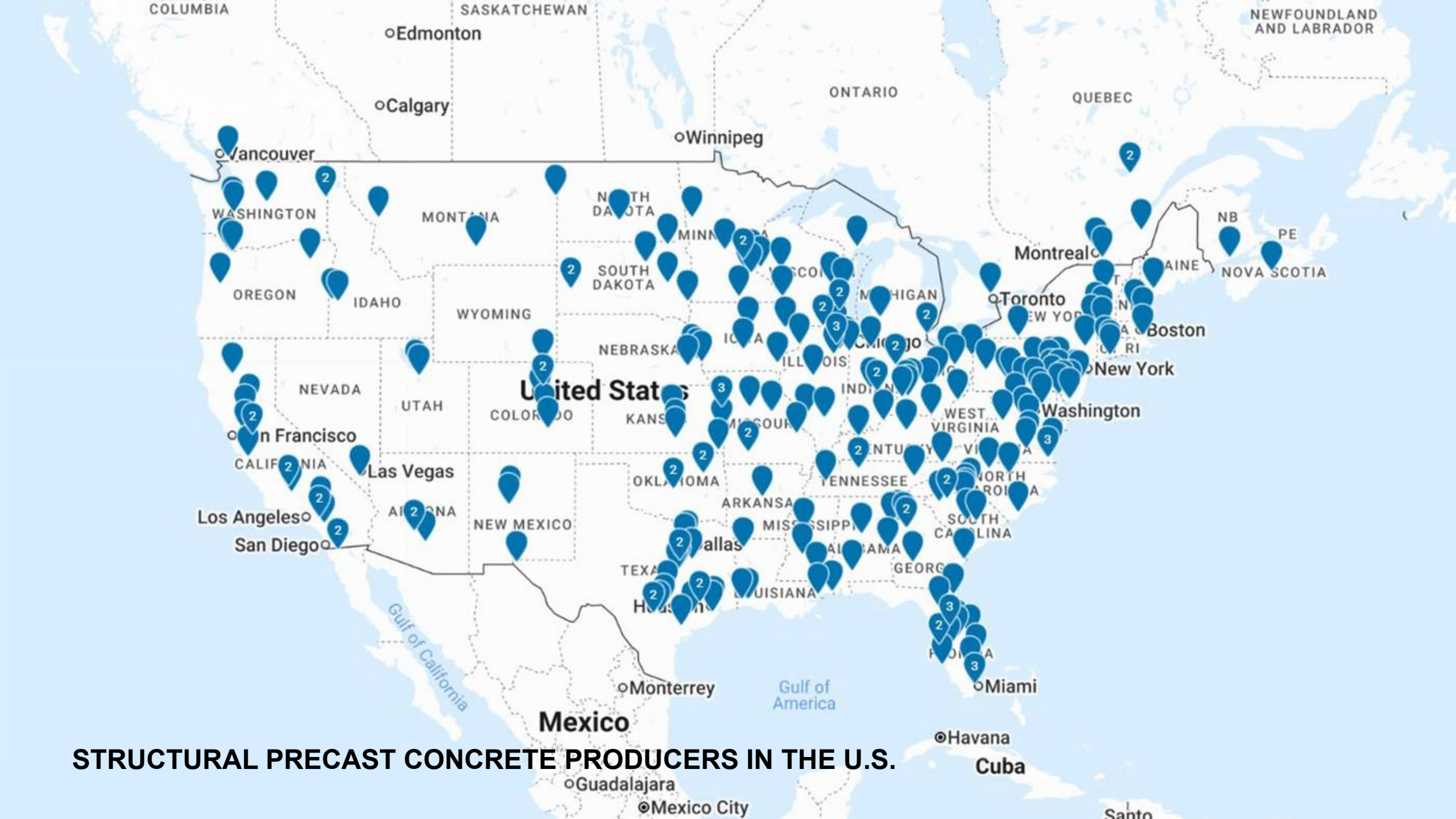
“WASTE-BLOCK” WALL

KERKSTRA PRECAST manufacturing plant near Grand Rapids, Michigan

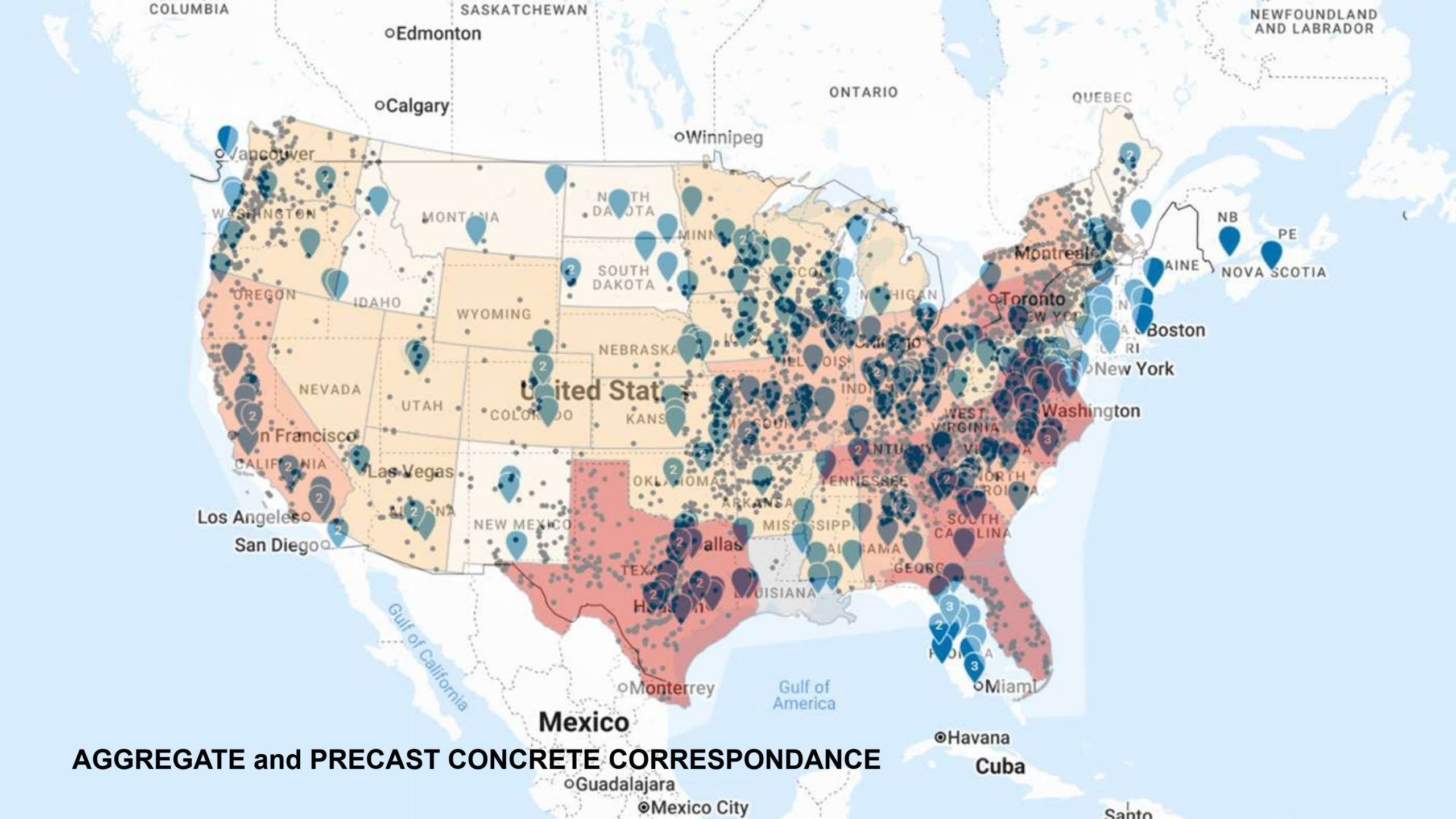
Gravel Mining near KERKSTRA PRECAST



AGGREGATE & GRAVEL MINING IN THE U.S.



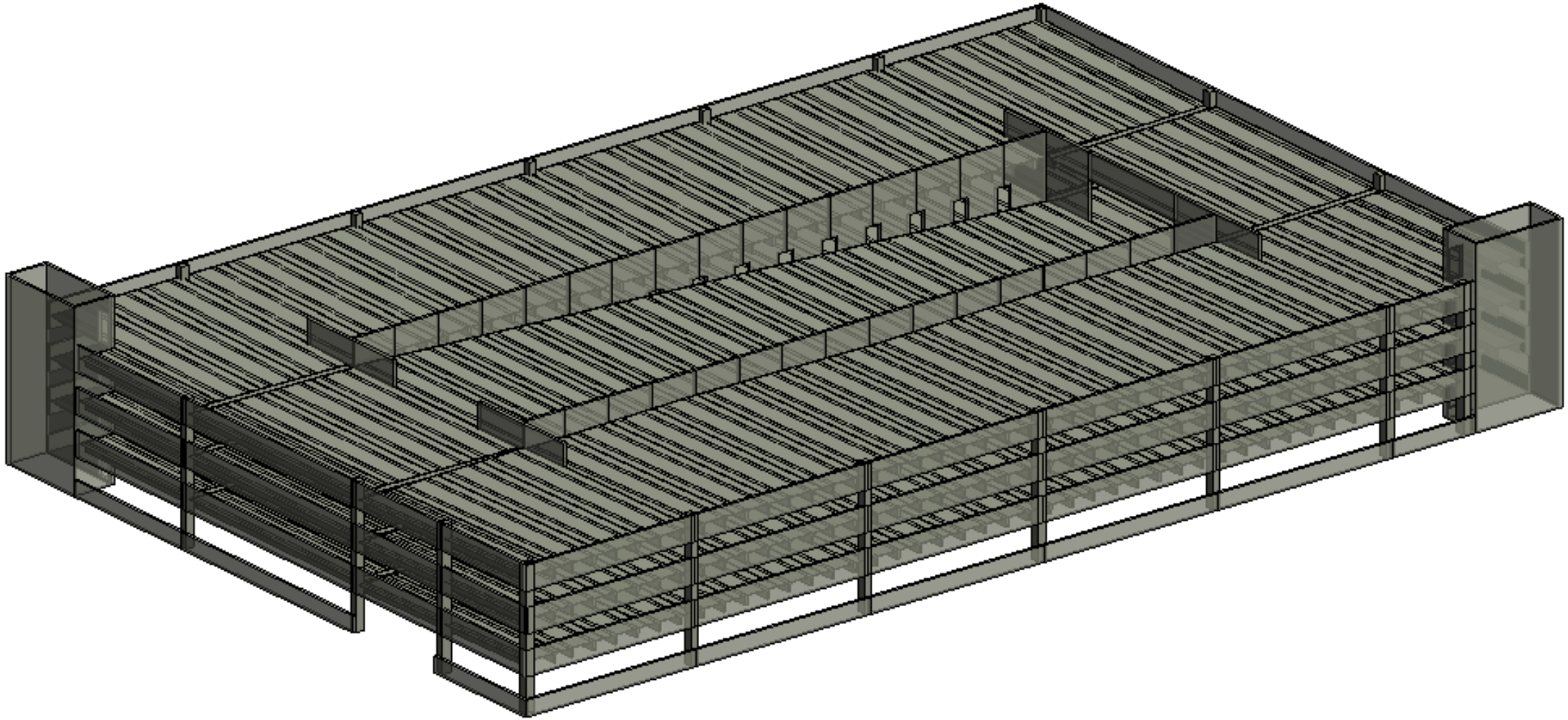
STRUCTURAL PRECAST CONCRETE PRODUCERS IN THE U.S.



AGGREGATE and PRECAST CONCRETE CORRESPONDANCE

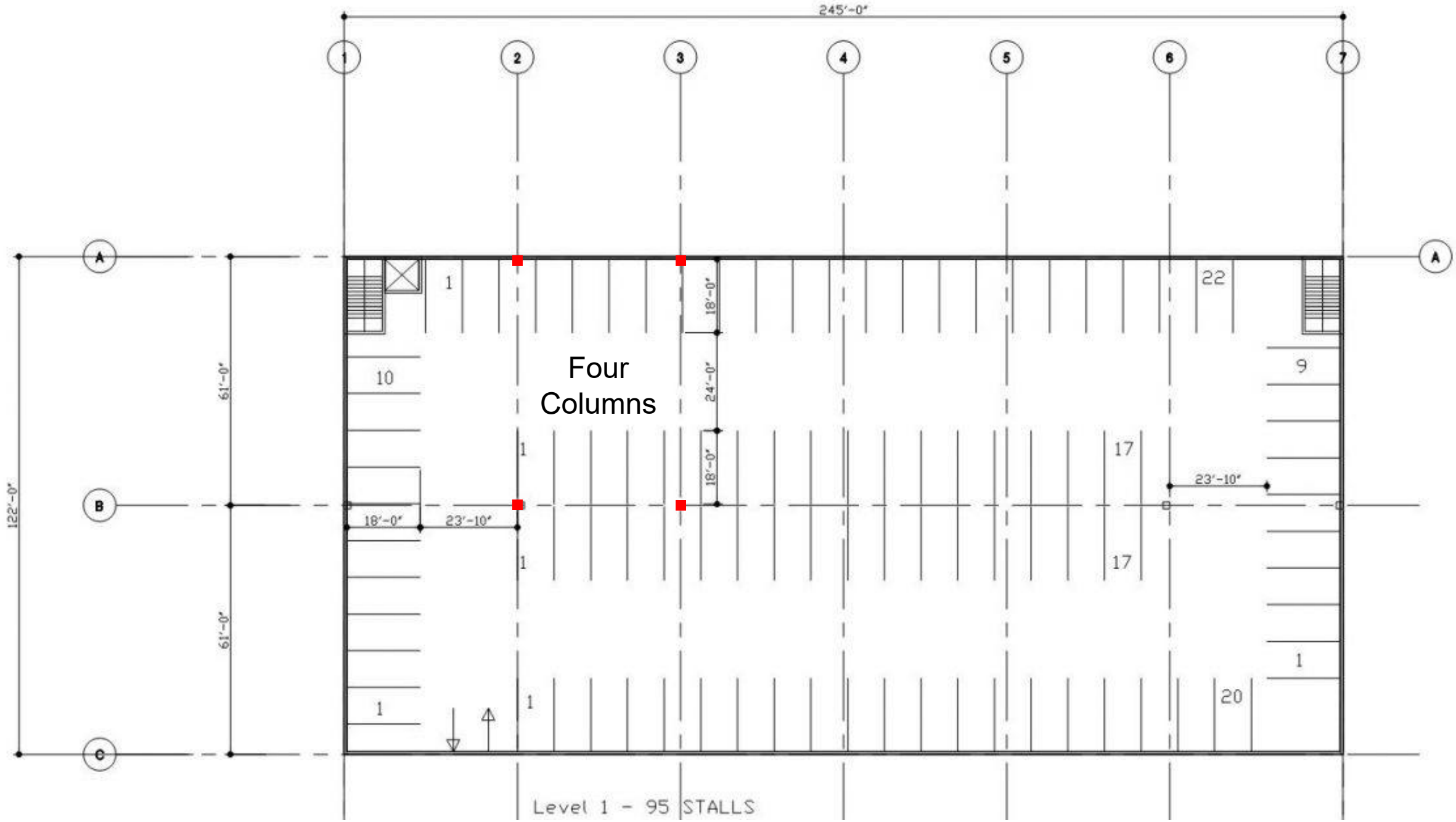
CONCRETE prestressed precast

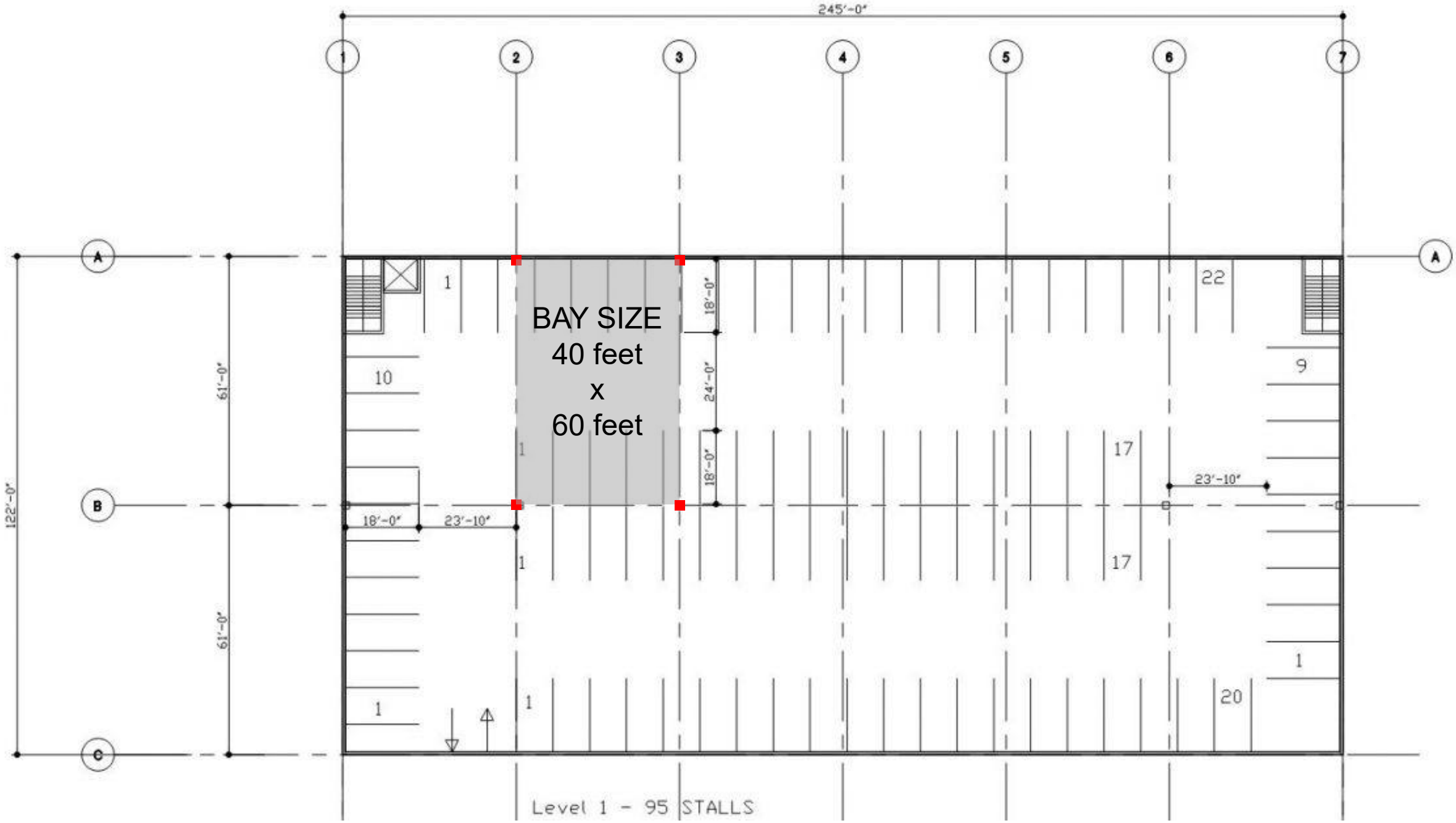
…how do you use a double tee in a building?

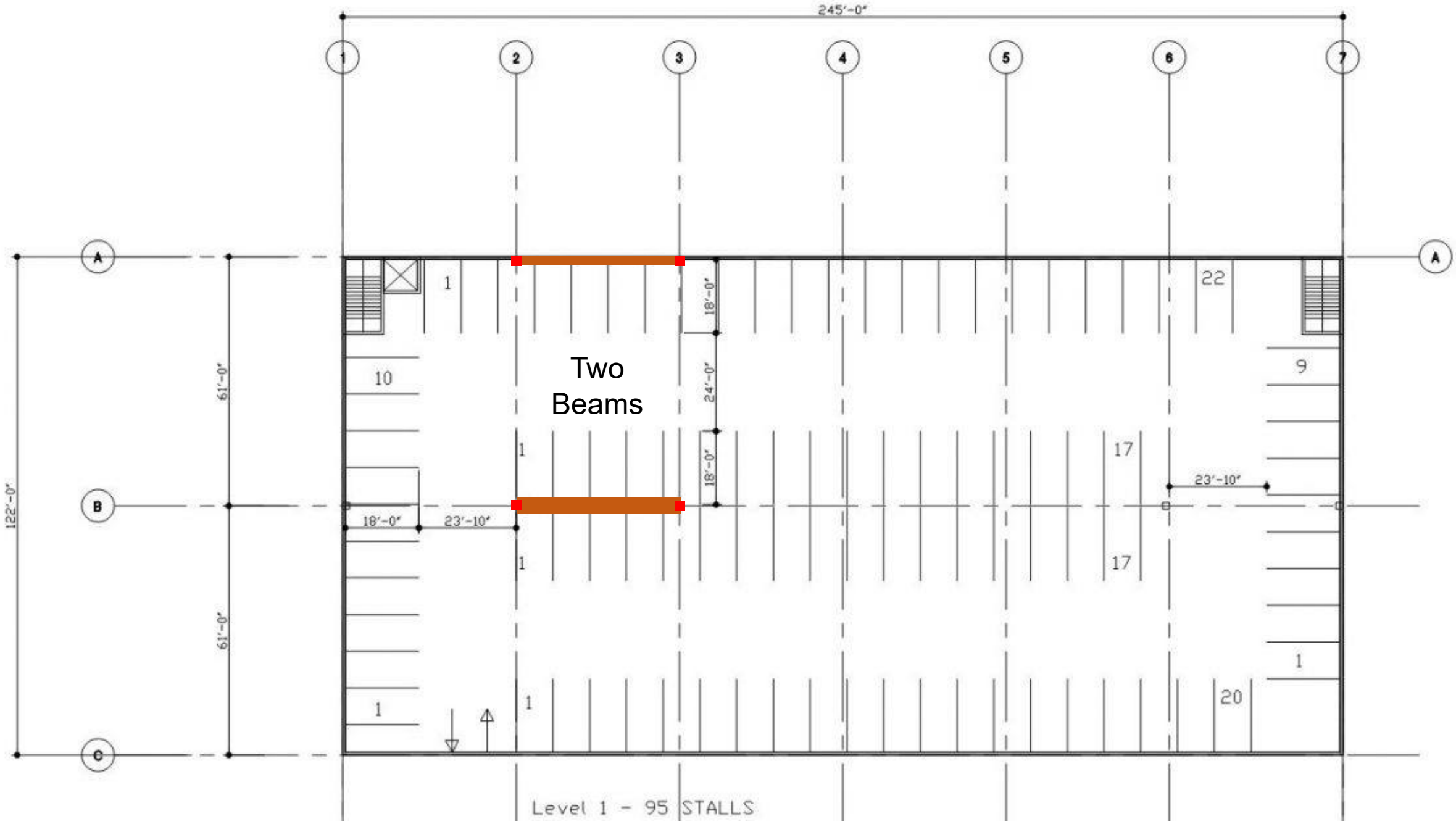


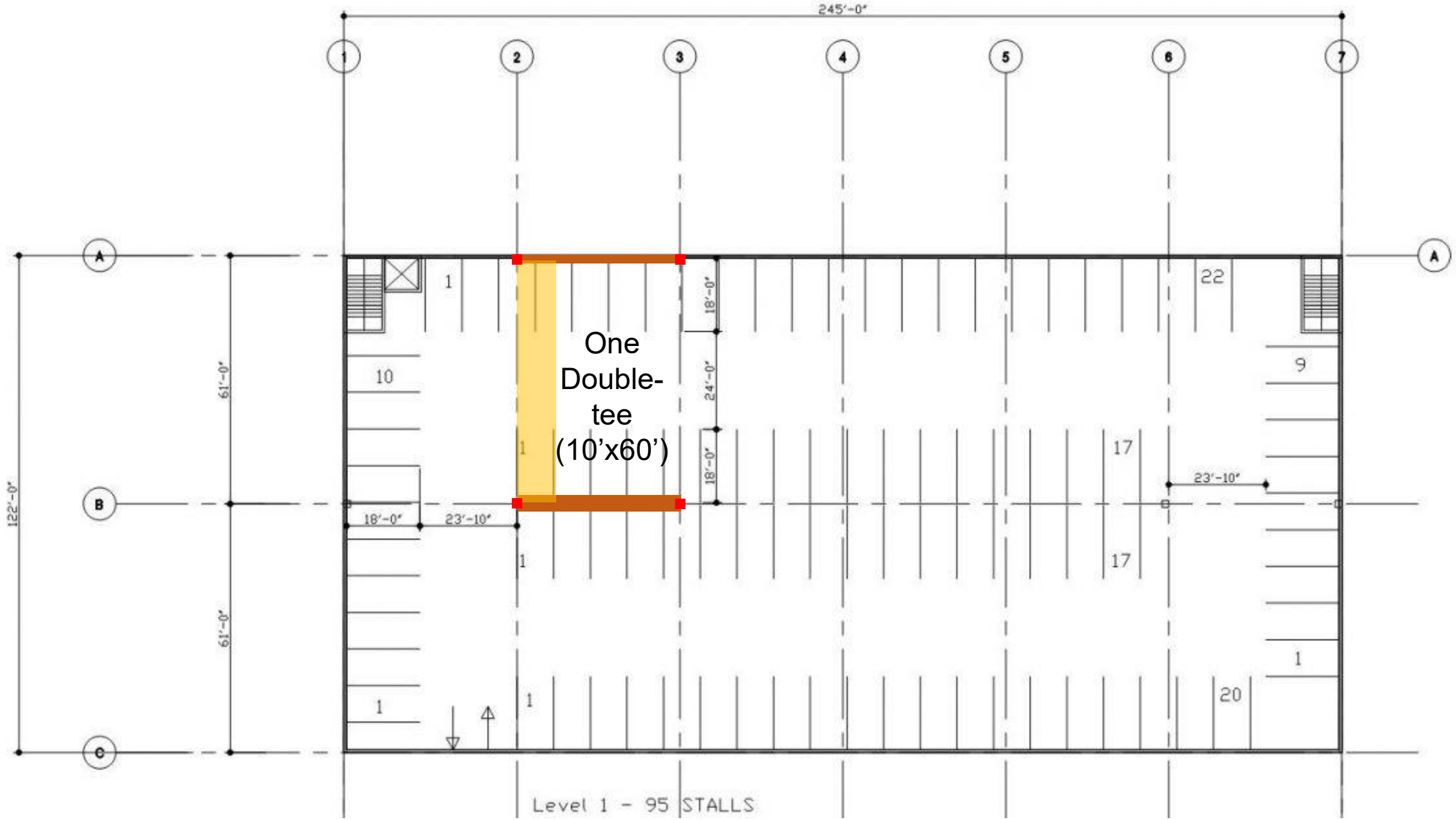
PARKING GARAGE AS A DEMONSTRATION MODEL

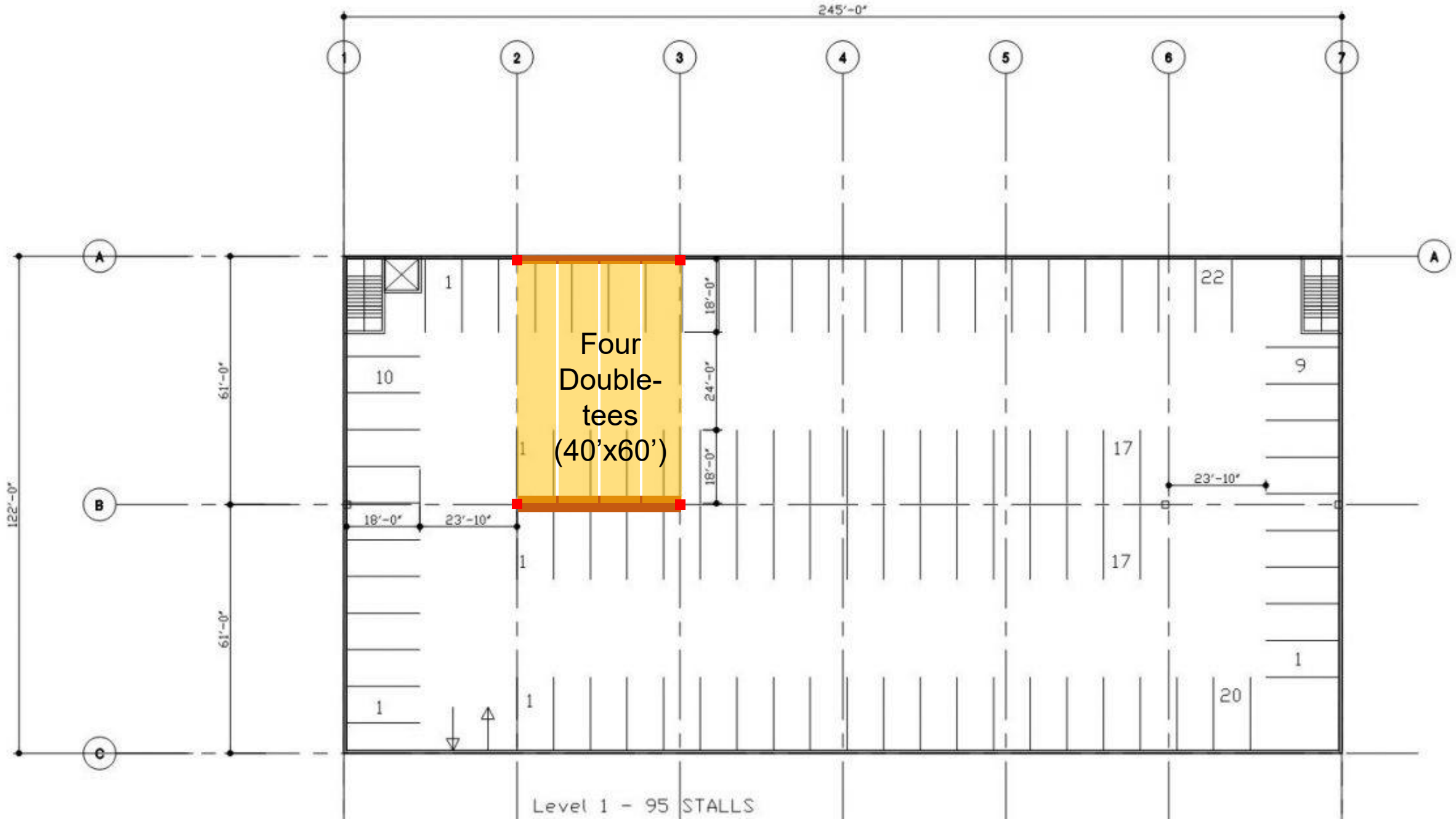
PRIMARY VERTICAL
Column
PRIMARY HORIZONTAL
Beam
SECONDARY HORIZONTAL
Double Tee

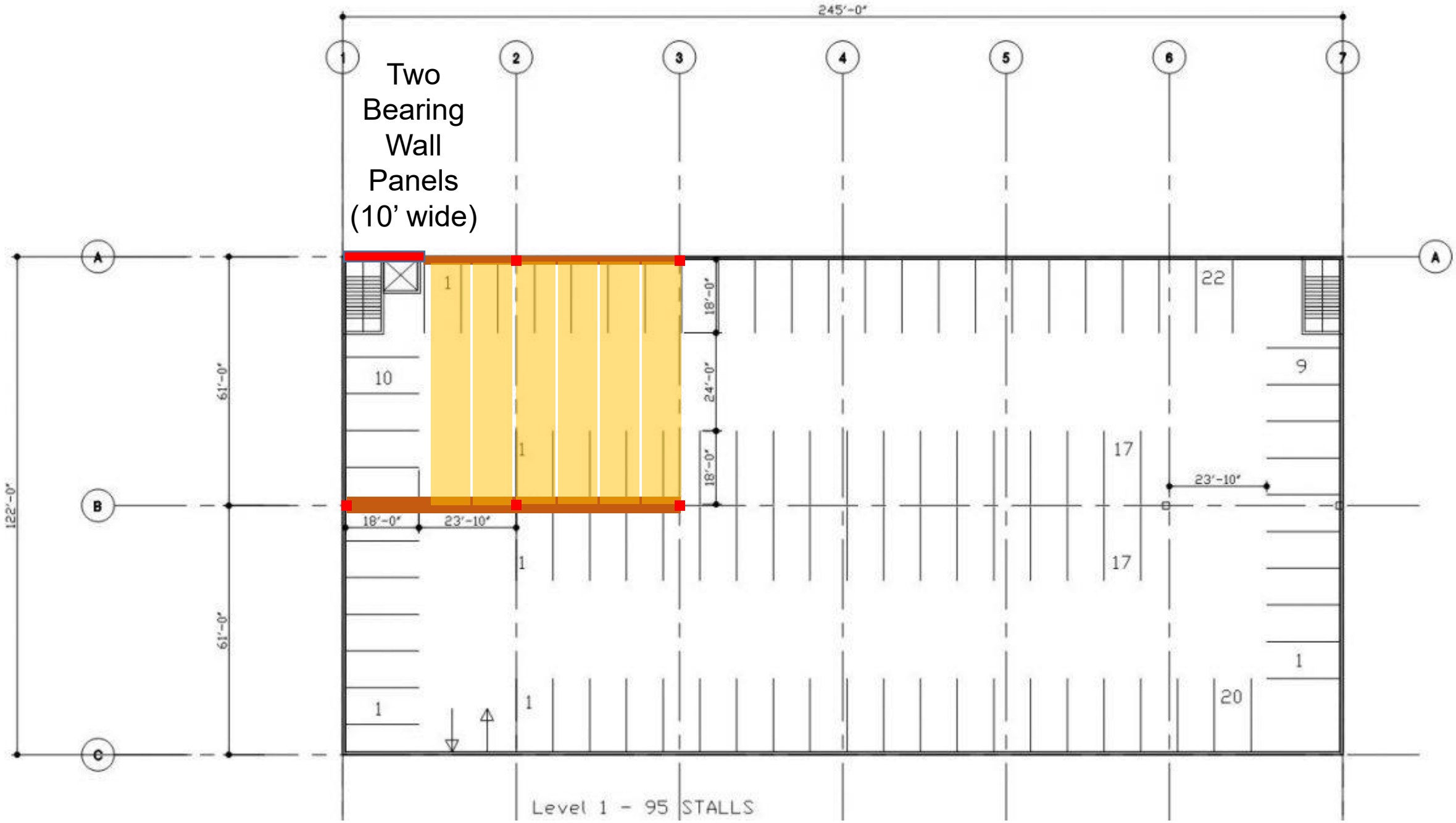

245'-0"
1
2
3
4
5
6
7
A
B
C
122'-0"
61'-0"
61'-0"
18'-0"
24'-0"
18'-0"
23'-10"
18'-0"
23'-10"
Four
Columns
1
10
1
1
1
1
22
9
17
17
1
20
Level 1 - 95 STALLS


245'-0"
1
2
3
4
5
6
7
A
B
C
122'-0"
61'-0"
61'-0"
BAY SIZE
40 feet
x
60 feet
18'-0"
24'-0"
18'-0"
18'-0"
23'-10"
23'-10"
10
1
1
1
1
1
22
9
17
17
1
20
Level 1 - 95 STALLS

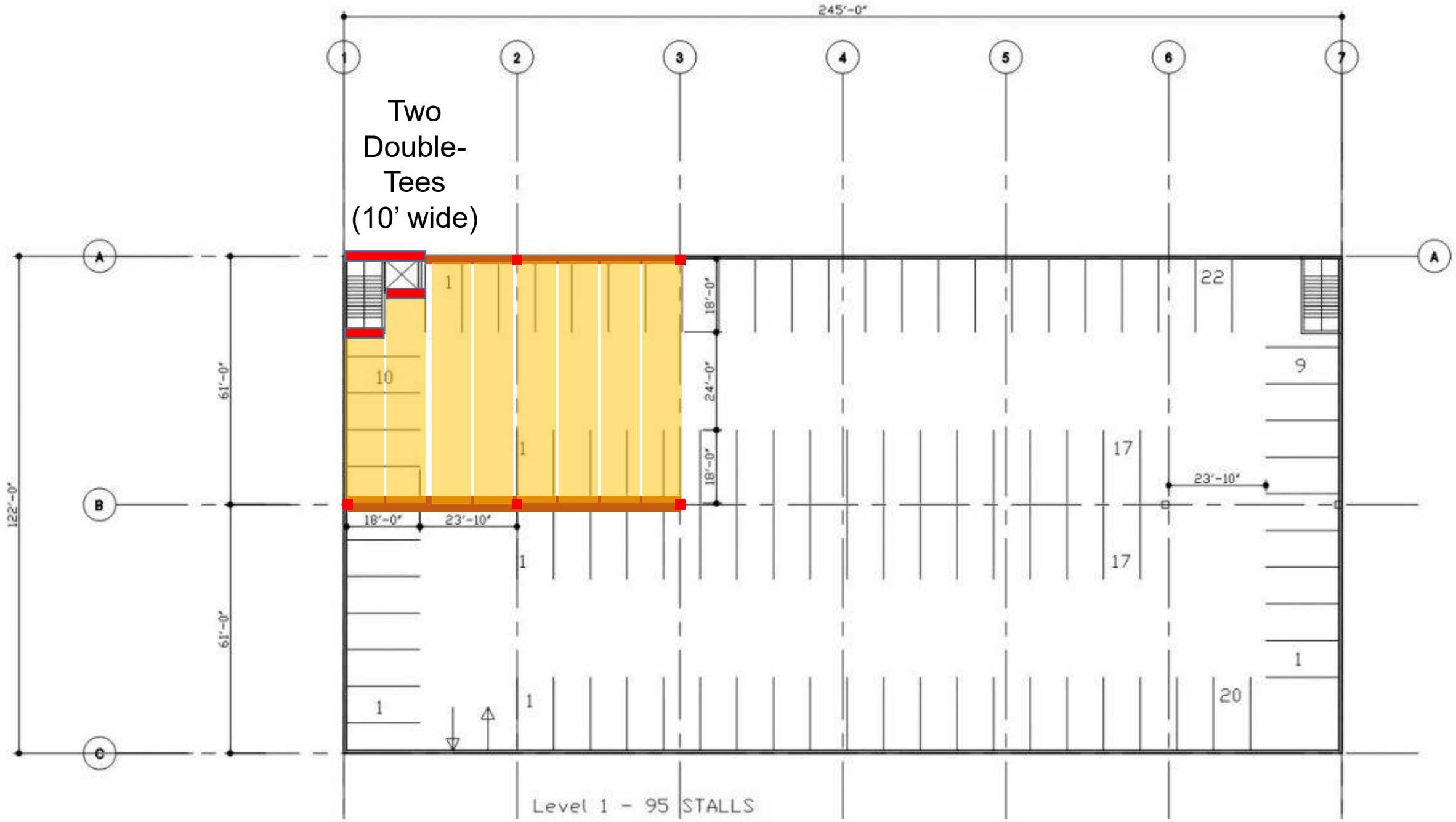

245'-0"
1
2
3
4
5
6
7
A
B
C
122'-0"
61'-0"
61'-0"
18'-0"
24'-0"
18'-0"
23'-10"
18'-0"
23'-10"
Two
Beams
1
10
1
1
1
22
9
17
17
1
20
Level 1 - 95 STALLS


245'-0"
1
2
3
4
5
6
7
A
B
C
122'-0"
61'-0"
61'-0"
18'-0"
24'-0"
18'-0"
23'-10"
18'-0"
23'-10"
One Double-tee (10'x60')
1
10
1
1
1
1
22
9
17
17
1
20
Level 1 - 95 STALLS

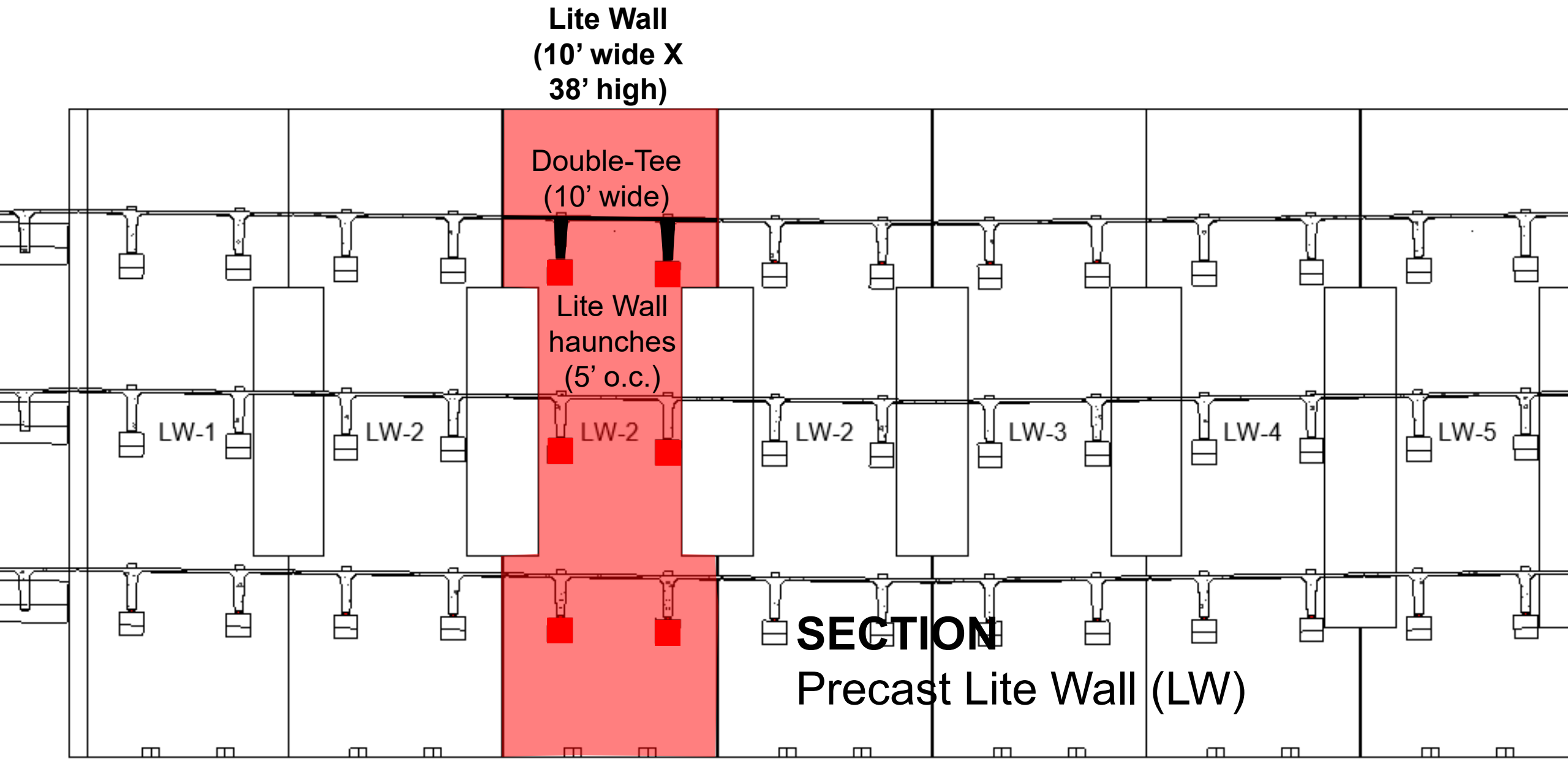

245'-0"
1
2
3
4
5
6
7
A
B
C
122'-0"
61'-0"
61'-0"
18'-0"
24'-0"
18'-0"
23'-10"
18'-0"
23'-10"
Four Double-tees (40'x60')
10
1
1
1
1
1
22
9
17
17
1
20
Level 1 - 95 STALLS


Two Bearing Wall Panels (10' wide)
245'-0"
122'-0"
61'-0"
61'-0"
18'-0"
24'-0"
18'-0"
18'-0"
23'-10"
23'-10"
1
2
3
4
5
6
7
A
B
C
10
1
1
1
1
22
9
17
17
1
20
Level 1 - 95 STALLS

Four Bearing Wall Panels (10' wide)
245'-0"
1
2
3
4
5
6
7
A
B
C
122'-0"
61'-0"
61'-0"
18'-0"
24'-0"
18'-0"
18'-0"
23'-10"
23'-10"
10
1
1
1
1
1
22
9
17
17
1
20
Level 1 - 95 STALLS


Two Double-Tees (10' wide)
245'-0"
122'-0"
61'-0"
61'-0"
18'-0"
24'-0"
18'-0"
18'-0"
23'-10"
23'-10"
1
2
3
4
5
6
7
A
B
C
10
1
1
1
1
1
22
9
17
17
1
20
Level 1 - 95 STALLS

Precast Lite Wall (LW): Structure that carries gravity load and resists lateral load

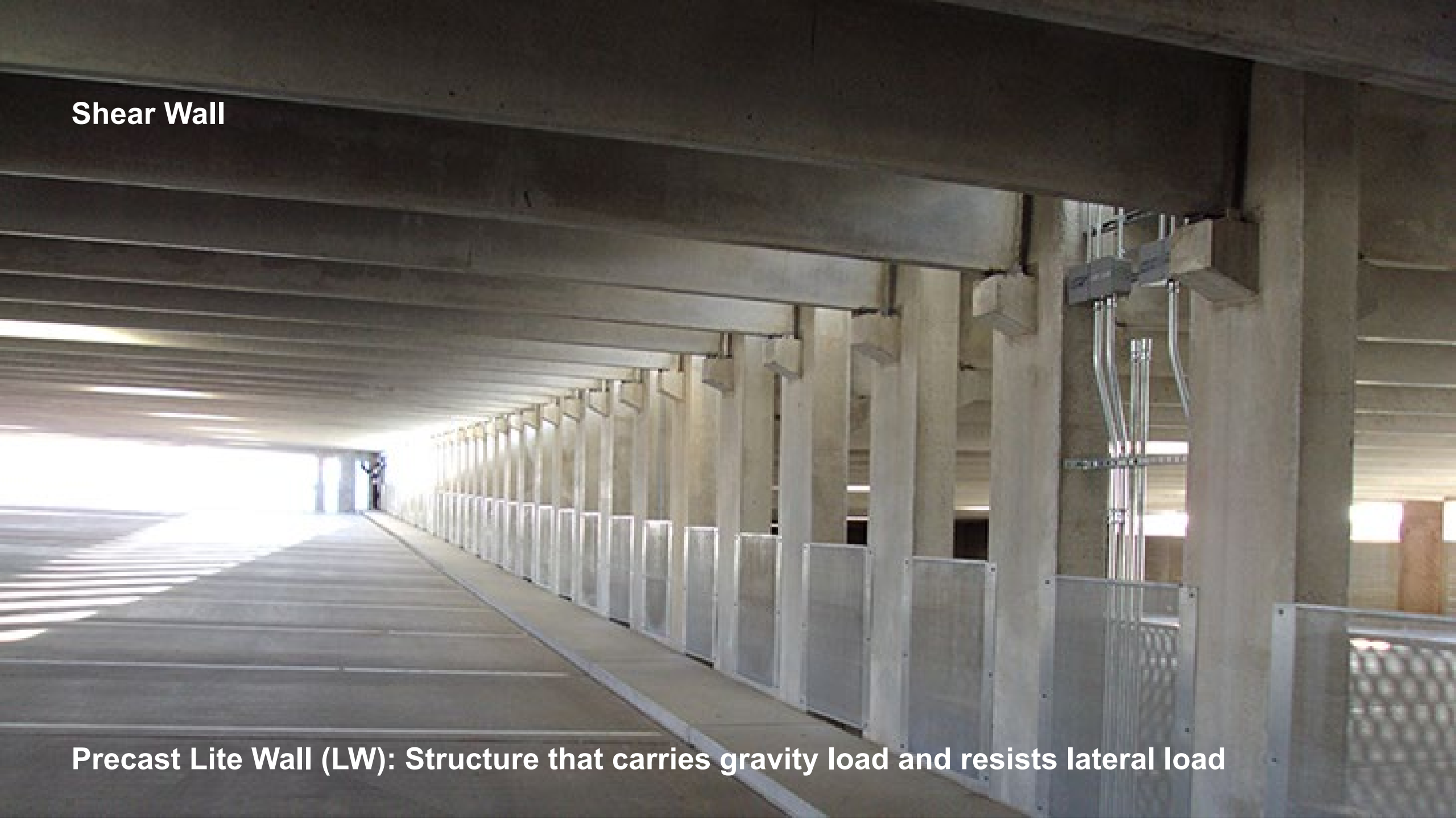


SECTION

Precast Lite Wall (LW)

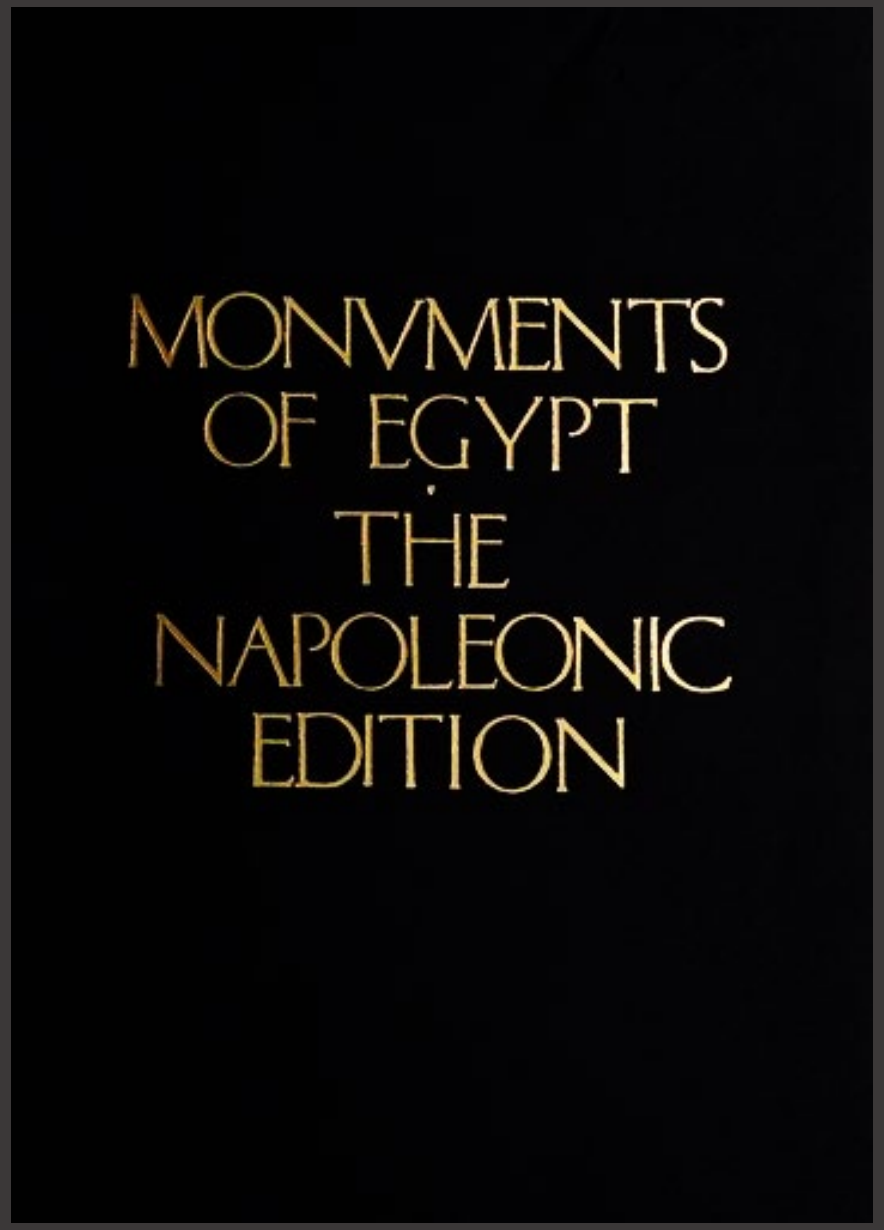
Shear Wall
Precast Lite Wall (LW): Structure that carries gravity load and resists lateral load

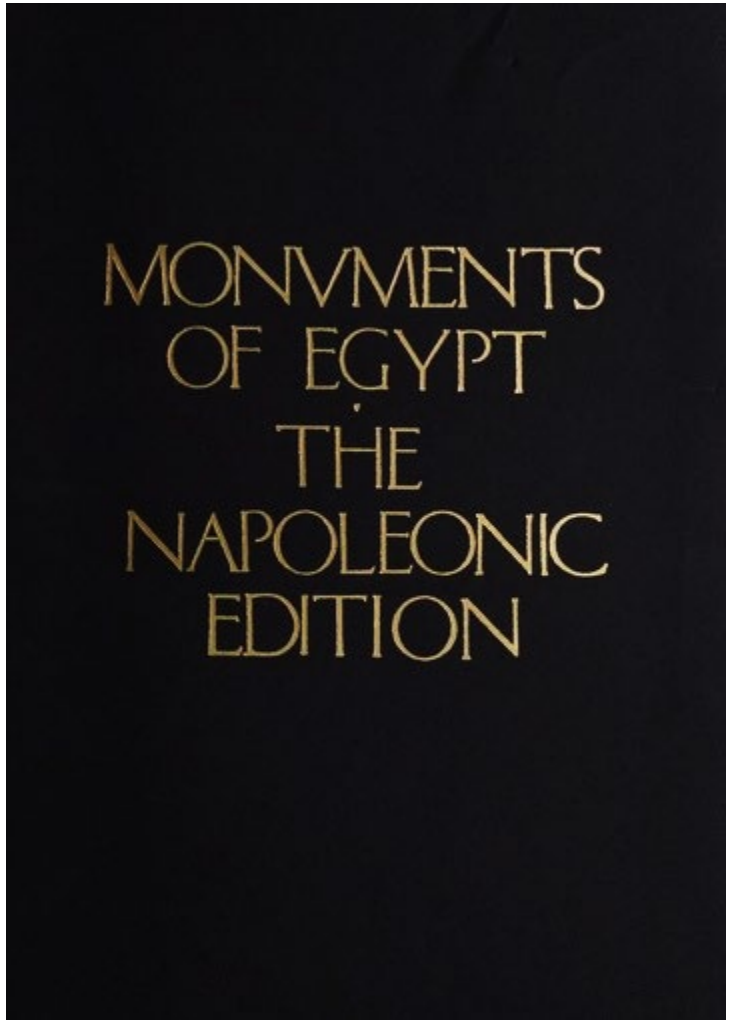
Shear Wall



Precast Lite Wall (LW): Structure that carries gravity load and resists lateral load

Shear Wall
Precast Lite Wall (LW): Structure that carries gravity load and resists lateral load


MONVMENTS
OF EGYPT
THE
NAPOLEONIC
EDITION



ANCIENT EGYPTIAN ARCHITECTURE structure of sculptural mass, light, and shadow

ESNA Temple of Khnum

ESNÉ (LATOPOLIS.)

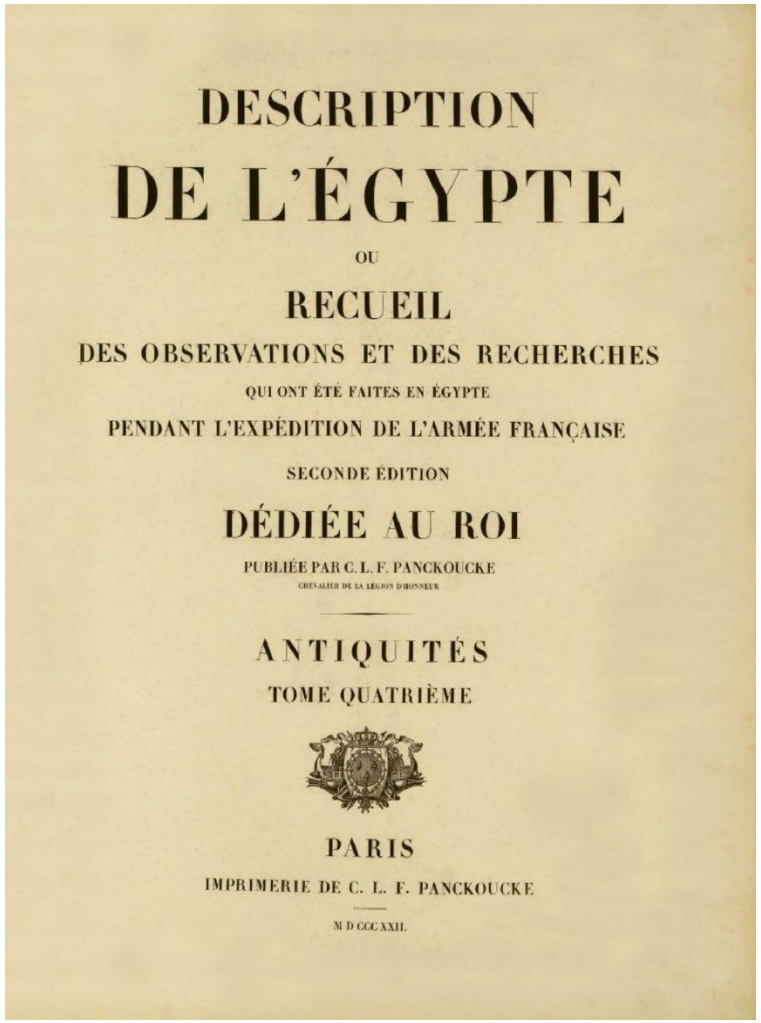
ESNA

ÉLÉVATION DU PORTIQUE.

ESNA

Minnesota Architectural Foundation
Rapson Traveling Study Competition
2026 Competion Dates
January 29 - February 10

EGYPT trip along the Nile


DESCRIPTION

DE L'ÉGYPTE

OU

RECUEIL

DES OBSERVATIONS ET DES RECHERCHES

QUI ONT ÉTÉ FAITES EN ÉGYPTE

PENDANT L'EXPÉDITION DE L'ARMÉE FRANÇAISE

SECONDE ÉDITION

DÉDIÉE AU ROI

PUBLIÉE PAR C. L. F. PANCKOUCKE

CHEVALIER DE LA LÉGION D'HONNEUR

ANTIQUITÉS

TOME QUATRIÈME

PARIS

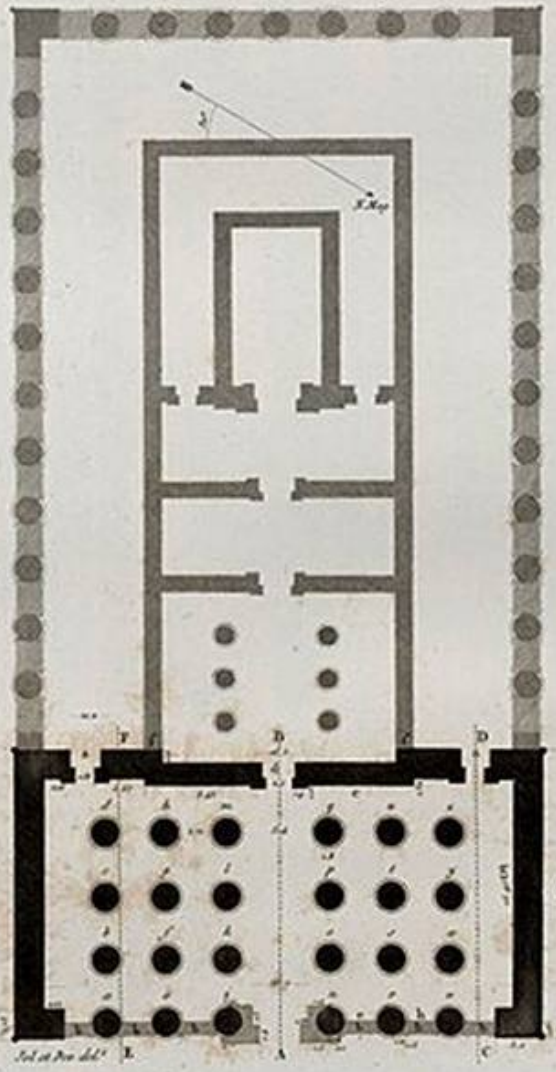
IMPRIMERIE DE C. L. F. PANCKOUCKE

M D CCC XXII.

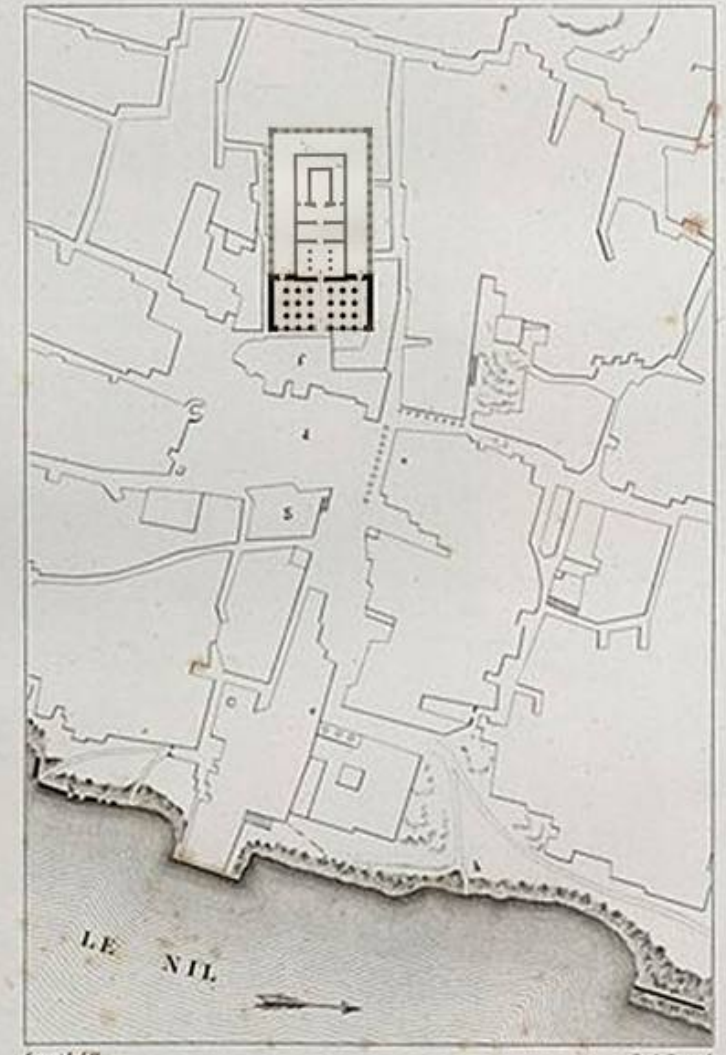
NAPOLEANIC ENGRAVINGS 1798 at the Cairo Library

1

3



2



Ech. de la fig. 1.

Ech. de la fig. 2.

1 . 2 PLANS DES ENVIRONS D'ESNÉ ET D'UNE PARTIE DE LA VILLE.

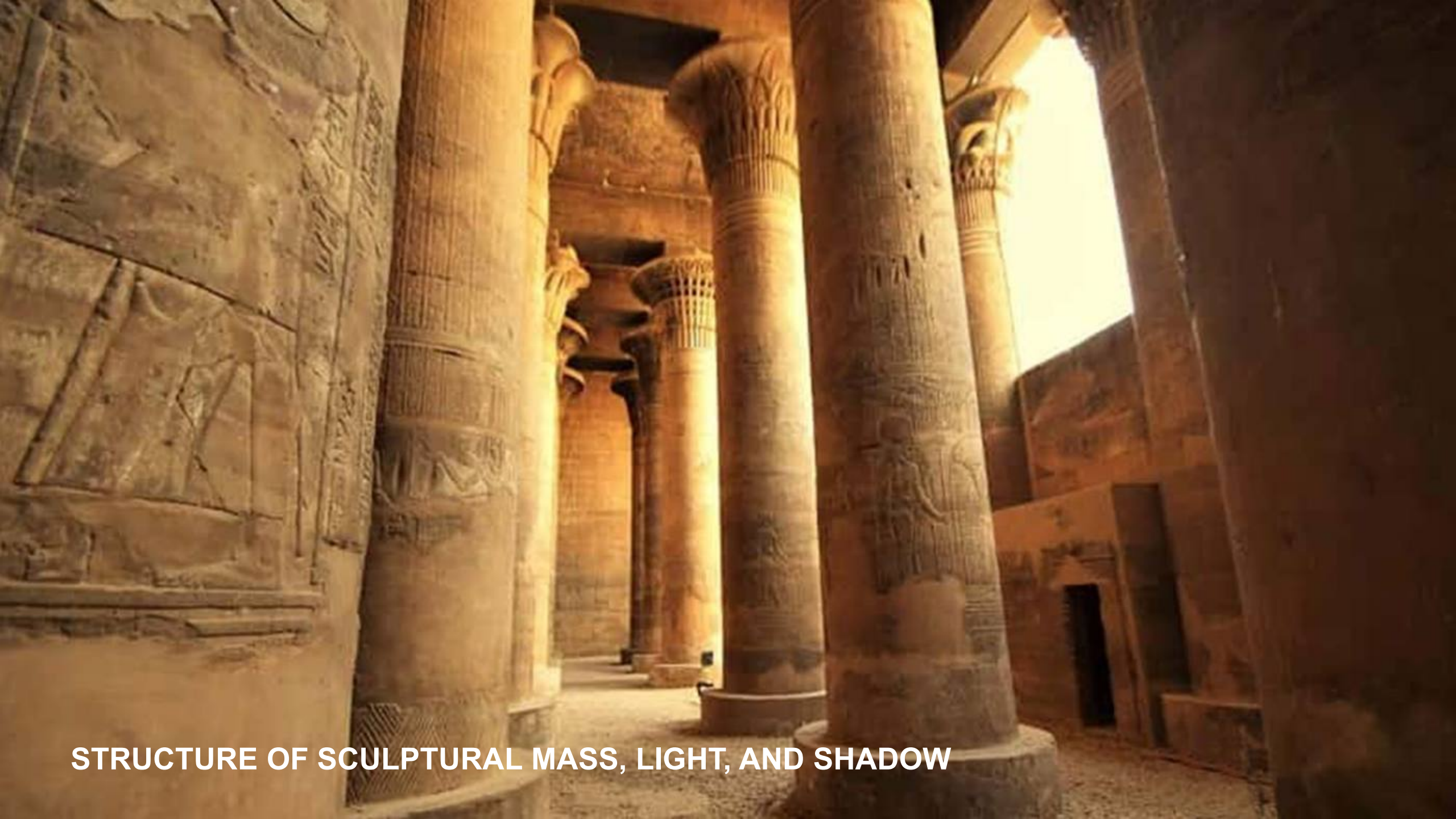
3 PLAN DU TEMPLE. 4 COUPE DU PORTIQUE.

Ech. de la fig. 3.

Ech. de la fig. 4.

ESNA: 1,800 year-old structure of limestone

ESNA


STRUCTURE OF SCULPTURAL MASS, LIGHT, AND SHADOW

STRUCTURE OF SCULPTURAL MASS, LIGHT, AND SHADOW

Steven Holl

Steven Holl: SAINT IGNATIOUS CHAPEL (precast structural wall panels)

Edward Larrabee Barnes

Edward Larrabee Barnes: WALKER ART CENTER (precast double-tee structure)

TRANSFORMATION and evolution

HISTORICAL EXPERIENCE to DESIGN RESEARCH

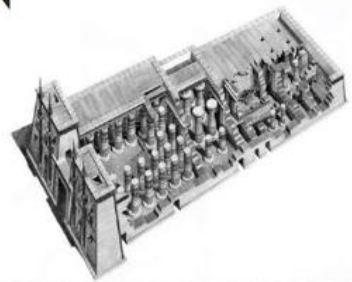
ANCIENT STRUCTURE to CONTEMPORARY STRUCTURE

STONE MATERIAL to PRECAST CONCRETE MATERIAL

PRECAST
PROTOTYPES

2021

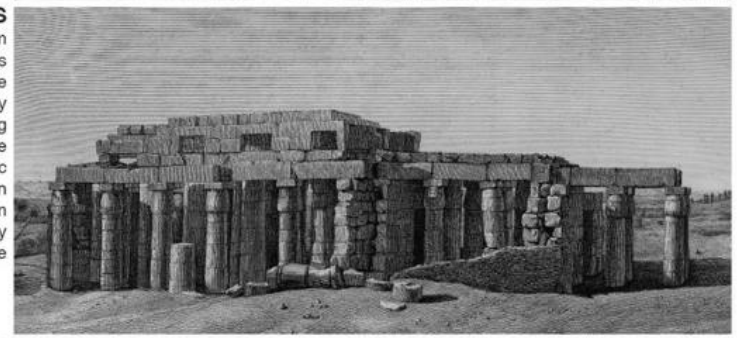
2021 BC CONSTRUCTION
CONVENTIONS



VIRTUES
ANCIENT

PRAGMATIC VIRTUES
Local Material
Local Technology
Modular Elements
Mass Production
Structural Expression
Resilient Construction
Adaptable Space
Fireproof
Finished Exterior
Integral Interior
Long Life Loose Fit

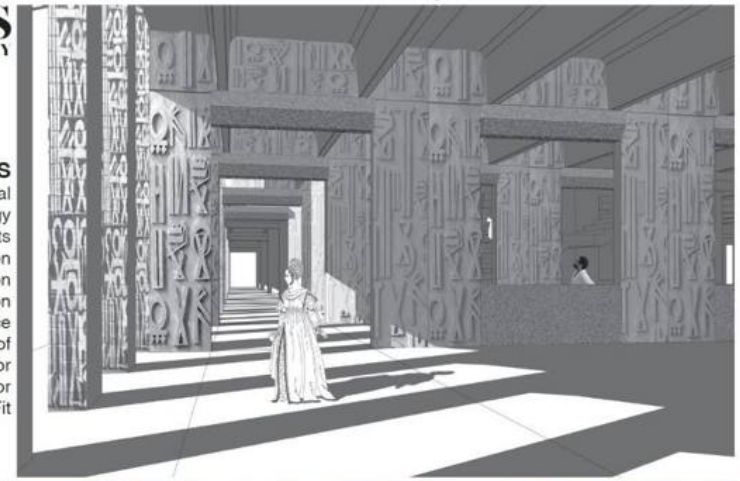
POETIC VIRTUES
Powerful Form
Intense Mass
Animated Surface
Material Integrity
Strategic Daylighting
Ritual Sequence
Spatially Dynamic
Elegant Proportion
Syncopated Rhythm
Age Gracefully
Primal Architecture



STONE
PROTOTYPES

2021 AD CONSTRUCTION
CONVENTIONS

VIRTUES
CONTEMPORARY



PRAGMATIC VIRTUES
Local Material
Local Technology
Modular Elements
Mass Production
Structural Expression
Resilient Construction
Adaptable Space
Fireproof
Finished Exterior
Integral Interior
Long Life Loose Fit

POETIC VIRTUES
Powerful Form
Intense Mass
Animated Surface
Material Integrity
Strategic Daylighting
Ritual Sequence
Spatially Dynamic
Elegant Proportion
Syncopated Rhythm
Age Gracefully
Primal Architecture

PRECAST
PROTOTYPES

MASS PRODUCTION AND CONVENTIONS

FABRICATION

Hollow-Core Plank Extruder

Hollow-Core Plank

Double Tee Formwork

Double Tee

Insulated Wall Panel Formwork

Insulated Wall Panel

"Waste Blocks" Formwork

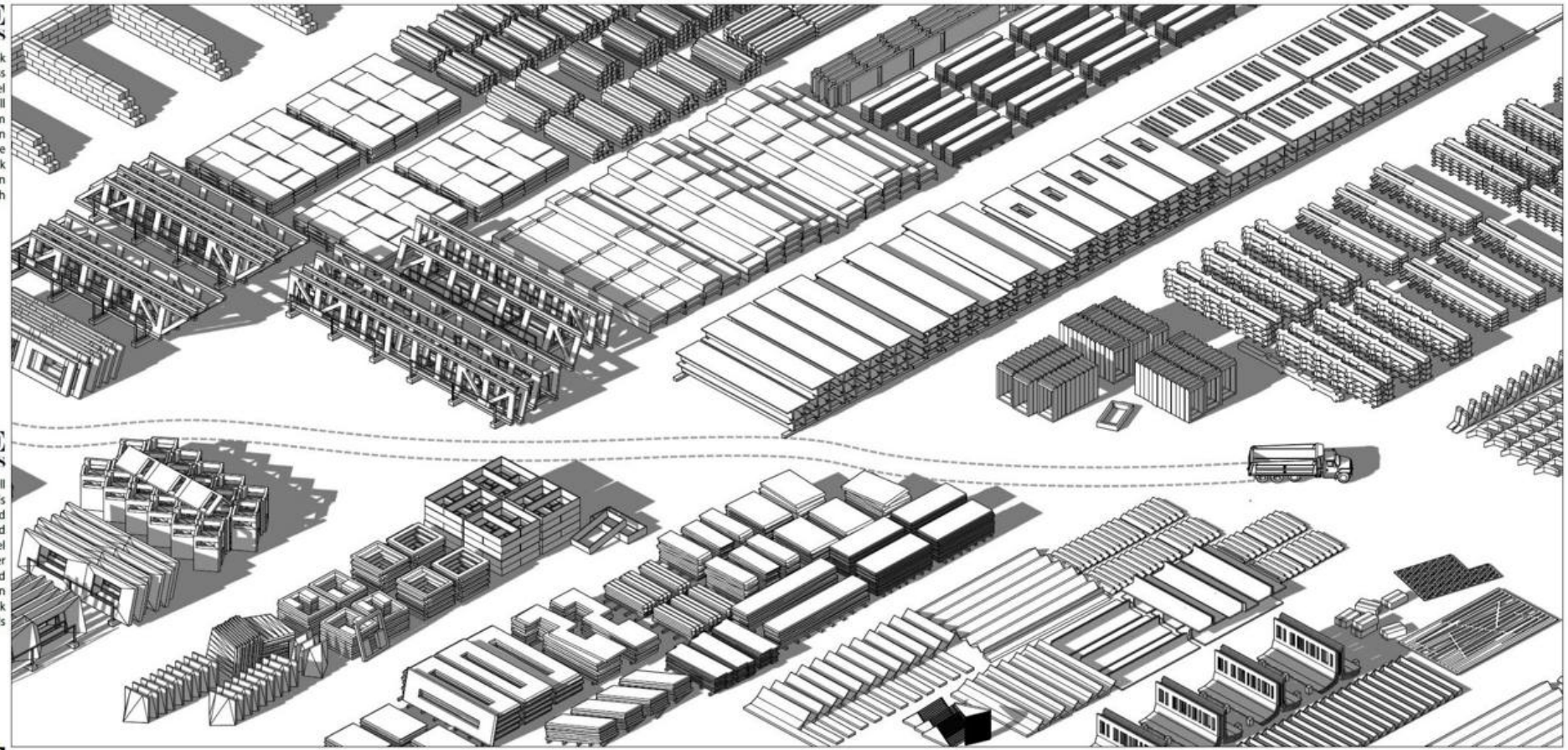
"Waste Blocks"

STRUCTURE
ELEMENTS

Waste Block
Vierendeel Truss
Structural Wall Panel
Lite Wall
Beam
Fin
Double Tee
Hollow-Core Plank
Column
Haunch

ENCLOSURE
ELEMENTS

Lite Wall
Insulated Panels
Shadow Hood
Sandwich Panel with Hood
Fin Panel
Insulated Louver
Hood
Screen
Waste Block
Tessellated Panels



PRECAST
PROTOTYPES

PROCESS AND METHODOLOGY FOR INNOVATION

12 WEEKS 36 BUILDINGS

RESEARCH

DESIGN

INTERVIEW

DISSEMINATE

	Week 1	OPTIMIZE STRUCTURE	Week 2	ASSEMBLE BAYS	Week 4	INTEGRATE ENCLOSURE	Week 8	NAME TYPES	Week 10	REPRESENT VARIANTS	Week 12	COMMUNICATE FINDINGS

PARAMETERS
INDUSTRY PARTNER

Develop Types and Variants based on compatible and optimized precast elements.

Employ "Total Precast Solutions" including structural and integrated thermal enclosure

Utilize 90% existing and conventional industry elements and 10% innovative variations on industry elements

Use a minimum of 100,000 square feet of precast elements per Variant

Limit the number of precast elements to 5 per Variant

Design for a variety of uses including office, warehouse, manufacturing, institutional and commercial.

PARAMETERS
DESIGN TEAM

Fully integrate and optimize daylighting and shadow generation

Express structural load paths in all elements and building form

Optimize and innovate elements

Create elegant character and proportion

Respond to cardinal orientation and assume local climate and context

Create spatial and formal diversity

Create adaptable space matrices for a range of functional layouts

Create Servant/Served space relationships

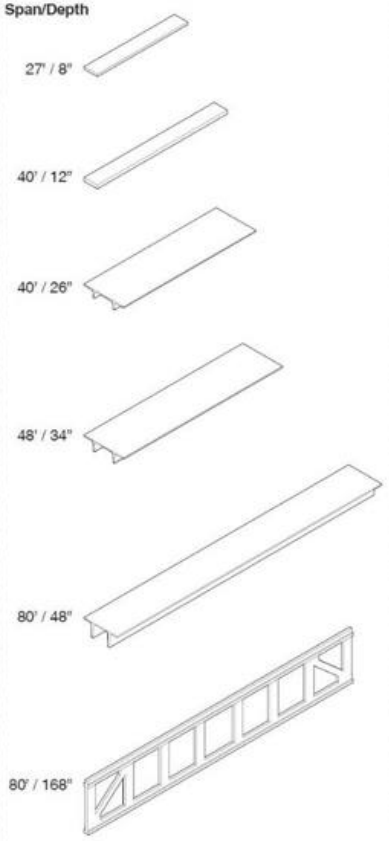
Provide covered parking options

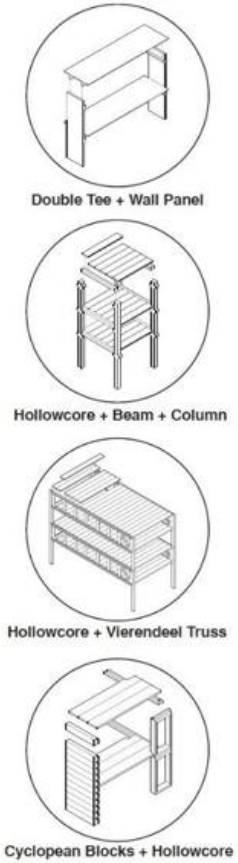
Vary floor heights based on 7-inch and 7-foot vertical modules (standard staircase adaptability)

Define a prominent, sheltered entry

PRECAST
PROTOTYPES

Span/Depth





▲ ***Innovation within convention*** *requires an understanding of the optimized spans of conventional precast elements. These optimal dimensions define each of the types, so as to make the most of precast concrete's inherent virtues.*

▲ ***Precast assemblies*** *utilize the virtues of each element to develop optimized structural bays. The complete structural systems lend a unique character to each type while providing flexibility in the configuration of interior space.*

▲ ***Precast enclosure*** *elements integrate insulation and daylight strategies. Vertical fins are optimized for east and west elevations, while louvers and hood*

▲ ***The NATO phonetic*** *alphabet was utilized to categorize the twelve precast types. Each type includes a two, three, and four*

▲ ***A novel representation strategy*** *was implemented to communicate the spirit of an anticipated and essential renaissance for precast concrete. Hybrid imagery synthesizes to express a vision for the future that*

▲ ***Diverse modes of dissemination*** *communicate the project to a range of audiences including academic sponsors, industry partners, potential clients, as well as students.*

PRECAST PROTOTYPES

PARAMETERS INDUSTRY PARTNER

Develop Types and Variants based on compatible and optimized precast elements.

Employ "Total Precast Solutions" including structural and integrated thermal enclosure

Utilize 90% existing and conventional industry elements and 10% innovative variations on industry elements

Use a minimum of 100,000 square feet of precast elements per Variant

Limit the number of precast elements to 5 per Variant

Design for a variety of uses including office, warehouse, manufacturing, institutional and commercial.

PARAMETERS DESIGN TEAM

Fully integrate and optimize daylighting and shadow generation

Express structural load paths in all elements and building form

Optimize and innovate elements

Create elegant character and proportion

Respond to cardinal orientation and assume local climate and context

Create spatial and formal diversity

Create adaptable space matrices for a range of functional layouts

Create Servant/Served space relationships

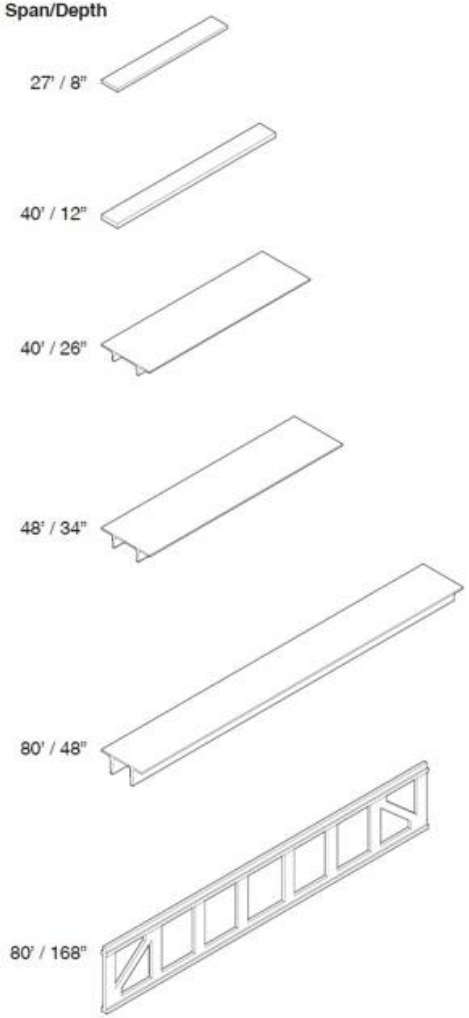
Provide covered parking options

Vary floor heights based on 7-inch and 7-foot vertical modules (standard staircase adaptability)

Define a prominent, sheltered entry

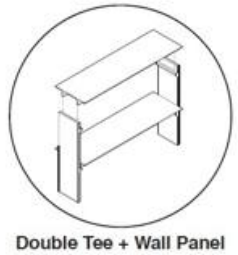
Week 1 OPTIMIZE STRUCTURE

Span/Depth



▲ ***Innovation within convention*** *requires an understanding of the optimized spans of conventional precast elements. These optimal dimensions define each of the types, so as to make the most of precast concrete's inherent virtues.*

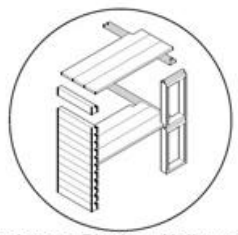
Week 2 ASSEMBLE BAYS



Double Tee + Wall Panel

Hollowcore + Beam + Column

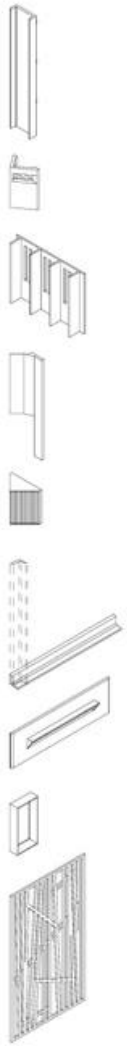
Hollowcore + Vierendeel Truss



Cyclopean Blocks + Hollowcore

▲ ***Precast assemblies*** *utilize the virtues of each element to develop optimized structural bays. The complete structural systems lend a unique character to each type while providing flexibility in the configuration of interior space.*

Week 4 INTEGRATE ENCLOSURE



▲ ***Precast enclosure*** *elements integrate insulation and daylight strategies. Vertical fins are optimized for east and west elevations, while louvers and hood elements shade south-facing elevations.*

Week 8 NAME TYPES

▲ ***The NATO phonetic*** *alphabet was utilized to categorize the twelve precast types. Each type includes a two, three, and four story variant.*

Week 10 REPRESENT VARIANTS

ALPHA
BRAVO
CHARLIE
DELTA
ECHO
FOXTROT
GOLF
HOTEL
INDIA
JULIET
KILO
LIMA

▲ ***A novel representation strategy*** *was implemented to communicate the spirit of an anticipated and essential renaissance for precast concrete. Hybrid imagery synthesizes to express a vision for the future that is both connected to the past and radically different from it.*

PRECAST
PROTOTYPES

PROJECT PARAMETERS

Develop 12 Types and 36 Variants of 2, 3 and 4 story height with adaptability for 1 story option.
Elements and Systems (85% Conventional and 15% Innovative)
Fully integrated Daylighting solutions
Minimum 100,000 sf of precast elements per Variant
Utilize existing and potential Kerkstra products
Innovate with variations on existing Kerkstra products
Optimize and innovate elements
Respond to Cardinal Orientation and assume SE/SW Michigan climate and context
Asymmetrical design for site and program adaptability
Range of spatial variations, including exterior.
Total Precast Solutions including structural and integrated thermal enclosure with high aesthetic value
Flexible for a variety of uses including office, warehouse, manufacturing, institutional and commercial.
Create Servant/Served space relationships
Covered Parking options
Variations on floor heights and 7-inch and 7-foot vertical modules (standard staircase adaptability)

PRECAST
PROTOTYPES

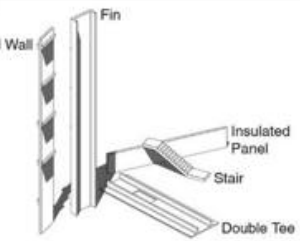
PROJECT PRINCIPLES

Minimum of Means to Maximum Effect
Optimization with Economy
Relevant and Creative Cardinal Orientation
Elegant Character
Maximum Daylight within Optimized System
Prominent Entry with Shelter
Clarity in Expression of Elements and Forces
Spatial and Formal Diversity

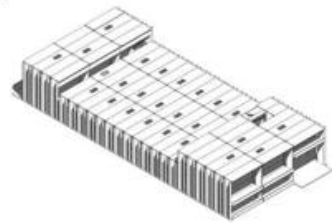
12 TYPES
56 VARIANTS

ALPHA TYPE A

PARTS



A.2



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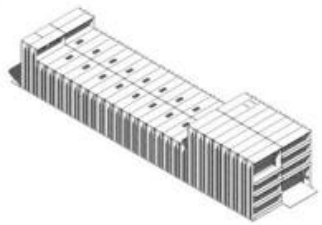

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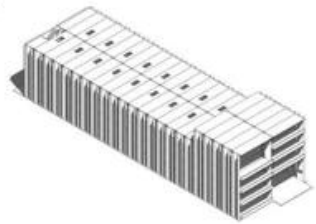

BRAVO TYPE B

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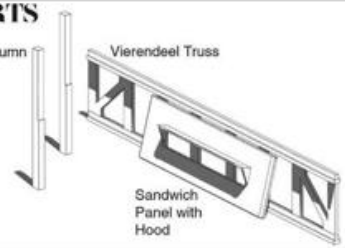

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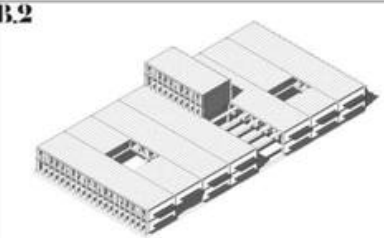

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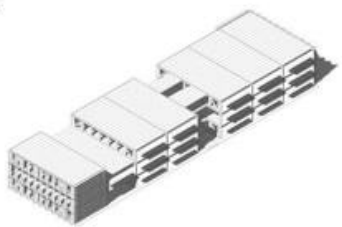

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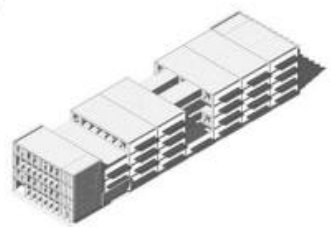

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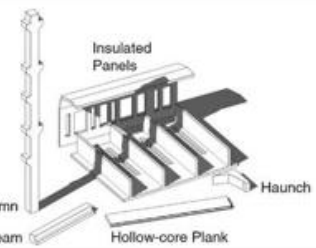
PARTS



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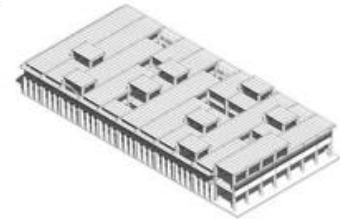

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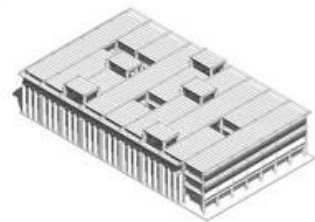

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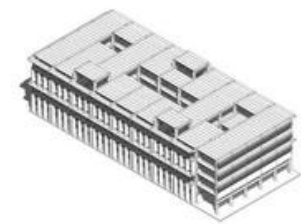

DELTA TYPE D

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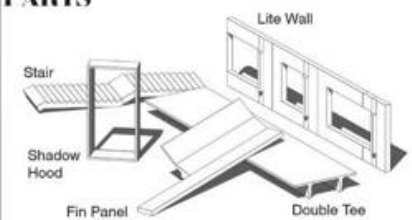


D.2

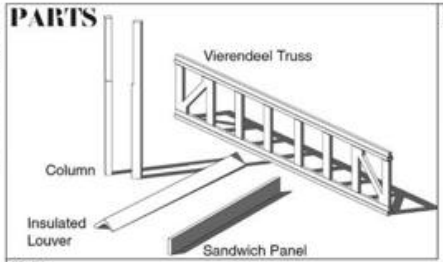
D.3

D.4

12 TYPES
56 VARIANTS

ECHO TYPE E

PARTS



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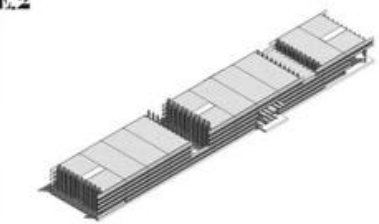

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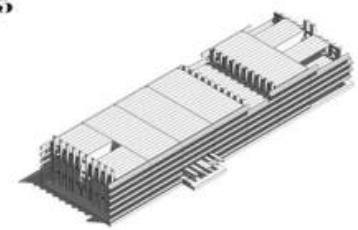

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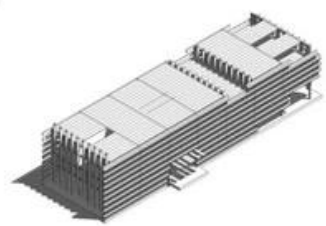
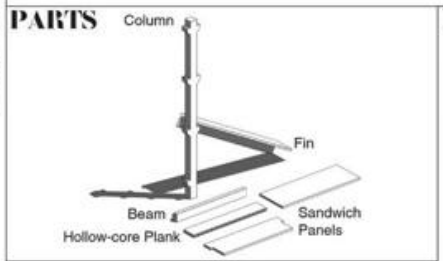


FOXTROT TYPE F

PARTS



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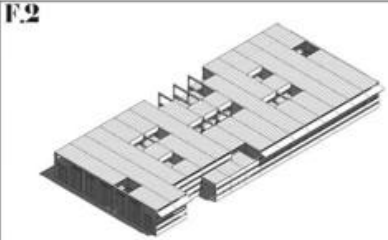

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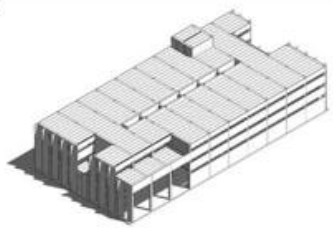
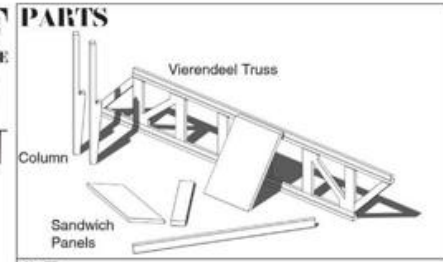


F.4

GOLF TYPE G

PARTS



G.2

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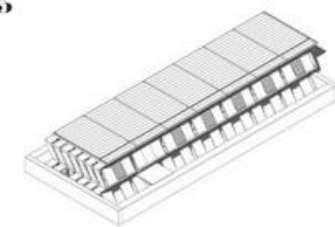

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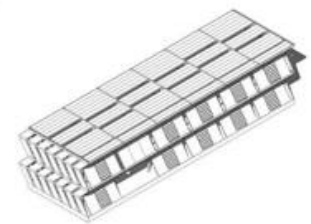
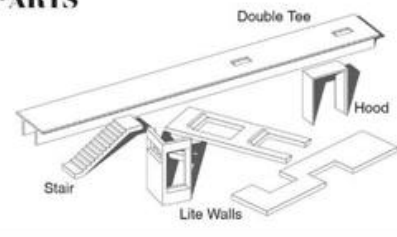
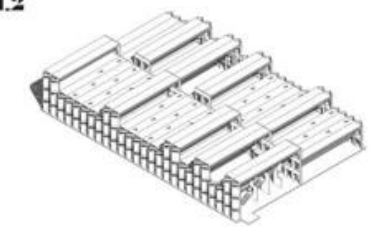


HOTEL TYPE H

PARTS

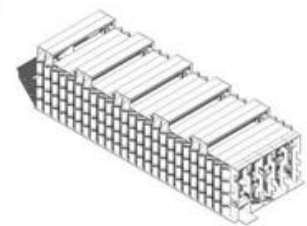


H.2



H.3

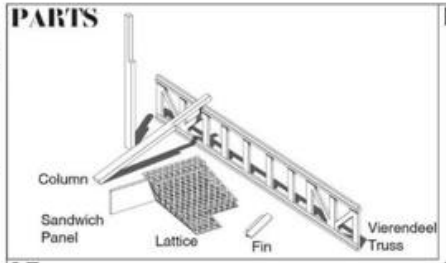
H.4



12 TYPES
56 VARIANTS

INDIA TYPE I

PARTS



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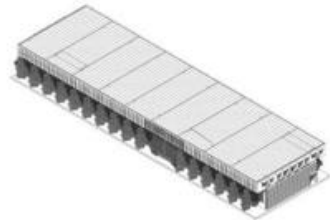

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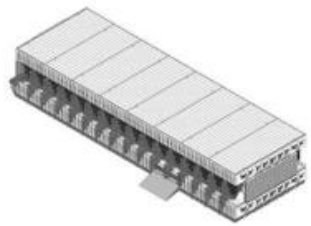

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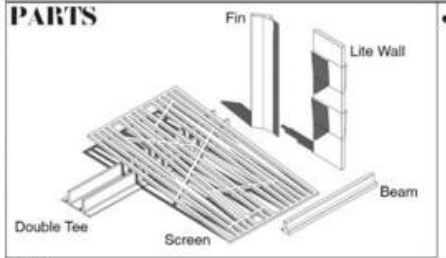
JULIET TYPE J

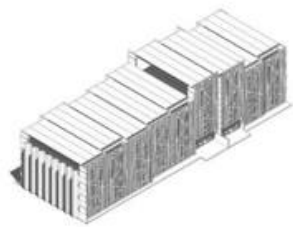
PARTS



J.2

J.3

J.4



KILO TYPE K

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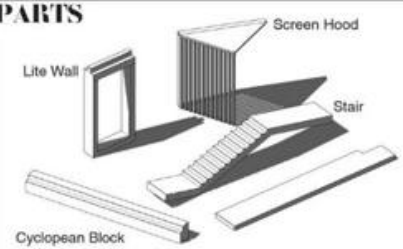

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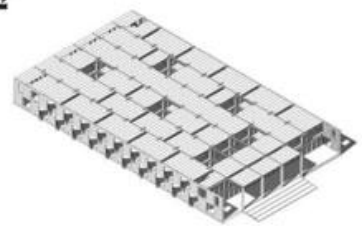

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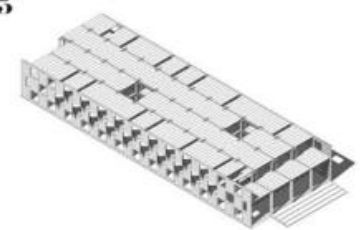

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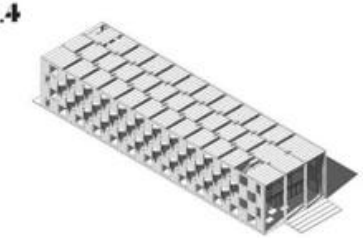
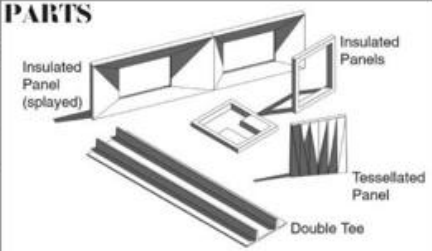
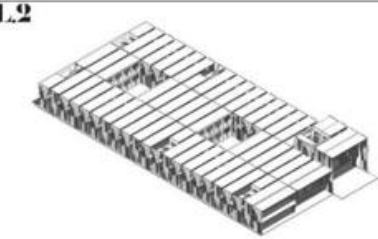


LIMA TYPE L

PARTS



L.2



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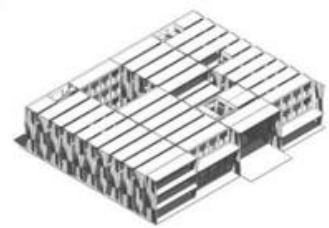

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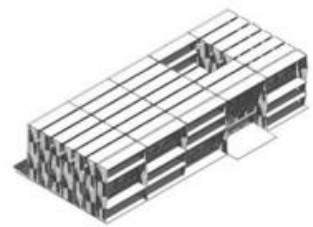


NATO
phonetic alphabet, codes & signals

The ability to communicate and make yourself understood can make a difference in life-threatening situations – imagine for example that you are trying to alert a search and rescue helicopter of the position of a downed pilot. To ensure clear communication, NATO uses a number of well-known formats which are in general use. NATO standardization agreements enable forces from many nations to communicate in a way that is understood by all.

Some standards can be found in everyday civilian and military life. "Bravo Zulu", typically signalled with naval flags on ships at sea and meaning "well done", is also commonly used in written communication by the military, for example by replying "BZ" to an email.

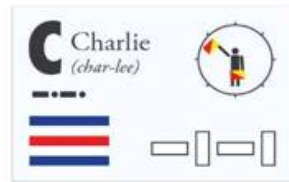
Phonetic alphabet
The NATO alphabet became effective in 1956 and, a few years later, turned into the established universal phonetic alphabet for all military, civilian and amateur radio communications.

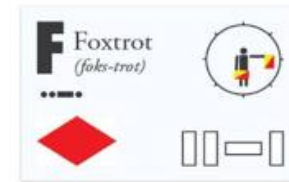
International Morse Code
Morse code transmits text through on-off tones, light-flashes or clicks. It was widely used in the 1890s for early radio communication, before it was possible to transmit voice.

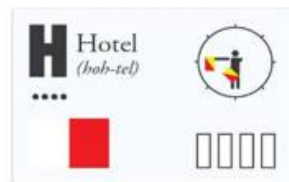
Flaghoist communication
Ships use flags as signals to send out messages to each other. The use of flags, known as flaghoist communication, is a fast and accurate way to send information in daylight.

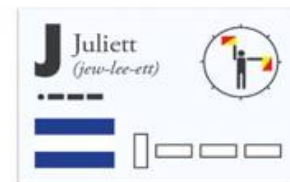
Semaphore
Semaphore is a system in which a person sends information at a distance using hand-held flags – depending on the position of the flags, the message will vary. The signaller holds the flag in different positions that represent letters or numbers.

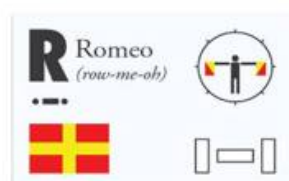
Panel signalling
Panels are visual signals for sending simple messages to an aircraft. Using a limited code, ground forces can send messages to pilots, for example to request medical supplies.











Numbers

More information on NATO's codes, signals and standards can be found on the NATO Standardization Office (NSO) website: http://nso.nato.int/nso/ – More on the history of the NATO phonetic alphabet: http://www.nato.int/docu/edito/

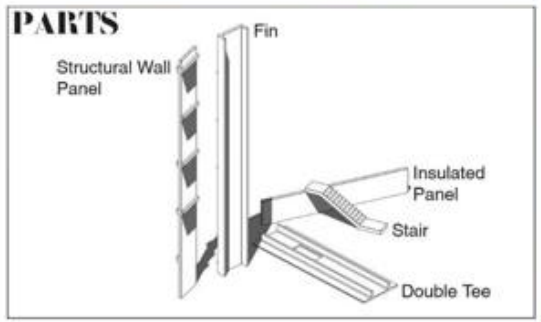
This poster can be downloaded at http://www.nato.int/alphabet

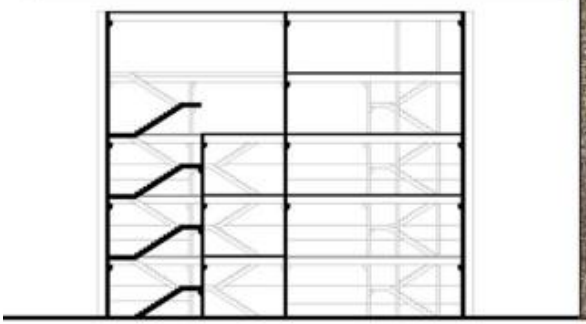
Created by Communications Services, Public Diplomacy Division, NATO HQ, Brussels, Belgium – NATO 2018 ©

Phonetic transcription is based on English pronunciation

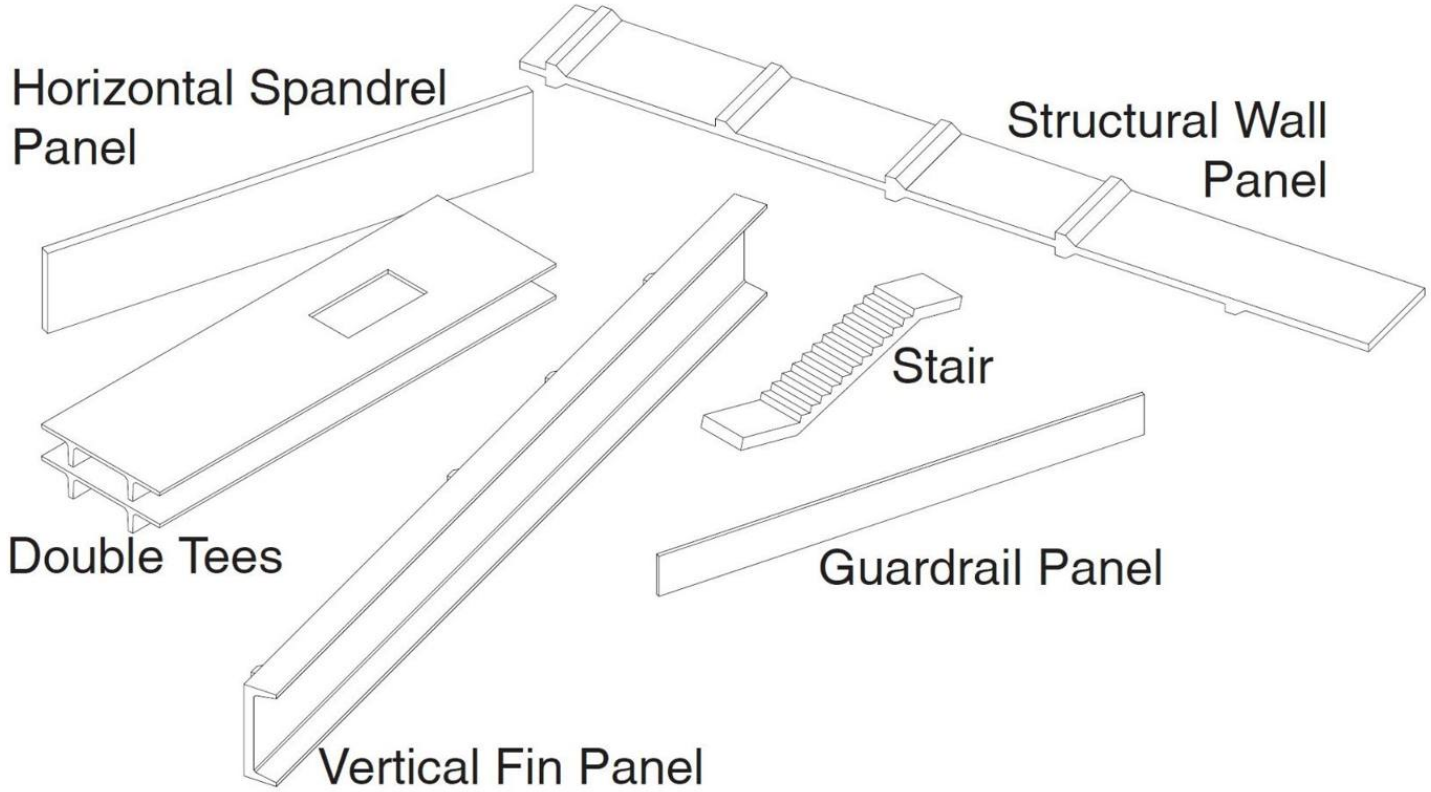
ALPHA
TYPE

A.3
VARIANT





FINITE ELEMENTS



MEASURES

Building Footprint

80 feet by 336 feet = 26,880 square feet

Structural Bay Size

12 feet by 40 feet

40 foot wide by infinity column-free space

Building Height

14 foot and 28 foot floor-to-floor height

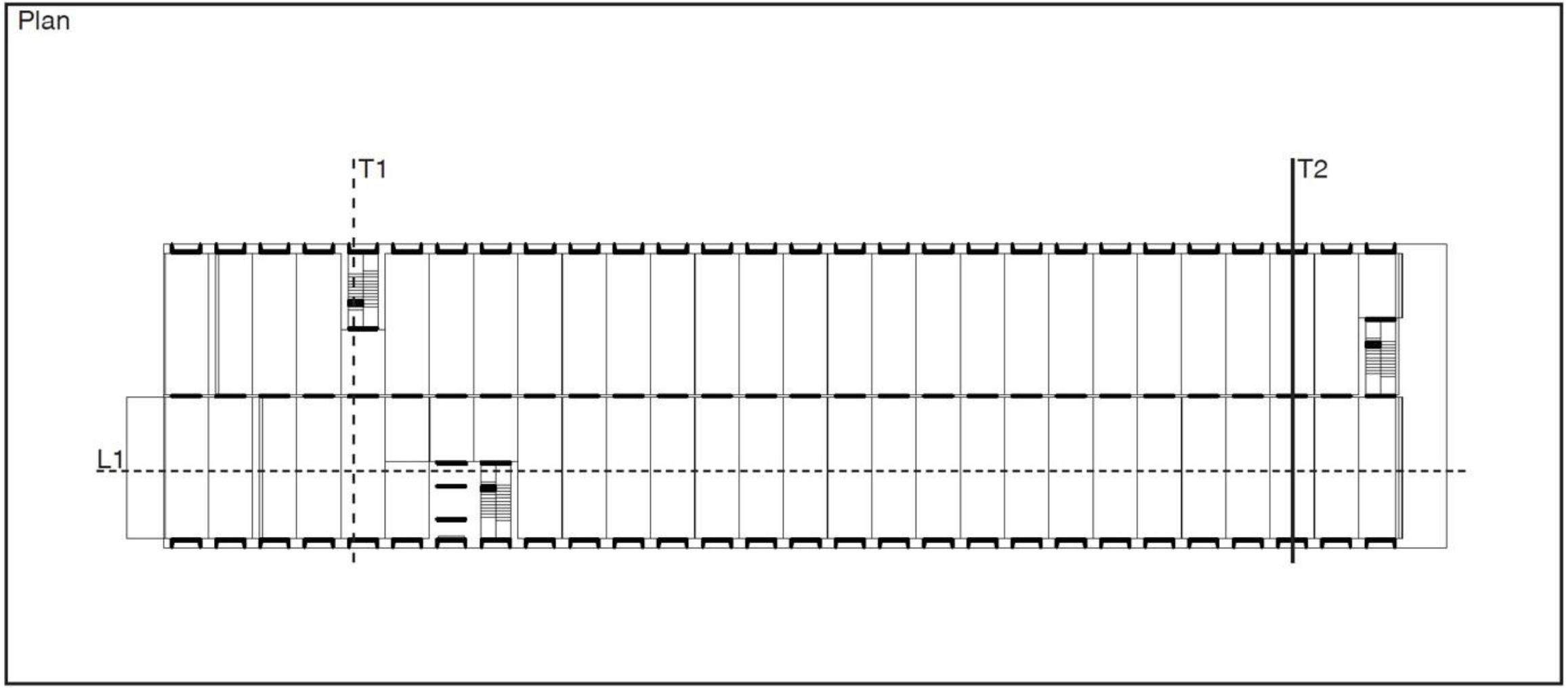
Three stories at 42 feet to 70 feet total height

Area of Precast Elements

129,080 square feet of structure and enclosure

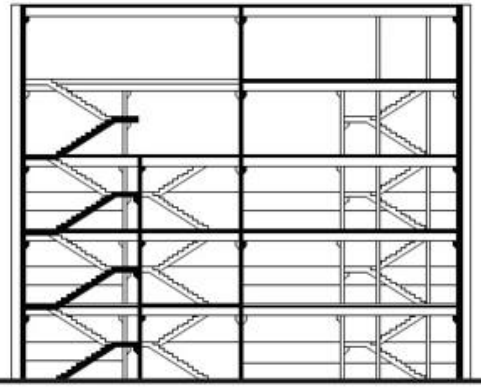
Enclosure Glazing Ratio

22% of total thermal enclosure

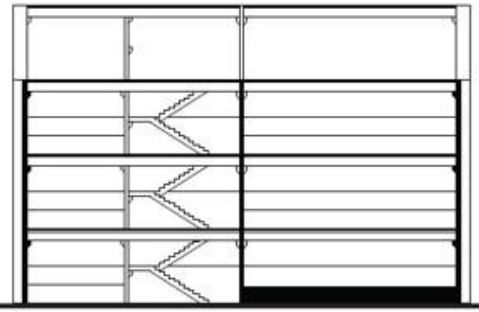


ALPHA

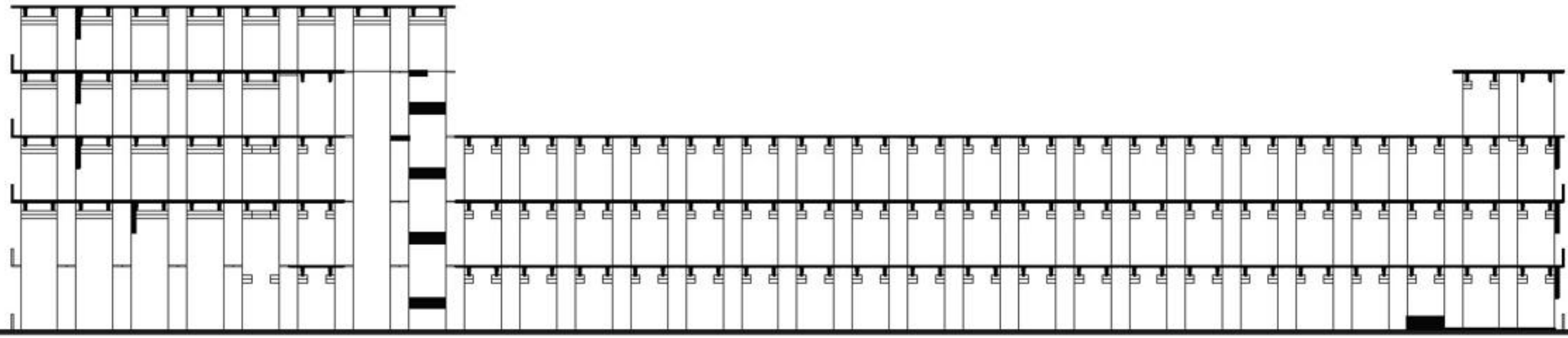
Section:
T1



Section:
T2



Section:
L1



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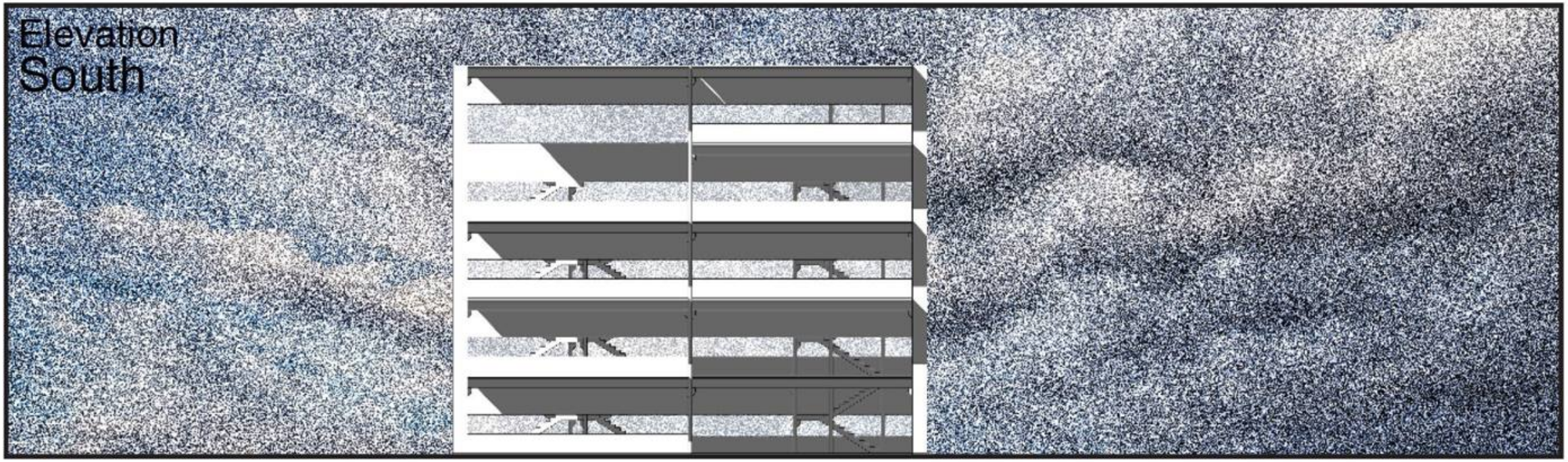


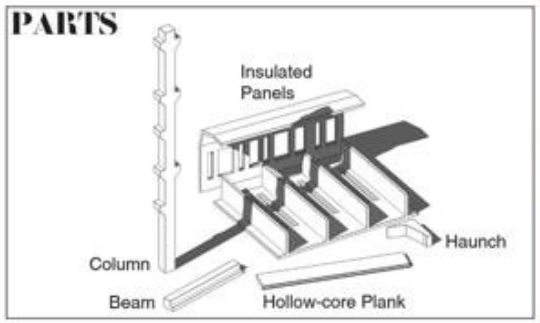
ALPHA

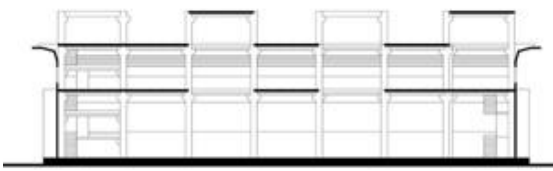
CHARLIE

TYPE

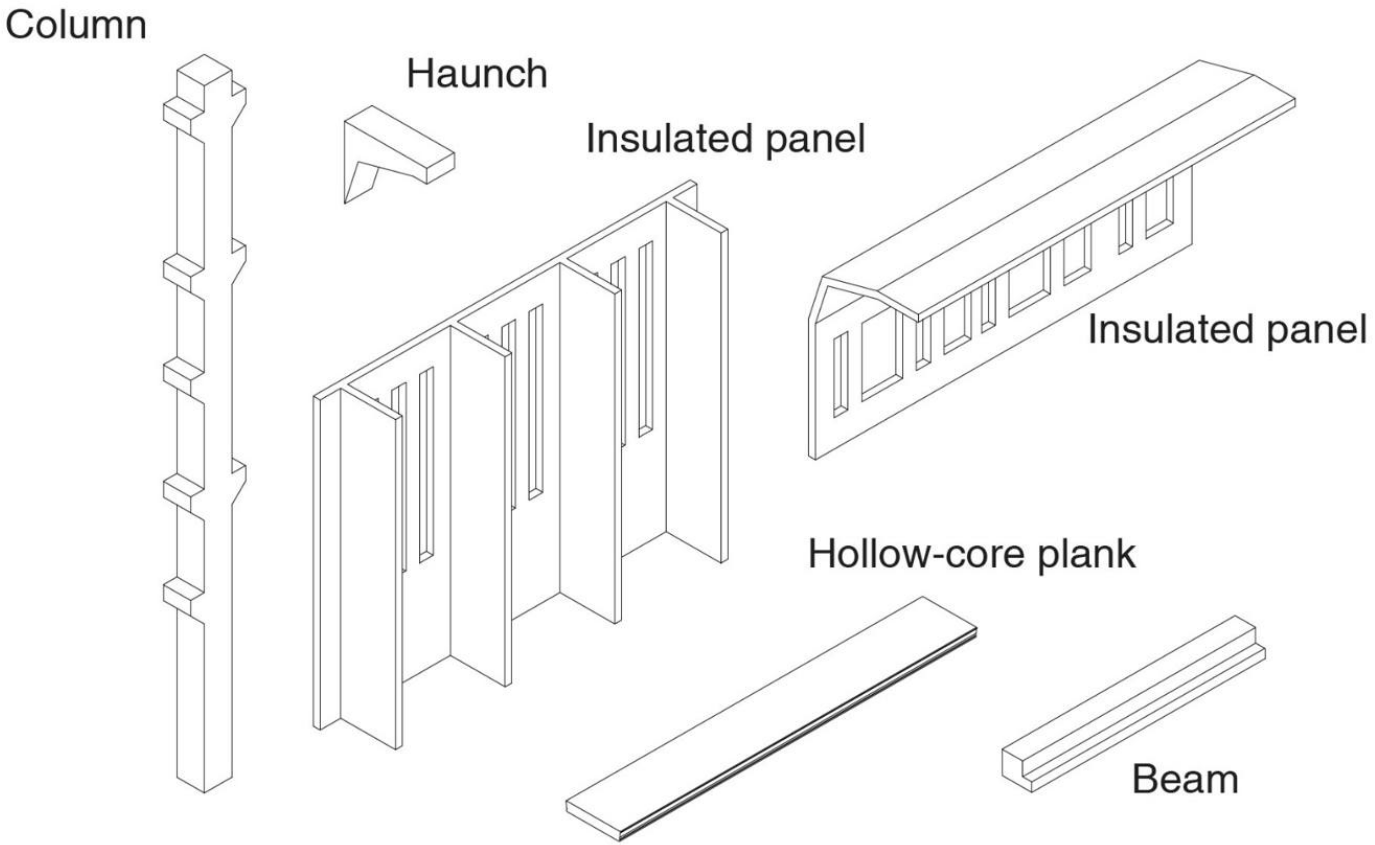
C.2

VARIANT





FINITE ELEMENTS



SPECIFICATIONS

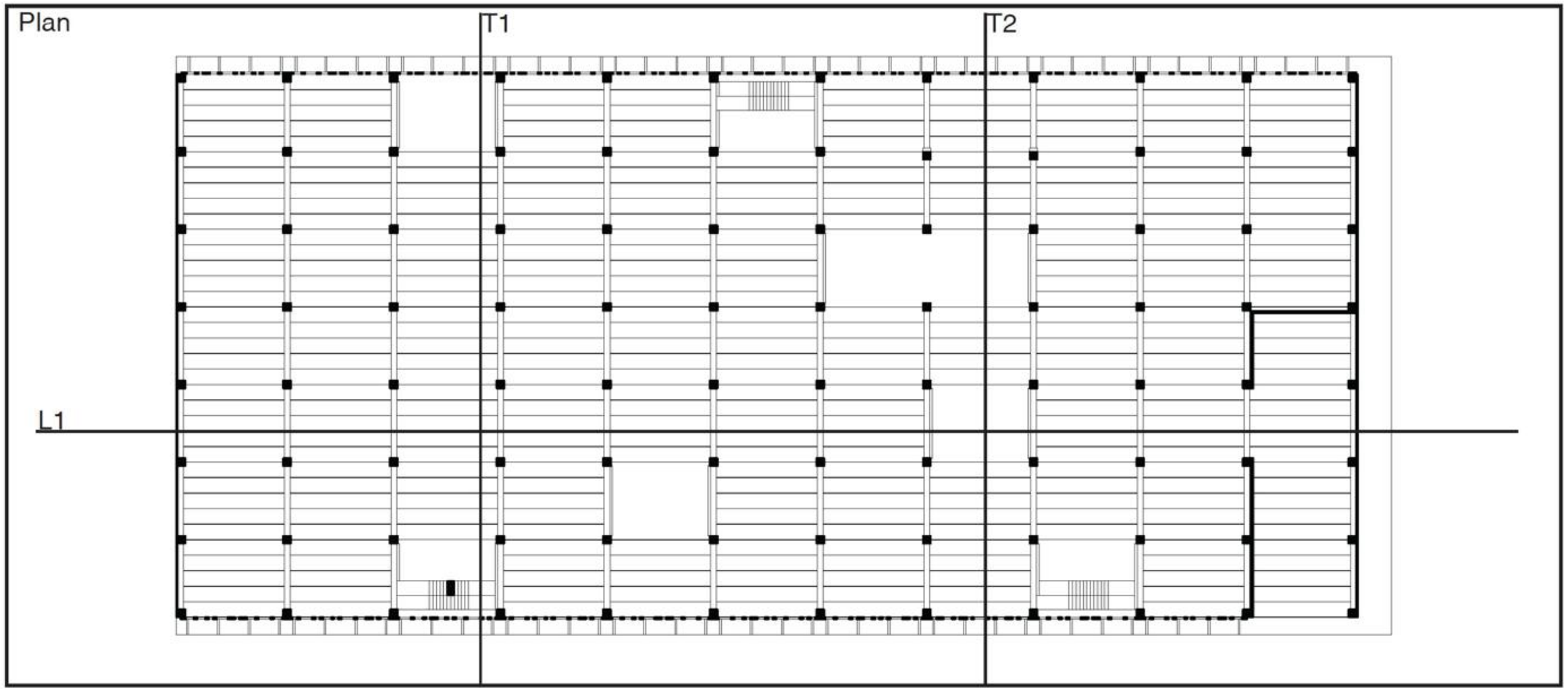
Building Footprint: 140 feet by 286 feet = 40,040 square feet

Structural Bay Size: 20 feet (beam span) by 26 feet (plank span), column-free space

Building Height: 14 foot floor-to-floor height. Two stories at 28 feet, with monitors at 42 feet total height

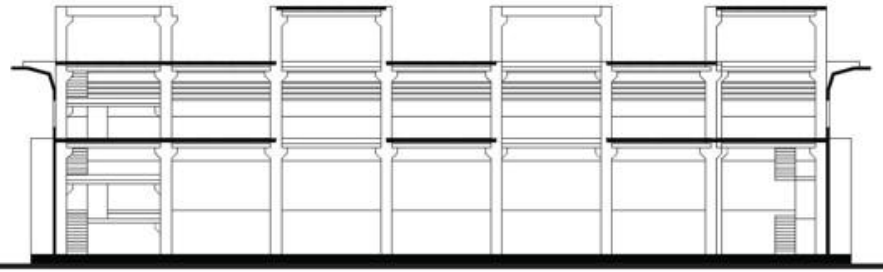
Area of Precast Elements: 96,044 square feet of structure and thermal enclosure

Enclosure Glazing Ratio: 21% of total thermal enclosure

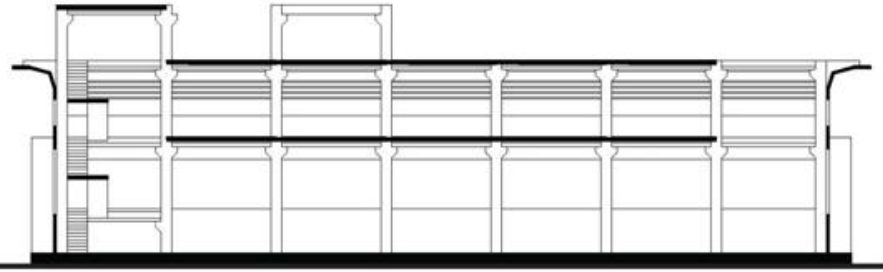


CHARLIE

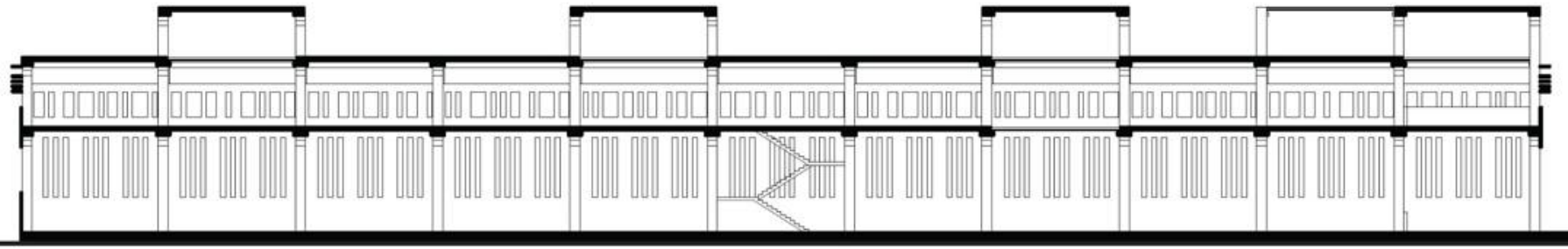
Section:
T1



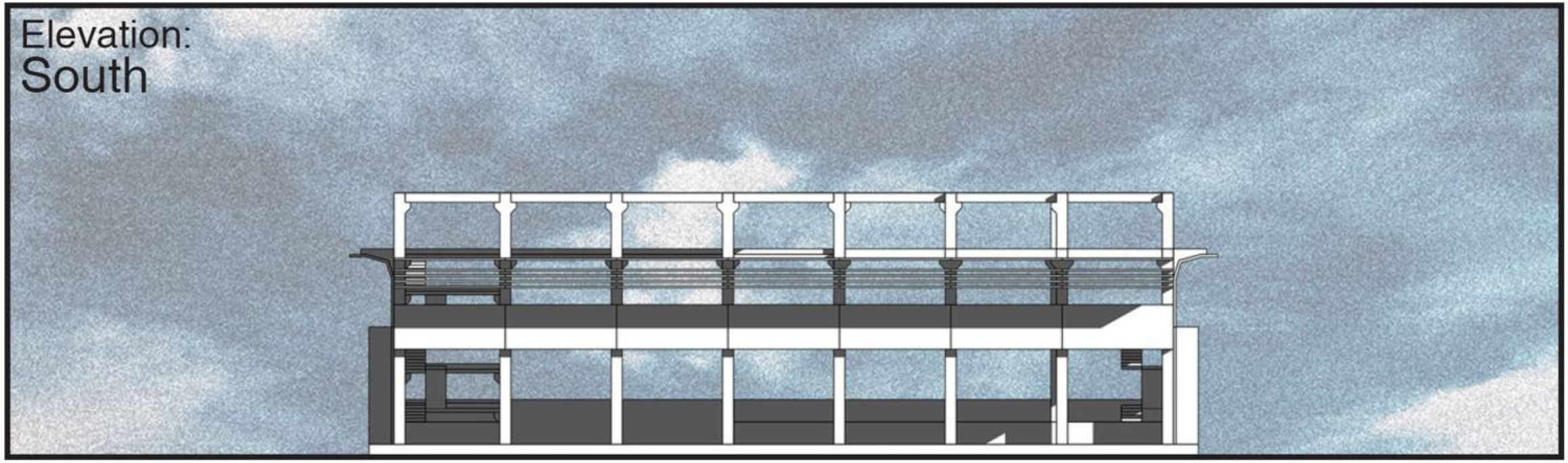
Section:
T2



Section:
L1



CHARLIE

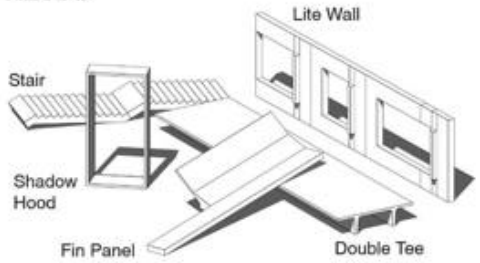


CHARLIE

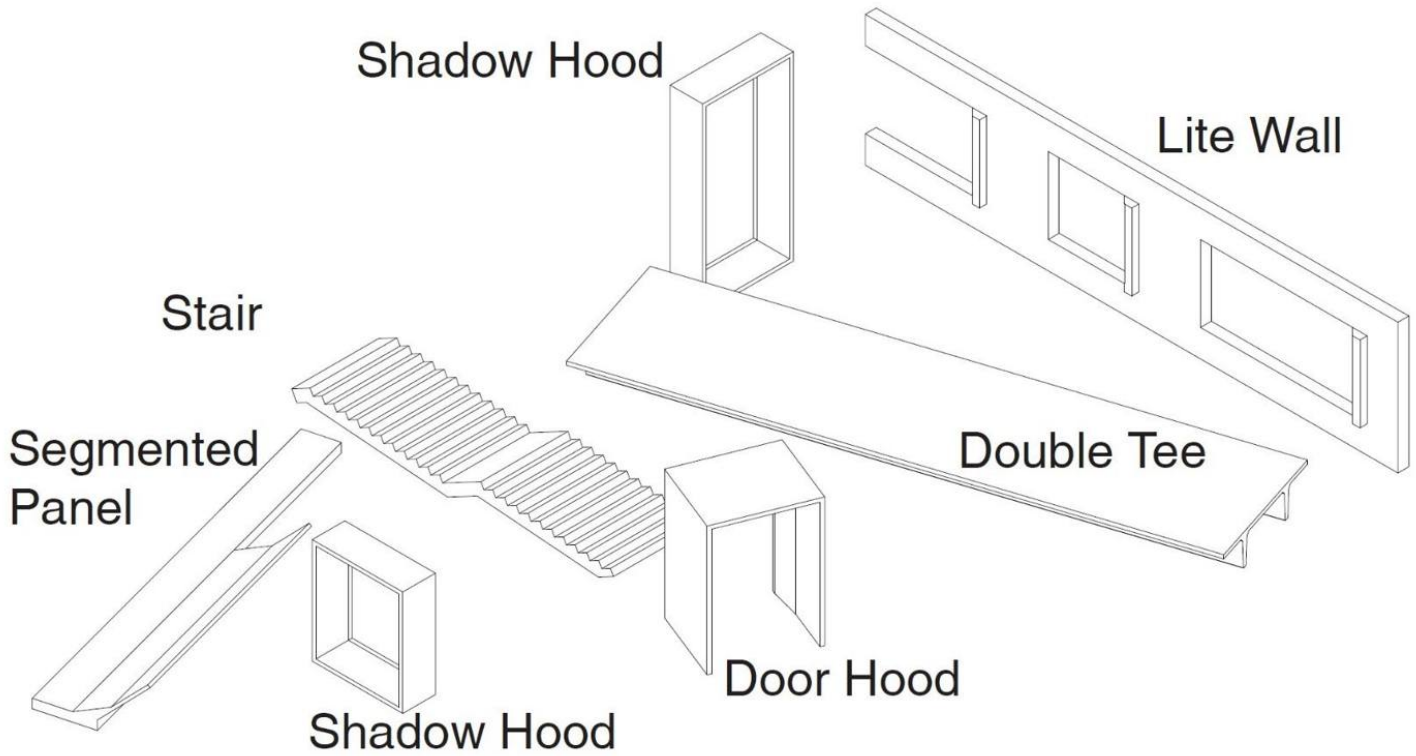
DELTA
TYPE

D.3
VARIANT

PARTS



FINITE ELEMENTS



MEASURES

Building Footprint

108 feet by 300 feet = 32,400 square feet

Structural Bay Size

12 feet by 50 feet

50 feet by 108 feet column-free space

Building Height

14 foot floor-to-floor height

Three stories at 42 feet to 49 feet total height

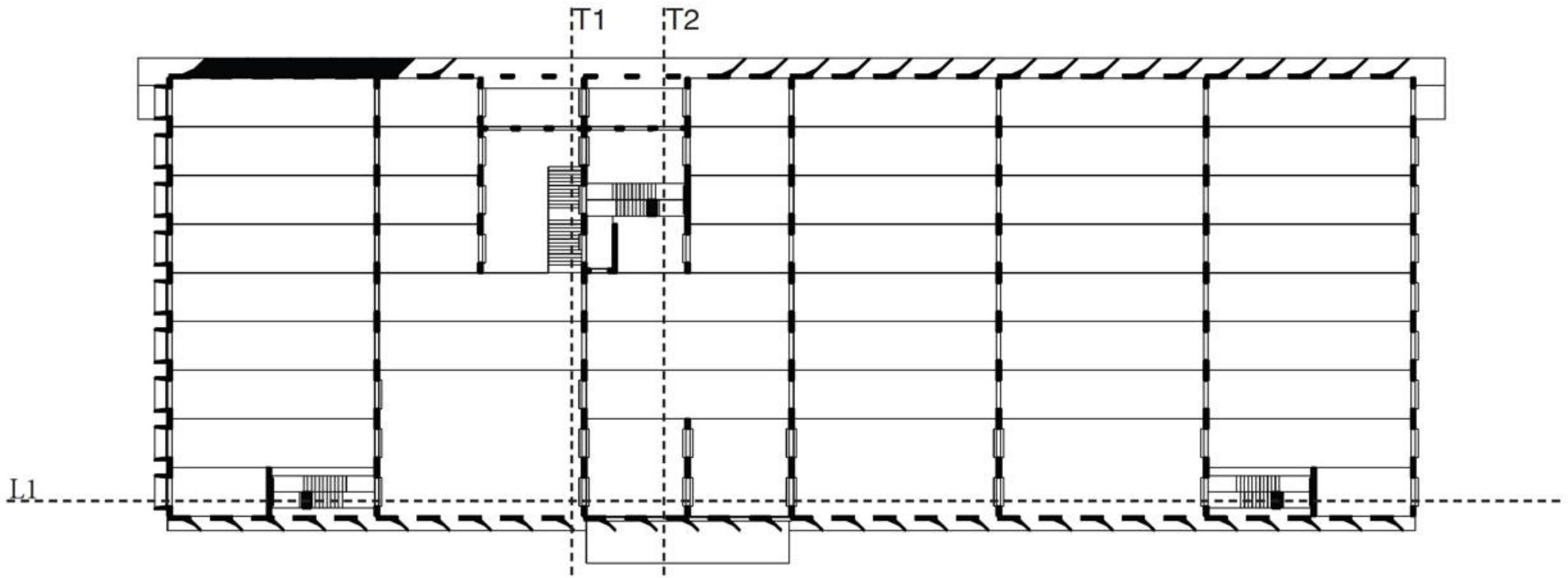
Area of Precast Elements

122,778 square feet of structure and enclosure

Enclosure Glazing Ratio

26% of total thermal enclosure

Plan

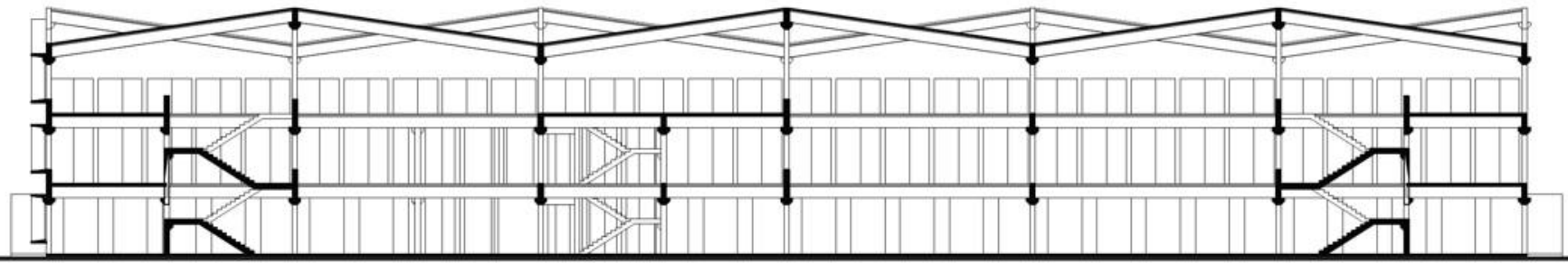


DELTA

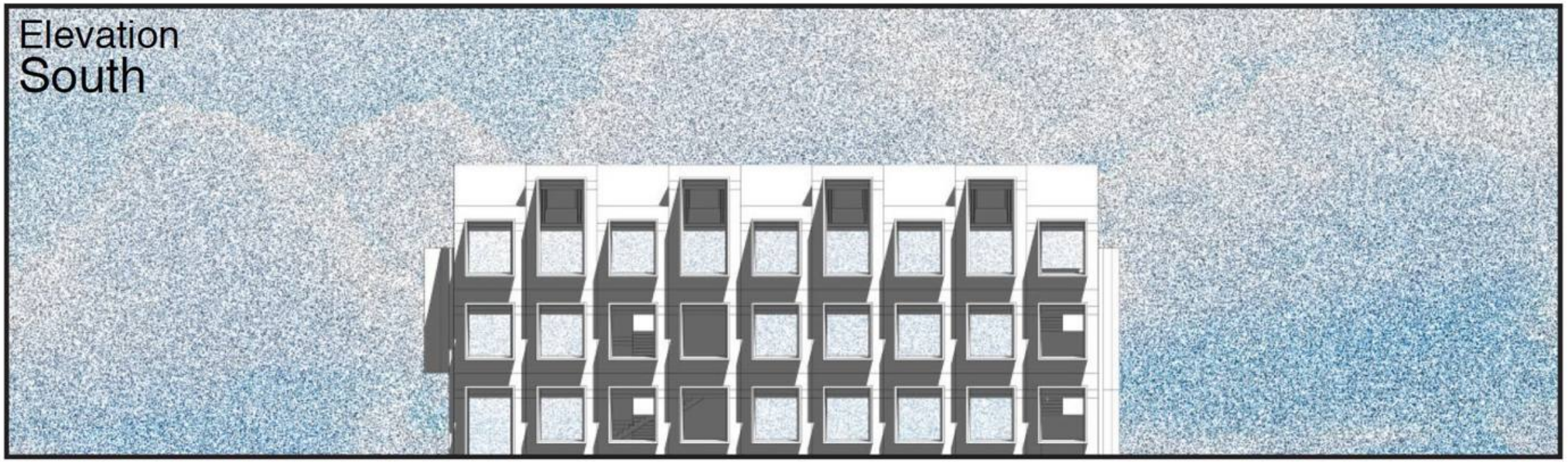
Section:
T1

Section:
T2

Section:
L1

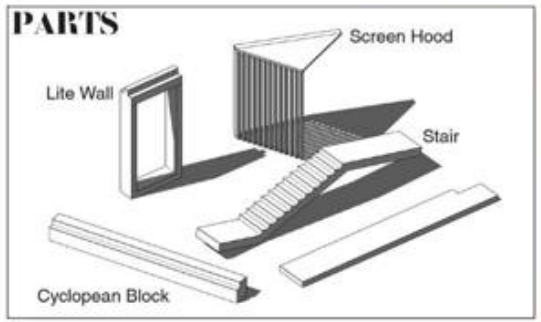


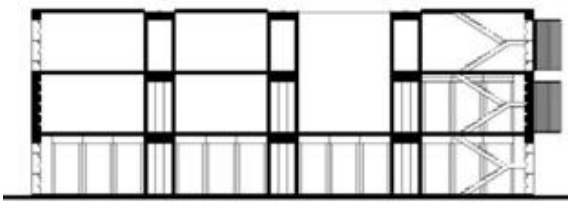
DELTA


Elevation
South

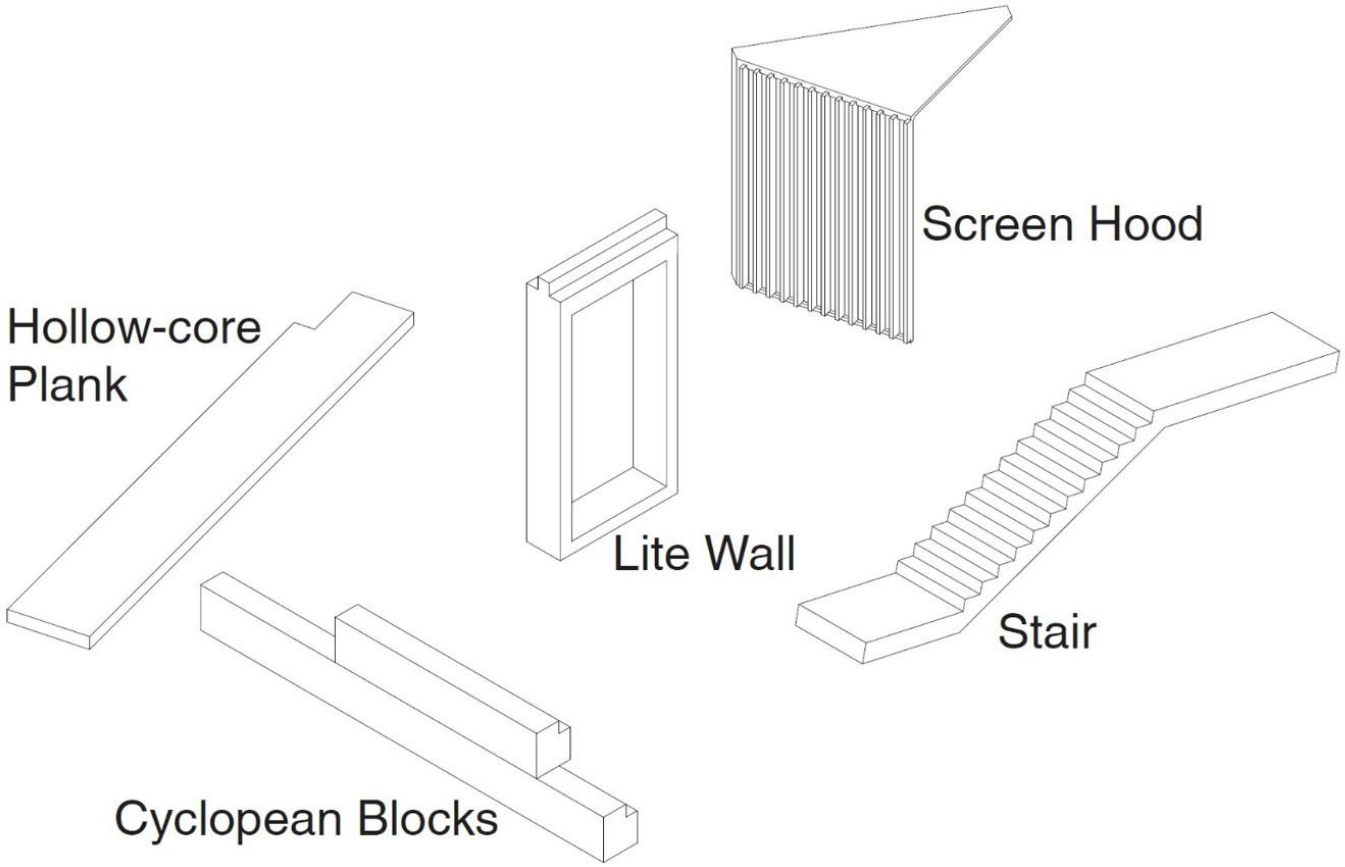
KILO
TYPE

K.3
VARIANT





FINITE ELEMENTS



MEASURES

Building Footprint

114 feet by 324 feet = 36,936 square feet

Structural Bay Size

24 feet by 28 feet

24-foot wide column-free space

Building Height

14-foot floor-to-floor height

Three stories at 42 feet total height

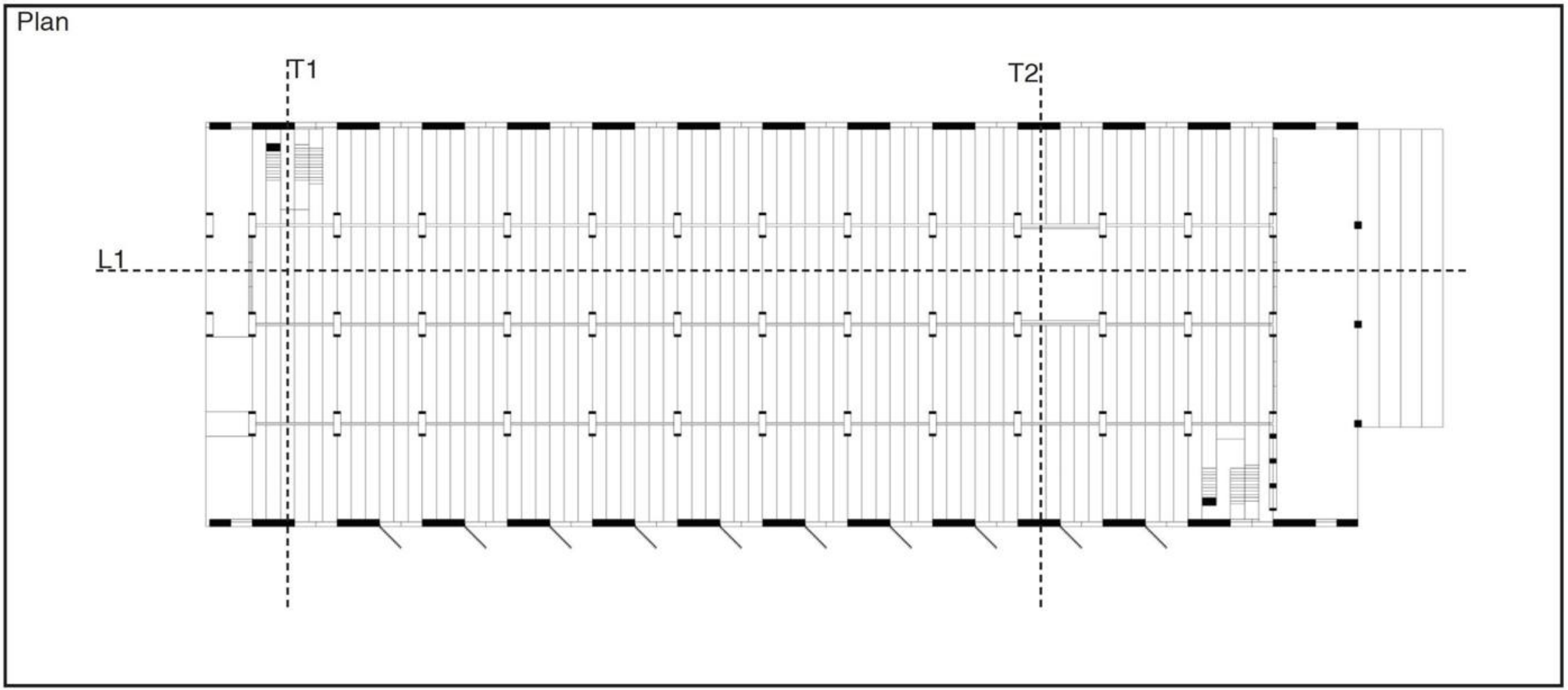
Area of Precast Elements

126,416 square feet of structure and enclosure

Enclosure Glazing Ratio

50% of total thermal enclosure

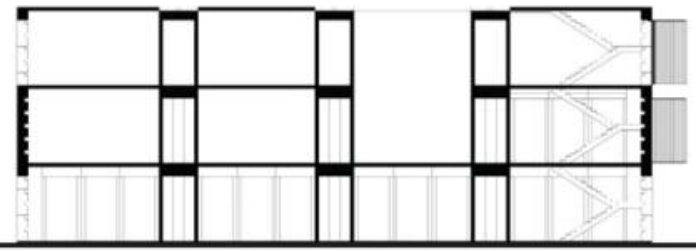
KILO



KILO

Section:
T1

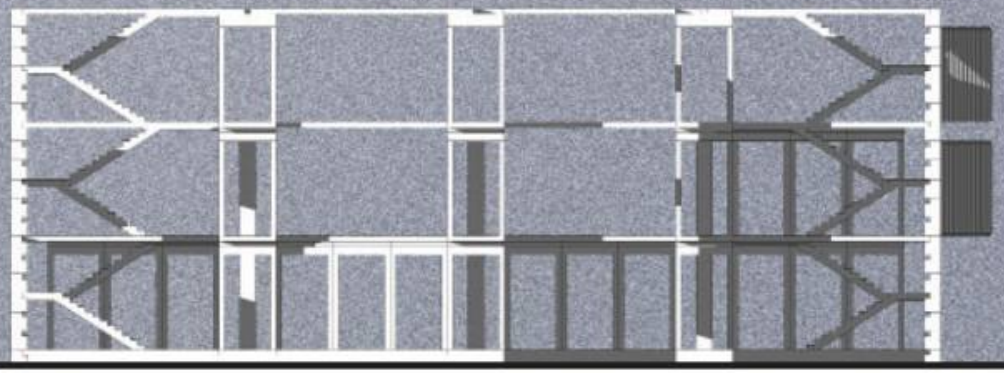
Section:
T2



Section:
L1

KILO

Elevation
North



KILO

TRANSFORMATION and evolution

INDUSTRY to ACADEMIA

RESEARCH to CLASSROOM

PRECAST PROTOTYPES to CONCRETE BUILDING

CONCRETE BUILDING project

PROJECT-APPLIED CONCEPTUAL PARAMETERS

***…Economy** of material and optimization of structural elements is the prime driver*

***…Functional** building size, structural bays, sloped roof, daylight apertures*

***…Aesthetics** guided by Economy, Function, and the "natural" aesthetic of material and structural elements*

CONCRETE BUILDING project

PROJECT PARAMETERS: Use, Size, and Structural Loads

ENGINEERING OFFICE: Ypsilanti, Michigan (60’ x 120’ nominal footprint) (mat slab raft foundation)
ROOF LOADS: Snow Load: 20 psf Roofing DL: 15 psf
FLOOR LOADS: Floor LL: 50 psf Floor 15: 15 psf

GROCERY STORE: Pinconning, Michigan (60’ x 120’ nominal footprint) (mat slab raft foundation)
ROOF LOADS: Snow Load: 40 psf Roofing DL: 15 psf
FLOOR LOADS: Floor LL: 75 psf Floor DL: 20 psf

PUBLIC LIBRARY: Houghton, Michigan (60’ x 120’ nominal footprint) (mat slab raft foundation)
ROOF LOADS: Snow Load: 100 psf Roofing DL: 15 psf
FLOOR LOADS: Floor LL: 150 psf Floor DL: 15 psf

Note: students rotate to the next building use & load criteria in line *(Grocery to Library etc…)*

CONCRETE BUILDING project

PROJECT PARAMETERS: For all buildings/all students

SLOPED ROOF: with pitch range of 3 degrees minimum up to 10 degrees maximum.
(use only single slope pitch, not double slope i.e. pitching with two slopes in one plane)

HEIGHTS: Main Floor to 2nd Floor: 14'-0", and 2nd Floor to Roof: 14'-0" to 21'-0"

FLOORS: 2nd Floor with Precast Structure with 2" concrete topping; Main Floor 2" concrete topping on Mat Slab Raft Foundation

TWO STRUCTURAL PRECAST STAIRS: to work with the logic & order of the floor structure, minimum width of 48" per flight and at landing

ONE ELEVATOR SHAFT OPENING: to work with the logic & order of the floor structure

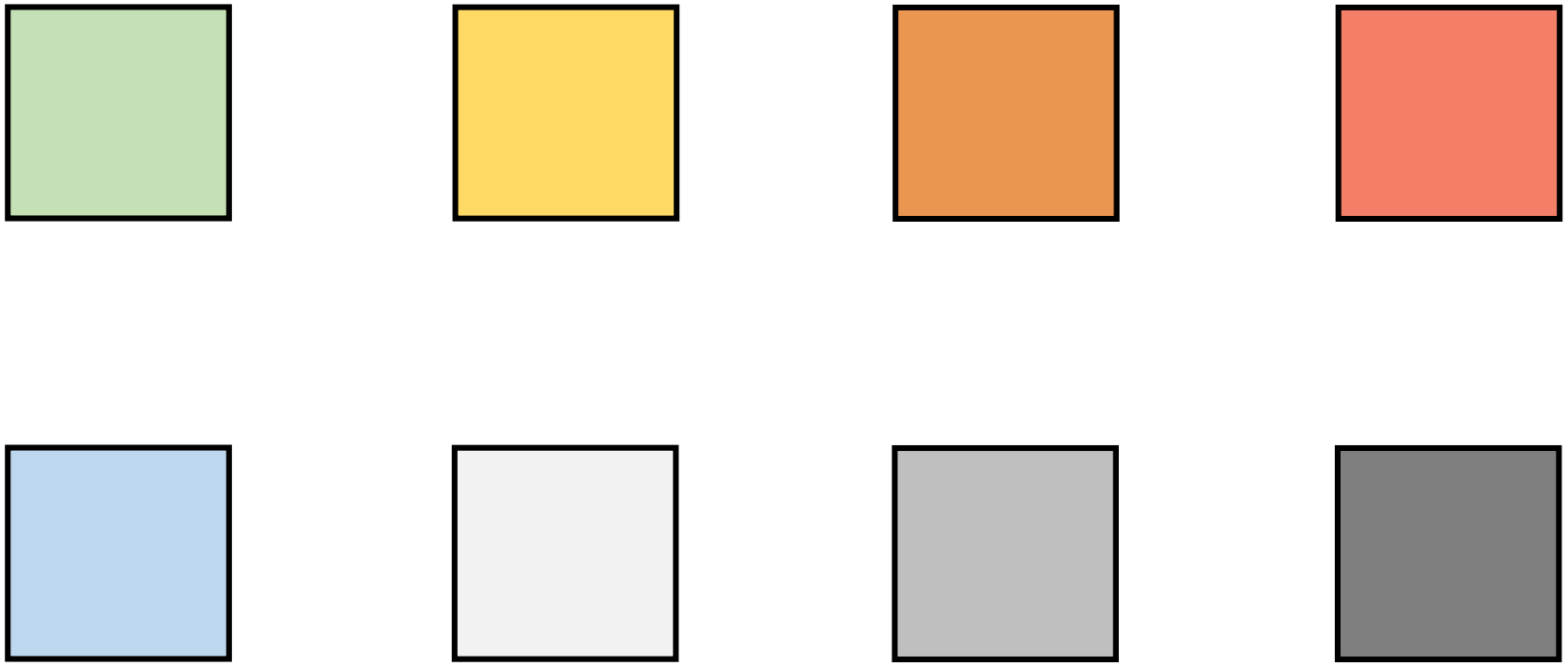
DOUBLE-HEIGHT SPACE(S): with open structural bays in 2nd Floor plan area, not to exceed 25% of 2nd Floor area

STRUCTURAL FOUNDATION: Mat Slab Raft Foundation (to resist high water-table uplift forces)

LATERAL-RESISTANT MECHANISM: Vertical Shear Walls in both directions & four sides + Horizontal Diaphragm at Floor & Wall

COLOR-CODED STRUCTURE ELEMENTS: with each element type a coded color (i.e. four elements = four colors)

CONCRETE BUILDING color pigment & coding



CMI
CONSPEC
materials inc
Dark Buff 2-lbs
Dark Buff 4-lbs
Dark Buff 6-lbs
Sandstone 1-lbs
Sandstone 3-lbs
735 Brown 1-lbs
735 Brown 3-lbs
735 Brown 5-lbs
Deep Black 3-lbs
Deep Black 5-lbs
10-285 Red 1-lbs
10-285 Red 3-lbs
10-285 Red 5-lbs
Irish Green 1-lbs
Irish Green 5-lbs
Jet Black 1-lbs
Jet Black 3-lbs
Jet Black 5-lbs
Rust Brown 1.5-lbs
Rust Brown 5-lbs
Terra Cotta 1-lbs
Terra Cotta 5-lbs
Yellow 3-lbs
Yellow 5-lbs
Surf Blue 5-lbs

EXPERTISE AND EXPERIENCE IN ARCHITECTURAL CONCRETE

SIKA IS A GLOBAL COMPANY with an enviable reputation for innovation, quality and experience. The ability to appreciate concrete as an aesthetically versatile construction material has led to many Sika innovations in architectural concrete.

The flexibility of concrete in design enables the designer to create structures not possible with other construction materials. Combining this flexibility with Sika innovation allows structures of true architectural value to be realized.

THE ADVANTAGES OF SIKA ARCHITECTURAL CONCRETE:

- Durable and low maintenance
- Cost effective in comparison with other systems
- Resists mechanical damage
- Eliminates the need for the application of coatings or paints
- Versatile in design

TRIBUTARY ELEMENTS

LOAD PATHS AND LOADS TO A COLUMN AND FOOTING

ROOF LOAD TO COLUMN

Roof Double-Tees (8DT12+2):
Roof LL + DL (45 psf) + DT self-weight (54 psf)
Roof Tributary Area = 15 ft x 24 ft = 360 sf
Roof Tributary Load to Beam = (99 psf) x 360 sf)
Roof Tributary Load to Beam = 35.640 Kips
Beam self-weight 250 plf x (12') x 2 = 6.0 Kips
Beam load in plf = 99 psf (15') + 250 plf = 1,735 plf
ROOF LOAD TO COLUMN = **41.640 Kips**

FLOOR LOAD TO COLUMN

Floor Double-Tees (8DT18+2):
Floor LL + DL (85 psf) + DT self-weight (59 psf)
Floor Tributary Area = 15 ft x 24 ft = 360 sf
Floor Tributary Load to Beam = (144 psf) x 360 sf)
Floor Tributary Load to Beam = 51.840 Kips
Beam self-weight 317 plf x (12') x 2 = 7.608 Kips
Beam load in plf = 144 psf (15') + 317 plf = 2,477 plf
FLOOR LOAD TO COLUMN = **59.448 Kips**

LOAD ON COLUMN

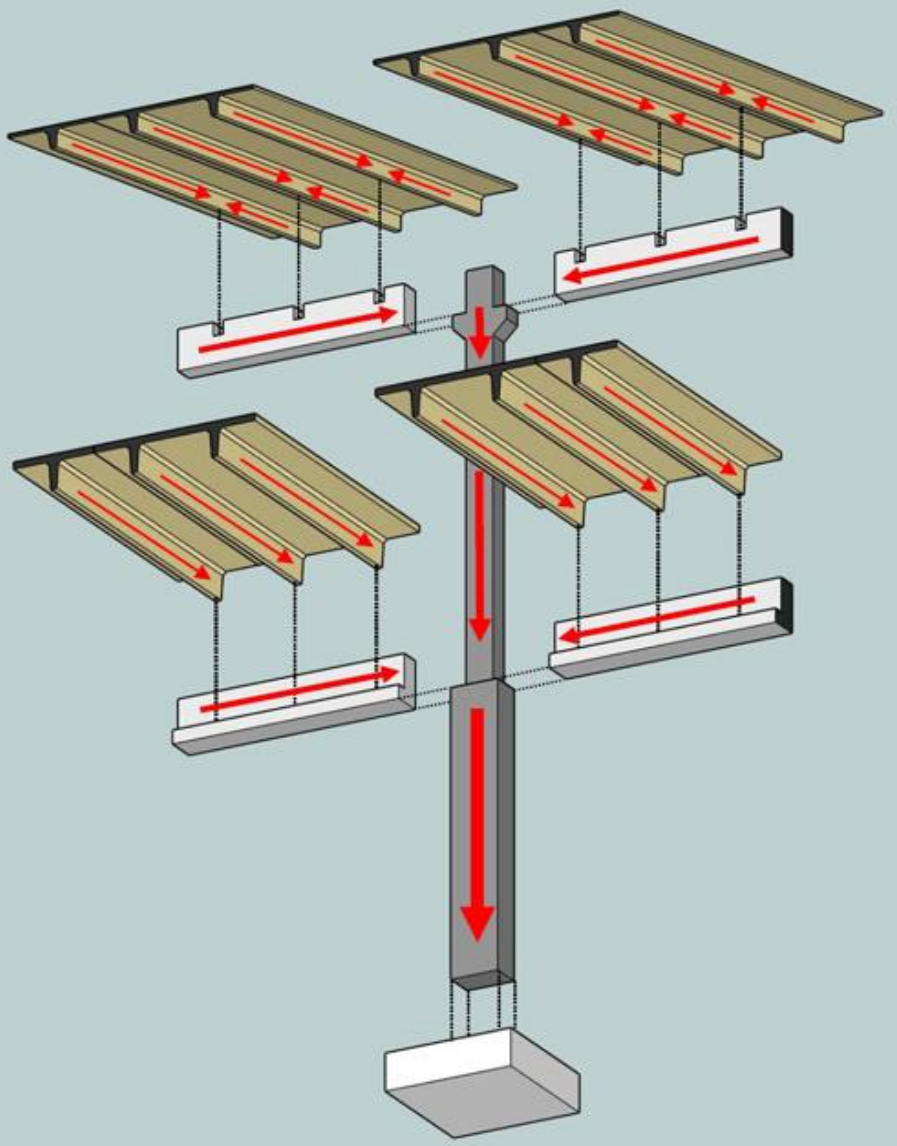
Roof 41.640 K + Floor 59.448 K = **101.088 Kips**

SELF-WEIGHT OF COLUMN

Column area x height x 150 lbs/cf concrete
Upper Column a x h = 1.78 sf x 20 ft = 35.6 cf
Upper weight = 35.6 cf x 150 lbs/cf = 5.34 Kips
Lower Column a x h = 3.5 sf x 14 ft = 49 cf
Lower weight = 49 cf x 150 lbs/cf = 7.35 Kips
COLUMN WEIGHT = 5.34 K + 7.35 K = **12.69 Kips**

LOAD ON FOOTING

Load on Column + Column self-weight
101.088 Kips + 12.69 Kips = **113.78 Kips**

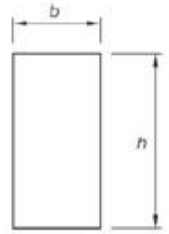


EXPLODED AXON OF TRIBUTARY ELEMENTS TRANSFERRING TRIBUTARY LOAD TO A COLUMN AND FOOTING

ROOF BEAM LOAD

12RB20 spans 24' and supports 2,620 plf
Roof Rectangular Beam load 1,735 plf < 2,620 plf allowable

3.9 Rectangular Beam Load Tables



f'_c = 5000 psi
f_{pu} = 270,000 psi
½ in. diameter, low-relaxation strand

Normalweight concrete

Designation	b in.	h in.	A in.²	I in.⁴	y_b in.	S in.³	wt lb/ft
Section Properties							
12RB16	12	16	192	4096	8	512	200
12RB20	12	20	240	8000	10	800	250

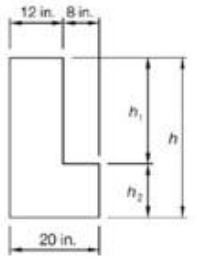
Table of superimposed service load capacity, lb/ft, and cambers, in.

Designation	Number strand	y_s in.	Span, ft 16	18	20	22	24	26	28	30	32	34
12RB16	5	3.00	3650	2770	2210	1790	[illegible]	1230	1040			
			0.4	0.5	0.6	0.8	0.9	1.0	1.1			
			0.2	0.2	0.2	0.2	0.3	0.3	0.3			
12RB20	8	3.00	6160	4820	3860	3150	2620	2260	1860	1600	1380	1190
			0.4	0.5	0.6	0.7	0.9	1.0	1.1	1.3	1.4	1.5
			0.2	0.2	0.3	0.3	0.4	0.4	0.4	0.5	0.5	0.5

FLOOR BEAM LOAD

20LB20 spans 24' and supports 2,760 plf
Floor L-Beam load 2,477 plf < 2,760 plf allowable

3.10 L-Beam Load Tables



f'_c = 5000 psi
f_{pu} = 270,000 psi
½ in. diameter, low-relaxation strand

Normalweight concrete

Designation	h in.	h_1/h_2 in.	A in.²	I in.⁴	y_b in.	S_b in.³	S_t in.³	wt lb/ft
Section Properties								
20LB20	20	12/8	304	10,160	8.74	1163	902	317

Table of superimposed service load capacity, lb/ft, and cambers, in.

Designation	Number strand	y_s in.	Span, ft 16	18	20	22	24	26	28	30	32	34	36
20LB20	9	2.44	6560	5130	4100	3340	2760	2310	1960	1670	1430	1240	1070
			0.3	0.4	0.5	0.6	[illegible]	0.8	0.9	1.0	1.0	1.1	1.2
			0.1	0.2	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.2

CONCRETE BUILDING project

PROJECT PARAMETERS: Precast Elements for ALPHA & CHARLIE Prototype Models

ALPHA

Primary Vertical: full-height Solid and/or Lite-Wall Panels 8’ wide, and one longitudinal grid-line of Columns supporting Beams.
Stair span support by Solid Panels as needed.
Primary Horizontal: Double Tees 8’ feet wide with 30’ span. Beams (L, T, or R) with 24’ span to support Double Tees.
Precast Stairs/Landings with 7” R & 11” T
Secondary Lateral: Concrete Shear Wall panels and/or Lite-Walls
Foundation: Mat slab/Raft Foundation

CHARLIE

Primary Vertical: full-height Columns to accept Beams, and three structural grid-lines of Lite-Wall Panels.
Stair span support by Solid Panels as needed.
Primary Horizontal: Beams (L, T, or R) with 20’ span to support Hollowcore Plank; Precast Stairs/Landings with 7” R & 11” T
Secondary Horizontal: Hollowcore Plank 4’ wide with 20’ span *or* 24’ span *or* 30’ span
Secondary Lateral: Concrete Shear Wall panels and/or Lite-Walls
Foundation: Mat slab/Raft Foundation

…use Precast Concrete span & load tables (on Canvas) to determine sizing of Precast structural elements

CONCRETE BUILDING project

PROJECT PARAMETERS: Precast Elements for DELTA & KILO Models

DELTA

Primary Vertical: full-height Lite-Wall Panels with option of some Solid Bearing-Wall Panels. Stair span support by Solid Panels as needed.
Primary Horizontal: Double Tees 8' *or* 10' feet wide with 40' span; Precast Stairs/Landings with 7" R & 11" T
Secondary Lateral: Concrete Shear Wall panels and/or Lite-Walls
Foundation: Mat slab/Raft Foundation

KILO

Primary Vertical: full-height Lite-Wall Panels to support Beams, with one longitudinal grid-line of continuous Lite-Wall to support Hollowcore. Stair spans supported by Solid Panels as needed.
Primary Horizontal: Beams (L, T, or R) with 24' span to support Hollowcore Plank; Precast Stairs/Landings with 7" R & 11" T
Secondary Horizontal: Hollowcore Plank 4' wide with 30' span
Secondary Lateral: Concrete Shear Wall panels and/or Lite-Walls
Foundation: Mat slab/Raft Foundation

…use Precast Concrete span & load tables (on Canvas) to determine sizing of Precast structural elements

CONCRETE BUILDING project

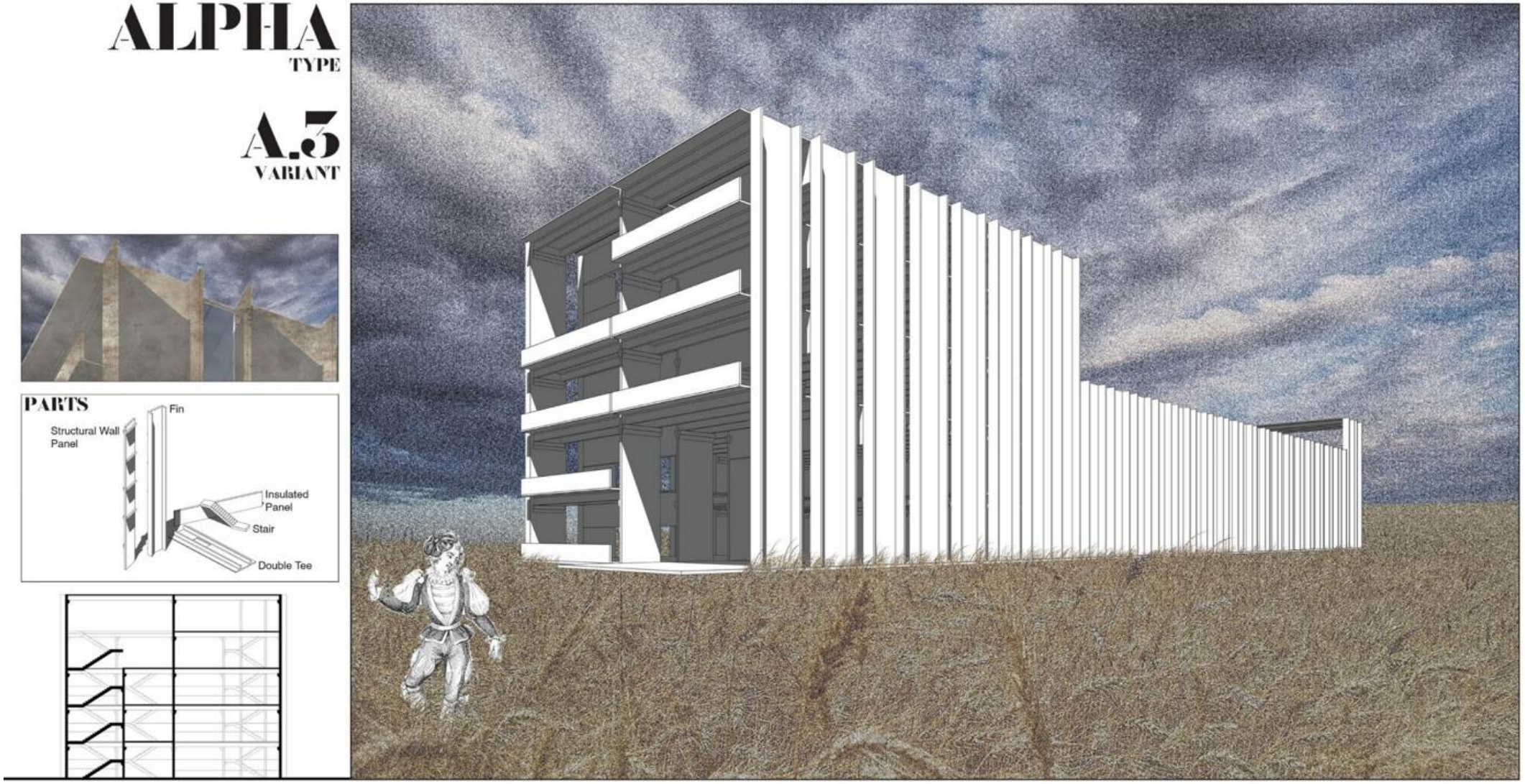
ALPHA or **CHARLIE** or **DELTA** or **KILO**

…you are assigned <u>one</u> prototype model to guide your project

CONCRETE BUILDING project

ALPHA or **CHARLIE** or **DELTA** or **KILO**

…how to use the prototype model to guide your project???

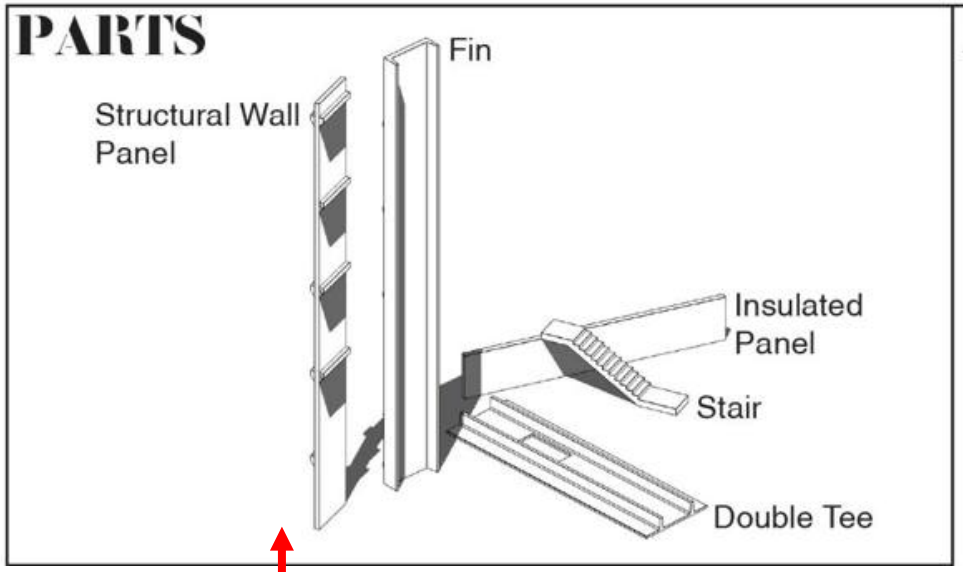


…study the prototype *(pdf poster and 3D digital model on Canvas)*

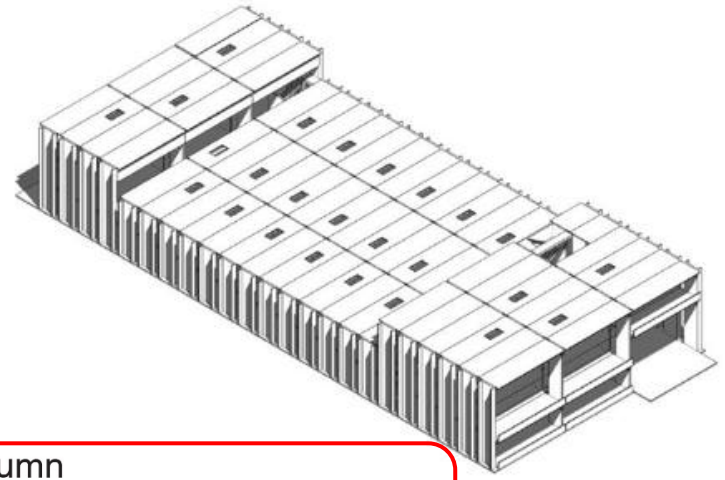
ALPHA TYPE

A

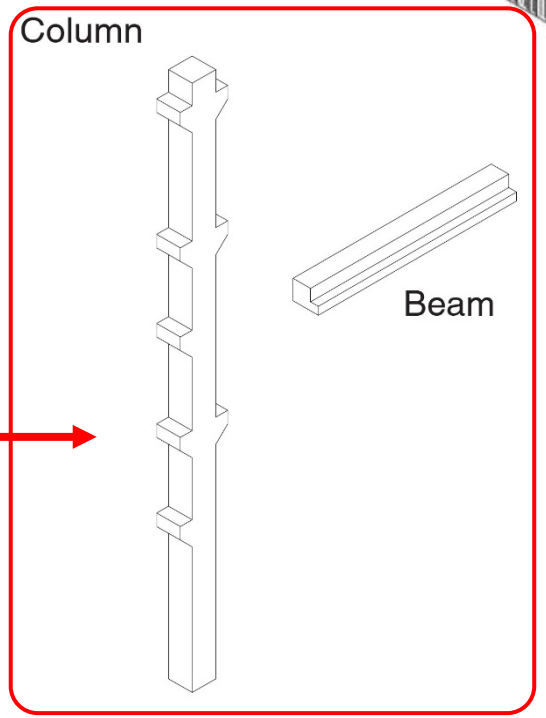
PARTS



A.2

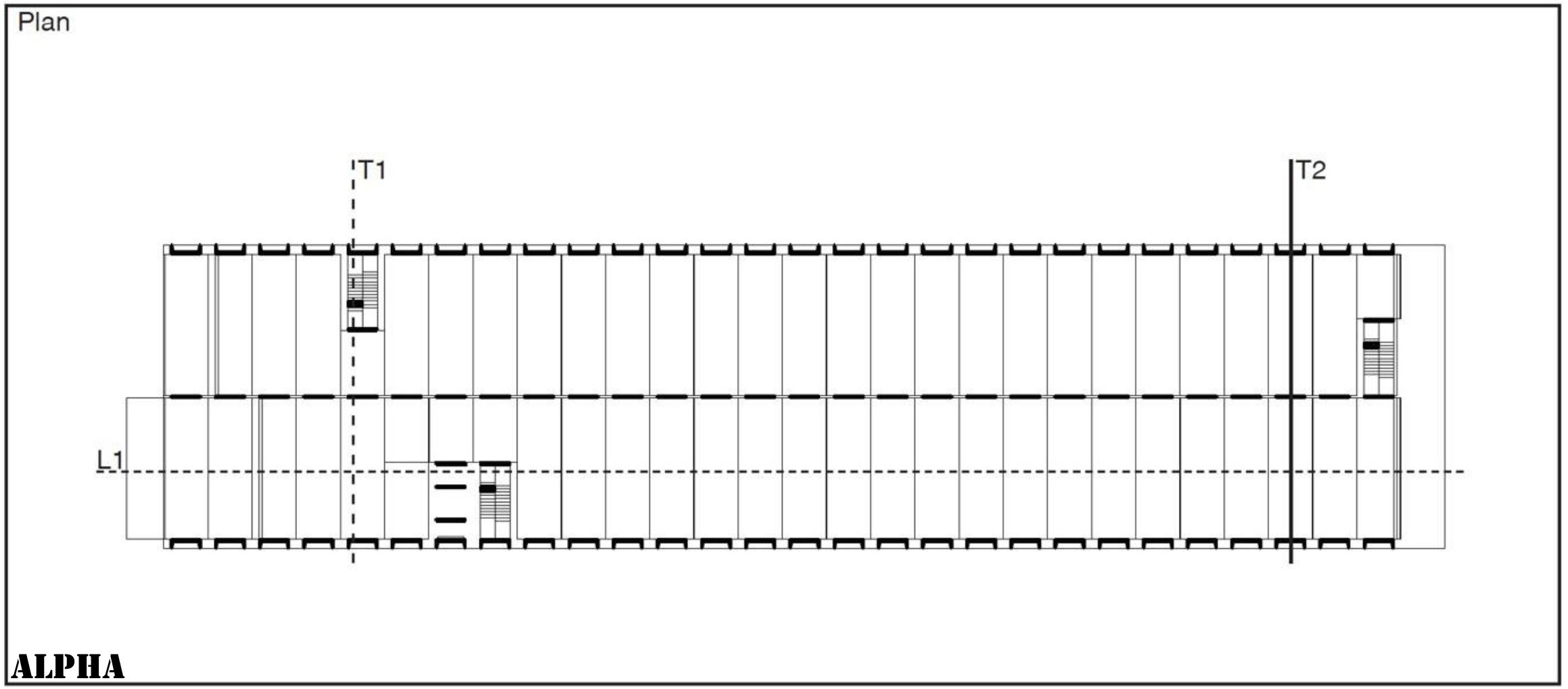


Available elements from prototype model

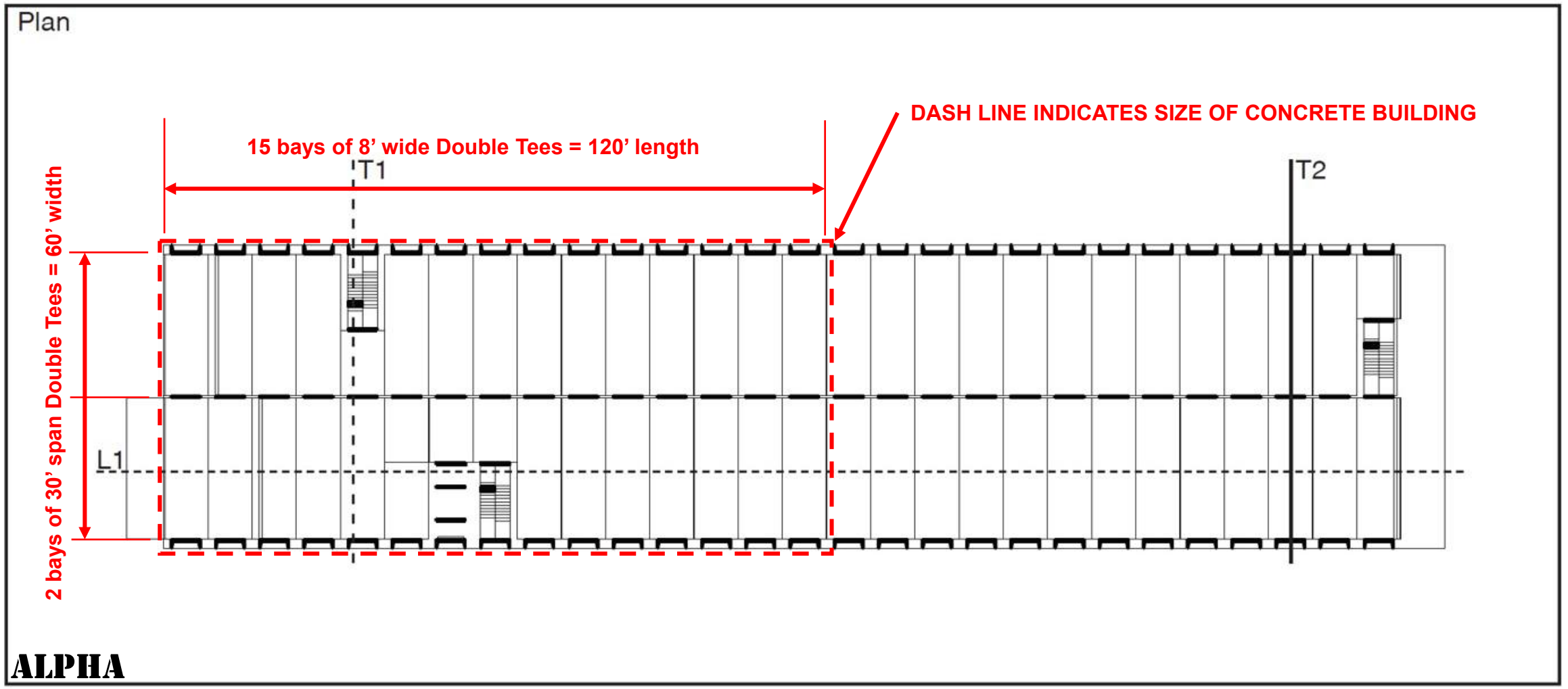


One new longitudinal grid-line of column and beams

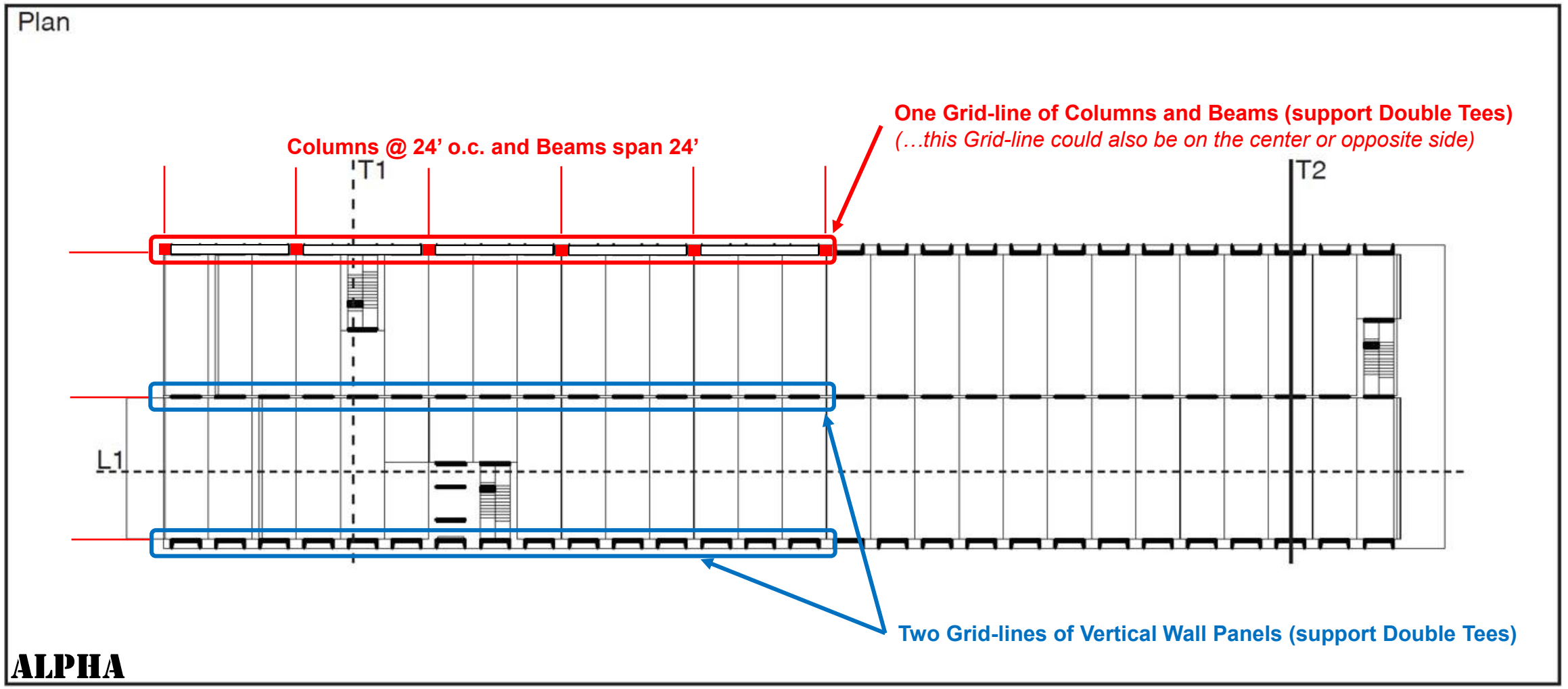
…clarify parameters for your Concrete Building



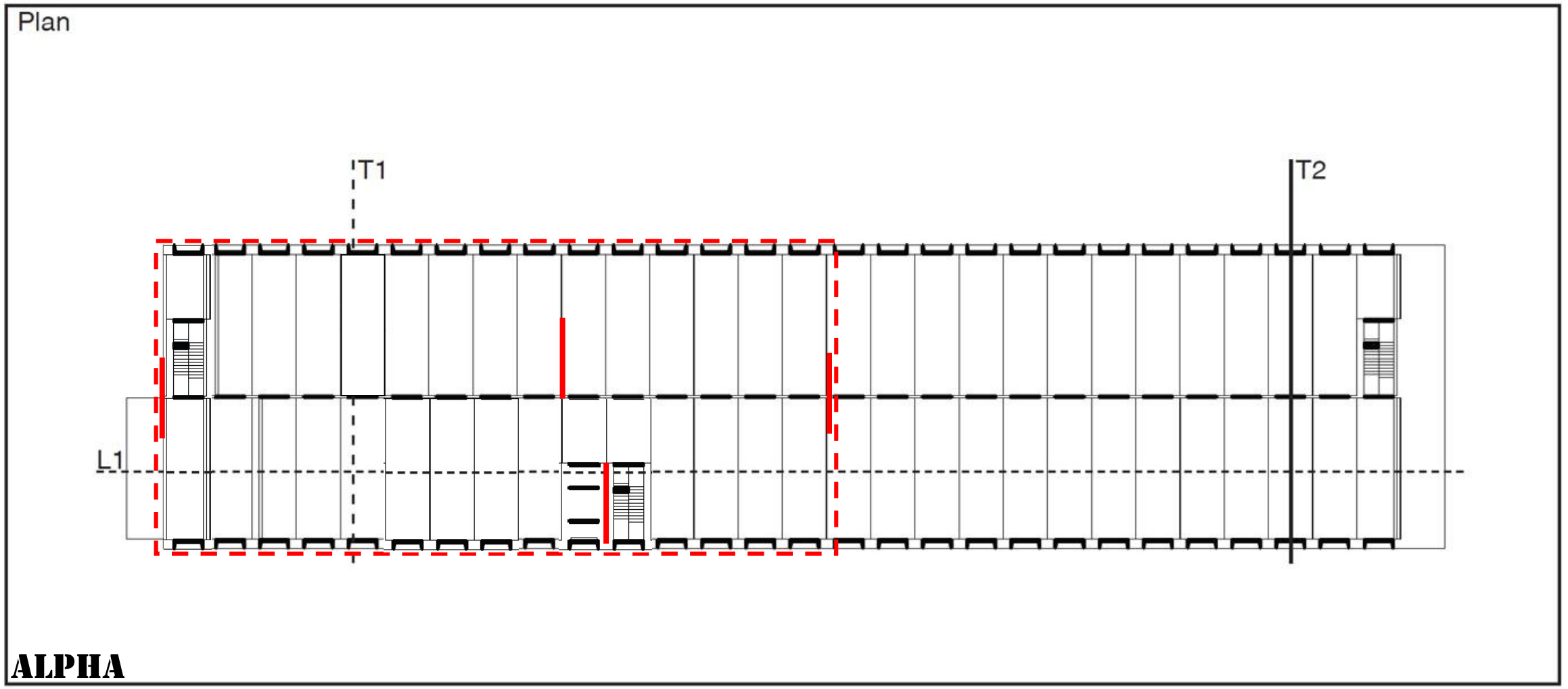
…interpret prototype model relative to Concrete Building parameters



…interpret dimensions/spans, elements, and structural bays

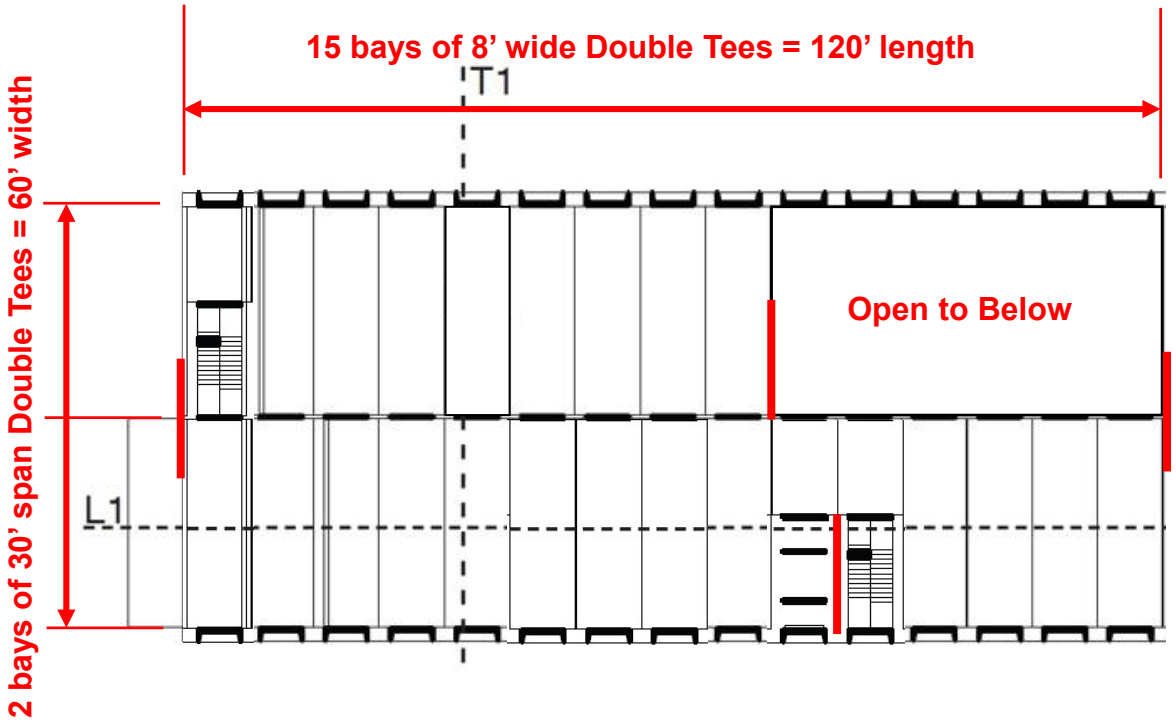


…ALPHA has two grid-lines of vertical wall panels, and one grid-line of columns/beams



…relocate (or reconfigure) stairs and elevator opening in floor, and add shear walls

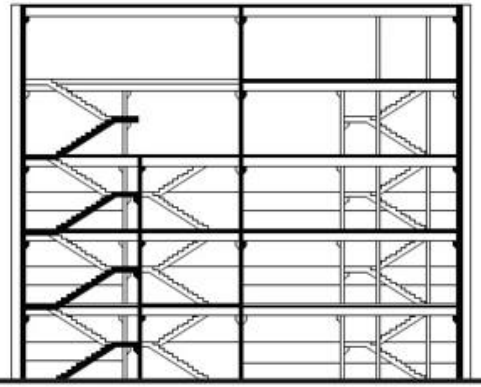
Plan



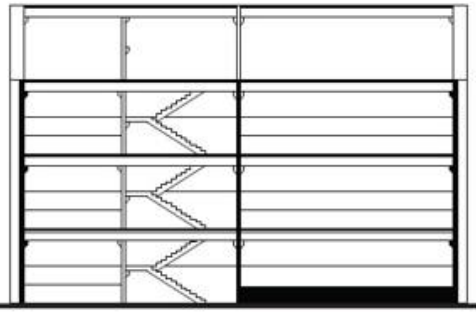
ALPHA

…define double-height space(s) (no more than 25% of 2nd Floor area)

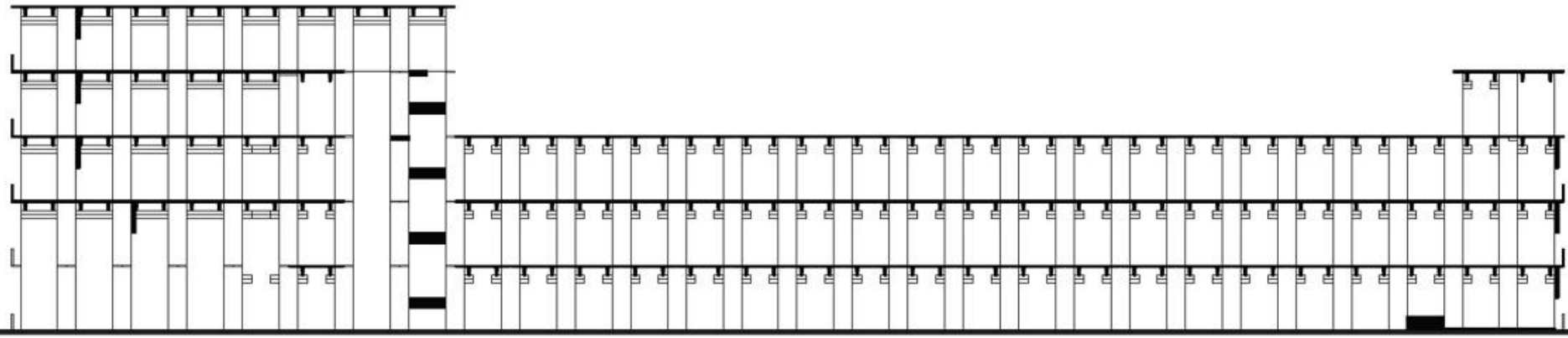
Section:
T1



Section:
T2



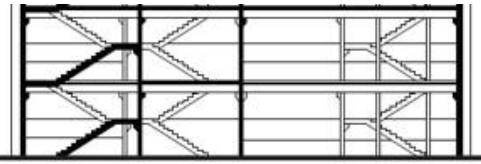
Section:
L1



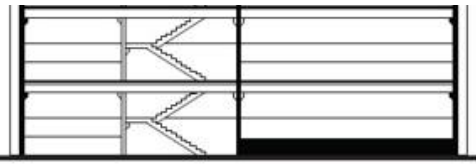
ALPHA

…interpretation of section to reflect CONCRETE BUILDING parameters?

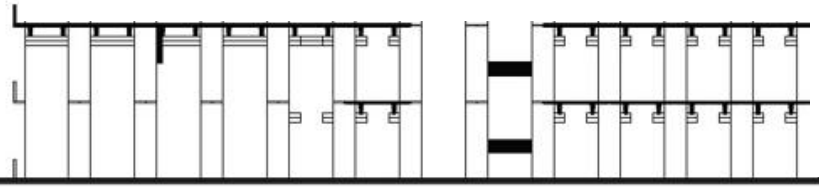
Section:
T1



Section:
T2



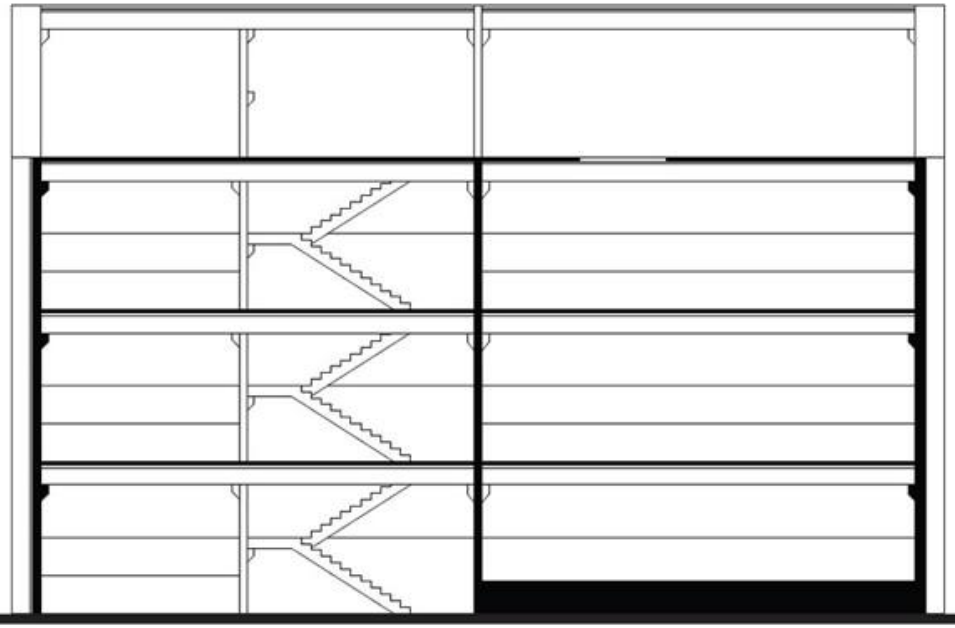
Section:
L1



ALPHA

…adjust height and stories

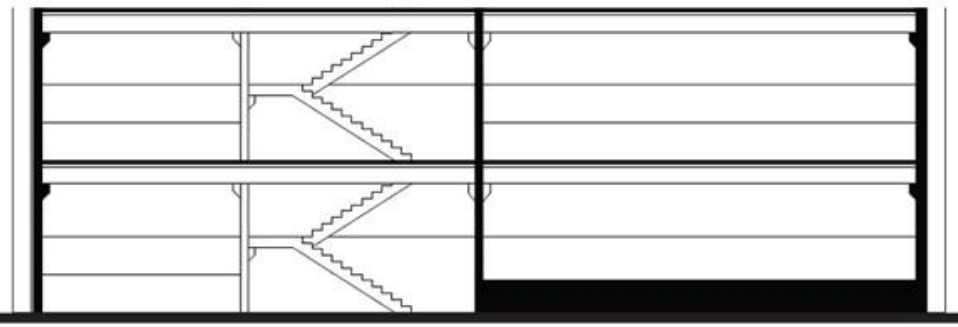
Section:
T2



ALPHA

…adjust height of 3-story prototype?

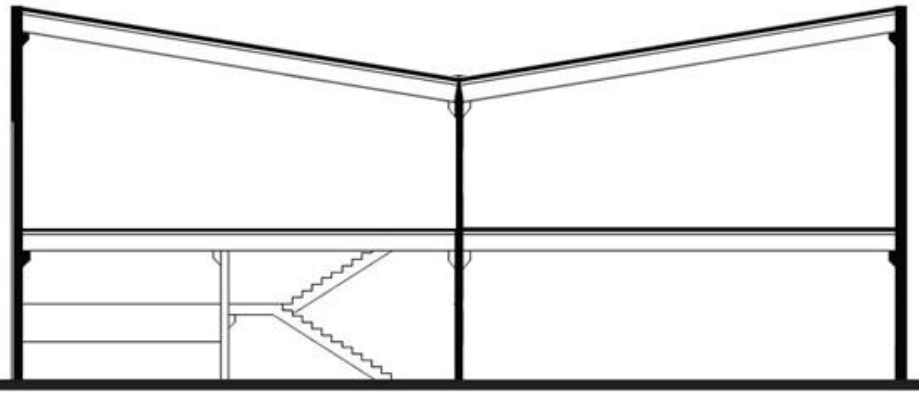
Section:
T2



ALPHA

…down to 2-story, and Main Floor to 2nd Floor at 14’-0”

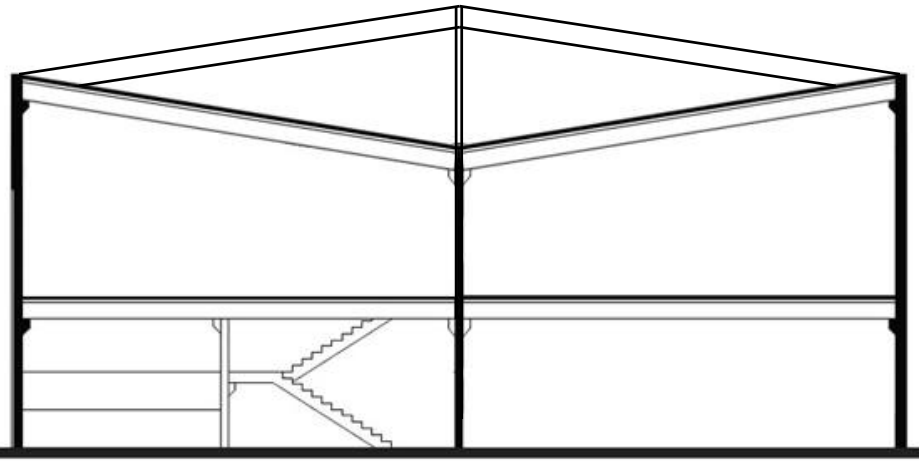
Section:
T2



CONCRETE BUILDING

…adjust slope of roof to meet CONCRETE BUILDING parameters and options

Section:
T2



CONCRETE BUILDING

…possible to have different slopes, as long as it follows the logic of structure

…learn from a parking garage?

…understand a gravity-load and lateral-load element?

CONCRETE BUILDING project

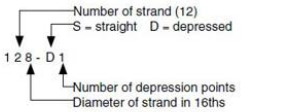
…how to size your elements using span and load tables?

8DT24 Span & Load Table

3.4 Double-Tee Load Tables

3

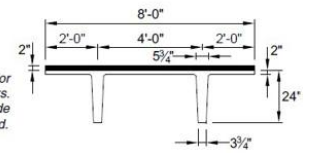
Strand Pattern Designation



Safe loads shown include dead load of 10 lb/ft² for untopped members and 15 lb/ft² for topped members. Remainder is live load. Long-time cambers include superimposed dead load but do not include live load.

Key
186 – Safe superimposed service load, lb/ft²
0.7 – Estimated camber at erection, in.
0.9 – Estimated long-time camber, in.

8'-0" × 24"
Normalweight Concrete



f'_c = 5000 psi
f_{pu} = 270,000 psi
1/2-in.-diameter regular strand

Check with regional producers for availability.

Section Properties

		No Topping	2 in. Topping
A	=	401 in.²	–
I	=	20,985 in.⁴	27,720 in.⁴
y_b	=	17.15 in.	19.27 in.
y_t	=	6.85 in.	6.73 in.
S_b	=	1224 in.³	1439 in.³
S_t	=	3064 in.³	4119 in.³
wt	=	418 lb/ft	618 lb/ft
DL	=	52 lb/ft²	77 lb/ft²
V/S	=	1.41 in.	

8DT24

Table of safe superimposed service load, lb/ft², and cambers, in. — **No Topping**

Strand pattern	y_s(end) y_s(center) in.	Span, ft																						
		32	34	36	38	40	42	44	46	48	50	52	54	56	58	60	62	64	66	68	70	72	74	76
68-S	4.00	186	161	140	122	106	93	81	71	62	55	48	42	36	31	27								
	4.00	0.7	0.8	0.8	0.9	0.9	1.0	1.0	1.0	1.0	1.0	0.9	0.8	0.7	0.6	0.5								
		0.9	1.0	1.1	1.1	1.1	1.1	1.1	1.1	1.0	0.9	0.8	0.6	0.4	0.1	–0.2								
88-S	5.00			185	162	143	126	112	99	88	78	70	62	55	49	43	38	33	29					
	5.00			1.1	1.2	1.3	1.3	1.4	1.4	1.5	1.5	1.5	1.5	1.4	1.4	1.3	1.1	1.0	0.8					
				1.4	1.5	1.6	1.6	1.7	1.7	1.7	1.7	1.6	1.5	1.4	1.2	0.9	0.6	0.3	–0.2					
108-S	6.00				197	174	155	138	123	110	98	88	79	71	64	57	51	45	39	34	29			
	6.00				1.4	1.5	1.6	1.7	1.7	1.8	1.9	1.9	1.9	1.9	1.9	1.9	1.8	1.7	1.5	1.3	1.1			
					1.8	1.9	2.0	2.1	2.1	2.1	2.1	2.1	2.1	2.0	1.9	1.8	1.6	1.3	0.9	0.5	0.0			
128-S	7.00							159	142	128	114	101	90	81	73	66	59	53	47	42	37	32	27	
	7.00							1.8	1.9	2.0	2.1	2.1	2.2	2.2	2.2	2.2	2.2	2.1	2.0	1.9	1.7	1.5	1.2	
								2.3	2.4	2.4	2.5	2.5	2.4	2.4	2.3	2.2	2.1	1.8	1.6	1.2	0.8	0.3	–0.3	
128-D1	11.67												114	103	92	83	74	66	59	53	48	43	39	35
	3.25												2.5	2.6	2.6	2.7	2.7	2.6	2.6	2.5	2.3	2.1	1.9	1.6
													2.9	2.9	2.9	2.8	2.6	2.4	2.2	1.8	1.5	1.1	0.6	0.0
148-D1	12.86																		71	64	58	52	46	42
	3.50																		3.1	3.1	3.0	2.9	2.7	2.5
																			3.0	2.7	2.4	2.0	1.6	1.1

8DT24 + 2

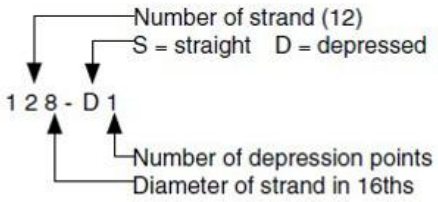
Table of safe superimposed service load, lb/ft², and cambers, in. — **2 in. Normalweight Topping**

Strand pattern	y_s(end) y_s(center) in.	Span, ft																						
		28	30	32	34	36	38	40	42	44	46	48	50	52	54	56	58	60	62	64	66	68	70	
48-S	3.00	169	141	117	97	81	67	55	45	36	29													
	3.00	0.4	0.4	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.4													
		0.4	0.4	0.4	0.4	0.3	0.3	0.2	0.0	–0.1	–0.3													
68-S	4.00			189	161	138	118	101	87	74	64	54	45	38	30									
	4.00			0.7	0.8	0.8	0.9	0.9	1.0	1.0	1.0	1.0	1.0	0.9	0.8									
				0.7	0.8	0.8	0.7	0.7	0.6	0.6	0.4	0.2	0.0	–0.3	–0.6									
88-S	5.00					188	163	142	124	108	94	82	71	62	52	42	33							
	5.00					1.1	1.1	1.3	1.3	1.4	1.4	1.5	1.5	1.5	1.5	1.4	1.4							
						1.1	1.2	1.1	1.1	1.1	1.0	0.9	0.7	0.5	0.3	–0.1	–0.5							
108-S	6.00							176	155	136	120	104	90	77	66	56	47	39	31					
	6.00							1.5	1.6	1.7	1.7	1.8	1.9	1.9	1.9	1.9	1.9	1.9	1.8					
								1.4	1.4	1.4	1.4	1.3	1.2	1.0	0.8	0.6	0.3	–0.1	–0.6					
128-S	7.00									160	140	121	104	90	77	67	57	48	41	33	27			
	7.00									1.8	1.9	2.0	2.1	2.1	2.2	2.2	2.2	2.2	2.2	2.1	2.0			
										1.6	1.6	1.6	1.5	1.3	1.1	0.9	0.6	0.3	–0.1	–0.6	–1.1			
128-D1	11.67														105	91	79	68	58	49	40	34	28	
	3.25														2.5	2.6	2.6	2.7	2.7	2.6	2.6	2.5	2.3	
															1.6	1.4	1.1	0.8	0.4	0.0	–0.6	–1.2	–1.8	

Strength is based on strain compatibility; bottom tension is limited to $12\sqrt{f'_c}$; see pages 3-8 through 3-11 for explanation.
Shaded values require release strengths higher than 3500 psi.

3.4 Double-Tee Load Tables

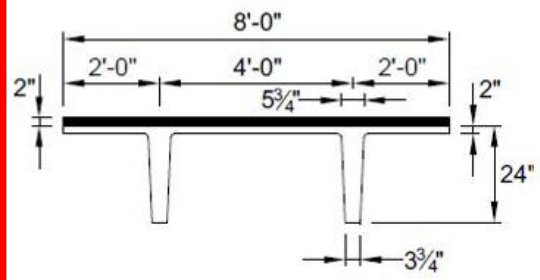
Strand Pattern Designation



Safe loads shown include dead load of 10 lb/ft² for untopped members and 15 lb/ft² for topped members. Remainder is live load. Long-time cambers include superimposed dead load but do not include live load.

Key
186 – Safe superimposed service load, lb/ft²
0.7 – Estimated camber at erection, in.
0.9 – Estimated long-time camber, in.

8'-0" × 24"
Normalweight Concrete



f'_c = 5000 psi

f_{pu} = 270,000 psi

$^1/_2$-in.-diameter regular strand

Section Properties

		No Topping		2 in. Topping	
A	=	401	in.²	–	
I	=	20,985	in.⁴	27,720	in.⁴
y_b	=	17.15	in.	19.27	in.
y_t	=	6.85	in.	6.73	in.
S_b	=	1224	in.³	1439	in.³
S_t	=	3064	in.³	4119	in.³
wt	=	418	lb/ft	618	lb/ft
DL	=	52	lb/ft²	77	lb/ft²
V/S	=	1.41	in.		

8DT24 Span & Load Table

8DT24 + 2

2 in. Normalweight Topping

Table of safe superimposed service load, lb/ft², and cambers, in.

Strand pattern	y_s (end) y_s (center) in.	Span, ft																					
		28	30	32	34	36	38	40	42	44	46	48	50	52	54	56	58	60	62	64	66	68	70
48-S	3.00 3.00	169 0.4 0.4	141 0.4 0.4	117 0.5 0.4	97 0.5 0.4	81 0.5 0.3	67 0.5 0.3	55 0.5 0.2	45 0.5 0.0	36 0.5 –0.1	29 0.4 –0.3												
68-S	4.00 4.00			189 0.7 0.7	161 0.8 0.8	138 0.8 0.8	118 0.9 0.7	101 0.9 0.7	87 1.0 0.6	74 1.0 0.6	64 1.0 0.4	54 1.0 0.2	45 1.0 0.0	38 0.9 –0.3	30 0.8 –0.6								
88-S	5.00 5.00					188 1.1 1.1	163 1.1 1.2	142 1.3 1.1	124 1.3 1.1	108 1.4 1.1	94 1.4 1.0	82 1.5 0.9	71 1.5 0.7	62 1.5 0.5	52 1.5 0.3	42 1.4 –0.1	33 1.4 –0.5						
108-S	6.00 6.00							176 1.5 1.4	155 1.6 1.4	136 1.7 1.4	120 1.7 1.4	104 1.8 1.3	90 1.9 1.2	77 1.9 1.0	66 1.9 0.8	56 1.9 0.6	47 1.9 0.3	39 1.9 –0.1	31 1.8 –0.6				
128-S	7.00 7.00									160 1.8 1.6	140 1.9 1.6	121 2.0 1.6	104 2.1 1.5	90 2.1 1.3	77 2.2 1.1	67 2.2 0.9	57 2.2 0.6	48 2.2 0.3	41 2.2 –0.1	33 2.1 –0.6	27 2.0 –1.1		
128-D1	11.67 3.25														105 2.5 1.6	91 2.6 1.4	79 2.6 1.1	68 2.7 0.8	58 2.7 0.4	49 2.6 0.0	40 2.6 –0.6	34 2.5 –1.2	28 2.3 –1.8

Strength is based on strain compatibility; bottom tension is limited to $12\sqrt{f'_c}$; see pages 3-8 through 3-11 for explanation.
Shaded values require release strengths higher than 3500 psi.

8DT24 Span & Load Table

CONCRETE BUILDING project

…how to size your elements using span and load tables?

…determine your Double-Tee or Hollowcore elements and how far they span
…add your loads for the roof, and loads for your floor (all in psf)
…refer to the Span & Load Tables on Canvas

CONCRETE BUILDING project

ALPHA

Primary Horizontal: Double Tees 8’ feet wide with 30’ span. Beams (L, T, or R) with 24’ span to support Double Tees.

PUBLIC LIBRARY: Houghton, Michigan (60’ x 120’ nominal footprint) (mat slab raft foundation)

ROOF LOADS: Snow Load: 100 psf Roofing DL: 15 psf **ROOF = 115 psf**

FLOOR LOADS: Floor LL: 150 psf Floor DL: 15 psf **FLOOR = 165 psf**

DOUBLE-TEE (in psf)

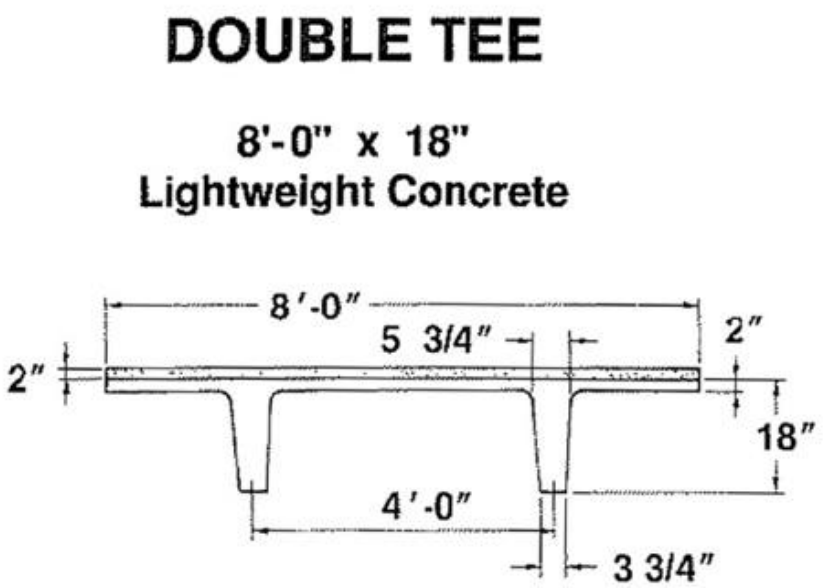
ALPHA

Primary Horizontal: Double Tees 8’ feet wide with 30’ span. Beams (L, T, or R) with 24’ span to support Double Tees.

PUBLIC LIBRARY: Houghton, Michigan (60’ x 120’ nominal footprint) (mat slab raft foundation)

ROOF LOADS: Snow Load: 100 psf Roofing DL: 15 psf **ROOF = 115 psf** **115 psf < 117 psf safe service load**

FLOOR LOADS: Floor LL: 150 psf Floor DL: 15 psf **FLOOR = 165 psf**



8LDT x 18” deep spans 30’ and supports 117 psf

ROOF STRUCTURE: 8LDT18 + 2

8LDT18 + 2
2" Normal Weight Topping

Table of safe superimposed service load (psf) and cambers

Strand Pattern	e_e / e_c	Span, ft. 16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	52
28-S	11.27 11.27	201 0.2 0.2	154 0.2 0.2	116 0.3 0.3	88 0.3 0.3	66 0.4 0.3	50 0.4 0.2	37 0.4 0.1												
48-S	10.27 10.27					164 0.8 0.8	133 0.9 0.8	108 1.0 0.8	88 1.1 0.6	72 1.2 0.8	59 1.2 0.7	47 1.3 0.6	38 1.3 0.4							
68-S	7.27 7.27						171 0.9 0.9	141 1.1 1.0	117 1.2 1.0	97 1.3 1.0	81 1.4 0.9	67 1.4 0.8	55 1.5 0.6	46 1.5 0.4	37 1.5 0.1					
68-D1	7.27 10.77							190 1.4 1.4	159 1.6 1.5	134 1.7 1.5	114 1.9 1.6	96 2.1 1.6	82 2.2 1.5	69 2.4 1.4	59 2.5 1.1	49 2.6 0.8	41 2.6 0.5	34 2.6 0.0		
88-D1	5.27 10.52														87 3.1 2.0	75 3.2 1.8	65 3.4 1.5	56 3.5 1.1	48 3.6 0.7	39 3.7 0.1

DOUBLE-TEE (in psf)

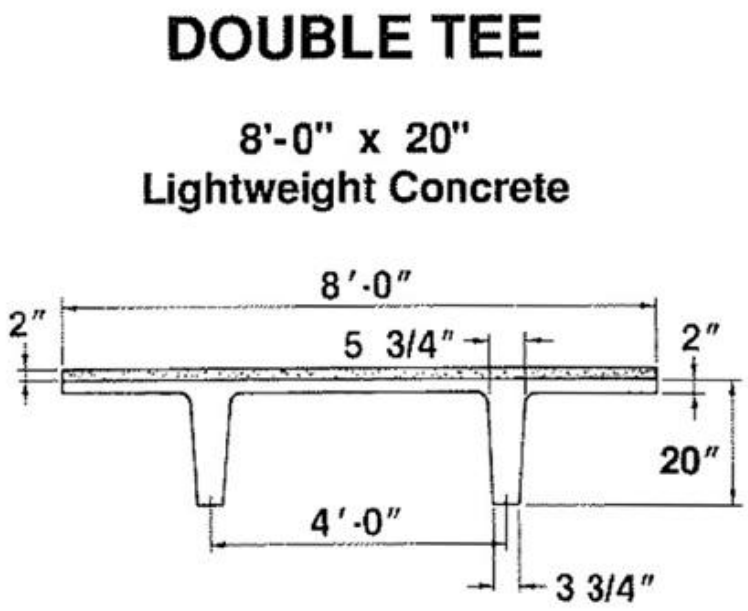
ALPHA

Primary Horizontal: Double Tees 8’ feet wide with 30’ span. Beams (L, T, or R) with 24’ span to support Double Tees.

PUBLIC LIBRARY: Houghton, Michigan (60’ x 120’ nominal footprint) (mat slab raft foundation)

ROOF LOADS: Snow Load: 100 psf Roofing DL: 15 psf **ROOF = 115 psf** **115 psf < 117 psf**

FLOOR LOADS: Floor LL: 150 psf Floor DL: 15 psf **FLOOR = 165 psf** **165 psf < 182 psf safe service load**



8LDT x 20” deep spans 30’ and supports 182 psf

FLOOR STRUCTURE: 8LDT20 + 2

8LDT20 + 2
2" Normal Weight Topping

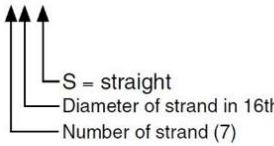
Table of safe superimposed service load (psf) and cambers

Strand Pattern	e_e / e_c	Span, ft. 24	26	28	30	32	34	36	38	40	42	44	46	48	50	52	54	56	58
48-S	11.59	188	153	125	[illegible]	85	70	57	46	31									
	11.59	0.7	0.7	0.8	0.9	1.0	1.1	1.1	1.2	1.2									
		0.7	0.7	0.8	0.8	0.8	0.7	0.7	0.6	0.4									
68-S	8.59			167	140	117	98	82	69	58	48	39							
	8.59			1.0	1.1	1.2	1.2	1.3	1.4	1.5	1.5	1.5							
				0.9	[illegible]	1.0	1.0	0.9	0.8	0.7	0.5	0.2							
68-D1	8.59				182	154	131	112	96	82	70	59	50	42	35				
	12.09				1.4	1.5	1.6	1.8	2.0	2.1	2.2	2.3	2.3	2.4	2.4				
					1.4	1.4	1.4	1.5	1.5	1.4	1.3	1.1	0.8	0.5	0.1				
88-D1	6.59								136	118	102	89	77	67	58	50	43	36	
	11.84								2.4	2.6	2.8	2.9	3.1	3.2	3.4	3.5	3.5	3.5	
									2.1	2.1	2.0	2.0	1.8	1.6	1.3	0.9	0.4	0.2	
108-D1	4.59																	53	45
	11.59																	4.3	4.3
																		0.8	0.2

HOLLOW-CORE PLANK (in psf)

3.6 Hollow-Core Load Tables

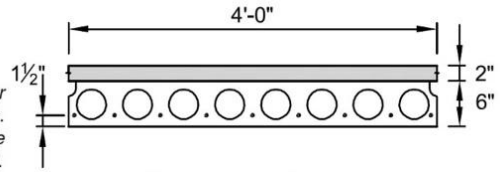
Strand Pattern Designation
76-S



Safe loads shown include dead load of 10 lb/ft² for untopped members and 15 lb/ft² for topped members. Remainder is live load. Long-time cambers include superimposed dead load but do not include live load.

Capacity of sections of other configurations are similar. For precise values, see local hollow-core manufacturer.

4'-0" × 6"
Normalweight Concrete



f'_c = 5000 psi

f_{pu} = 270,000 psi

Section Properties

		No Topping		2 in. topping	
A	=	187	in.²	-	
I	=	763	in.⁴	1640	in.⁴
y_b	=	3.00	in.	4.14	in.
y_t	=	3.00	in.	3.86	in.
S_b	=	254	in.³	396	in.³
S_t	=	254	in.³	425	in.³
wt	=	195	lb/ft	295	lb/ft
DL	=	49	lb/ft²	74	lb/ft²
V/S	=	1.73	in.		

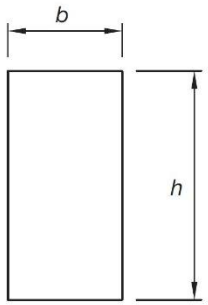
4HC6 + 2

Table of safe superimposed service load, lb/ft², and cambers, in. **2 in. Normalweight Topping**

Strand designation code	Span, ft																		
	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
66-S	333	300	272	248	227	210	182	158	136	113	93	75	59	46	34				
	0.2	0.2	0.2	0.2	0.2	02	0.2	0.2	0.2	0.2	0.1	0.1	0.0	–0.1	–0.2				
	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.0	–0.1	–0.2	–0.3	–0.5	–0.7	–0.9	–1.2				
76-S		357	324	296	272	248	216	188	163	137	115	95	78	63	50	38	27		
		0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.1	0.1	–0.0	–0.1	–0.3		
		0.2	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.0	–0.2	–0.3	–0.5	–0.7	–0.9	–1.2	–1.5		
96-S			418	384	353	319	279	245	216	186	160	137	116	98	82	68	55	43	33
			0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.4	0.3	0.3	0.1	0.0	–0.1
			0.4	0.4	0.4	0.4	0.4	0.4	0.3	0.3	0.2	0.1	–0.1	–0.3	–0.5	–0.7	–1.0	–1.4	–1.7
87-S			422	388	359	332	309	288	258	224	195	169	147	127	109	94	80	67	55
			0.5	0.5	0.6	0.6	0.7	0.7	0.7	0.7	0.8	0.8	0.7	0.7	0.7	0.6	0.5	0.4	0.3
			0.5	0.5	0.5	0.6	0.6	0.6	0.5	0.5	0.4	0.4	0.2	0.1	–0.1	–0.3	–0.5	–0.8	–1.2
97-S			473	433	400	371	346	323	288	251	219	192	168	146	127	110	95	82	70
			0.5	0.6	0.7	0.7	0.8	0.8	0.9	0.9	0.9	0.9	1.0	0.9	0.9	0.9	0.8	0.7	0.6
			0.6	0.6	0.7	0.7	0.7	0.7	0.7	0.7	0.6	0.6	0.5	0.4	0.2	0.0	–0.2	–0.5	–0.8

RECTANGULAR BEAM (in plf)

3.9 *Rectangular Beam Load Tables*



f'_c = 5000 psi
f_{pu} = 270,000 psi
½ in. diameter,
low-relaxation strand

Normalweight concrete

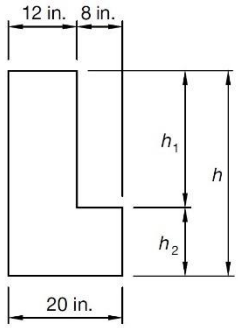
Section Properties							
Designation	*b* in.	*h* in.	*A* in.^{2}	*I* in.^{4}	y_b in.	*S* in.^{3}	*wt* lb/ft
12RB16	12	16	192	4096	8	512	200
12RB20	12	20	240	8000	10	800	250
12RB24	12	24	288	13,824	12	1152	300
12RB28	12	28	336	21,952	14	1568	350
12RB32	12	32	384	32,768	16	2048	400
12RB36	12	36	432	46,656	18	2592	450
16RB24	16	24	384	18,432	12	1536	400
16RB28	16	28	448	29,269	14	2091	467
16RB32	16	32	512	43,691	16	2731	533
16RB36	16	36	576	62,208	18	3456	600
16RB40	16	40	640	85,333	20	4267	667

Table of superimposed service load capacity, lb/ft, and cambers, in.

Designation	Number strand	y_s in.	Span, ft																		
			16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	
12RB16	5	3.00	3550 0.4 0.2	2770 0.5 0.2	2210 0.6 0.2	1790 0.8 0.2	1480 0.9 0.3	1230 1.0 0.3	1040 1.1 0.3												
12RB20	8	3.00	6160 0.4 0.2	4820 0.5 0.2	3860 0.6 0.3	3150 0.7 0.3	2620 0.9 0.4	2200 1.0 0.4	1860 1.1 0.4	1600 1.3 0.5	1380 1.4 0.5	1190 1.5 0.5	1040 1.7 0.5								
12RB24	10	3.60	8950 0.4 0.2	7010 0.4 0.2	5630 0.5 0.3	4610 0.7 0.3	3830 0.8 0.3	3230 0.9 0.4	2740 1.0 0.4	2360 1.1 0.5	2040 1.3 0.5	1780 1.4 0.6	1560 1.5 0.6	1370 1.6 0.6	1210 1.8 0.6	1070 1.9 0.7	960 2.0 0.6				
12RB28	12	4.00		9780 0.4 0.2	7860 0.5 0.2	6440 0.6 0.3	5370 0.7 0.3	4530 0.8 0.4	3860 0.9 0.4	3320 1.0 0.5	2890 1.2 0.5	2520 1. 3 0.6	2220 1.4 0.6	1960 1.5 0.7	1740 1.7 0.7	1550 1.8 0.7	1380 1.9 0.8	1240 2.0 0.8	1110 2.1 0.8	1000 2.2 0.8
12RB32	13	4.77				8320 0.5 0.2	6930 0.6 0.3	5850 0.7 0.3	5000 0.8 0.3	4310 0.9 0.4	3750 1.0. 0.4	3280 1.1 0.4	2890 1.2 0.5	2560 1.3 0.5	2270 1.4 0.5	2030 1.5 0.5	1820 1.6 0.6	1630 1.7 0.6	1470 1.8 0.6	1330 1.9 0.6
12RB36	15	5.07					9010 0.5 0.2	7620 0.6 0.3	6520 0.7 0.3	5630 0.8 0.4	4900 0.9 0.4	4290 1.0 0.4	3790 1.1 0.5	3360 1.2 0.5	2990 1.3 0.6	2680 1.4 0.6	2410 1.5 0.6	2170 1.6 0.6	1960 1.7 0.7	1780 1.8 0.7
16RB24	13	3.54		9390 0.4 0.2	7540 0.5 0.2	6170 0.6 0.3	5130 0.8 0.3	4320 0.9 0.4	3680 1.0 0.4	3160 1.1 0.5	2730 1.2 0.5	2380 1.4 0.5	2090 1.5 0.5	1840 1.6 0.6	1620 1.7 0.6	1440 1.8 0.6	1280 1.9 0.6	1140 2.0 0.6	1020 2.1 0.5	
16RB28	14	3.71				8730 0.5 0.2	7270 0.6 0.2	6130 0.7 0.3	5230 0.8 0.3	4510 0.9 0.3	3910 1.1 0.4	3420 1.2 0.4	3010 1.3 0.4	2660 1.4 0.4	2360 1.5 0.4	2100 1.6 0.4	1880 1.7 0.4	1680 1.8 0.4	1510 1.9 0.4	1360 1.9 0.3
16RB32	18	4.67					9340 0.6 0.3	7890 0.7 0.3	6740 0.8 0.4	5810 0.9 0.4	5050 1.0 0.5	4420 1.1 0.5	3890 1.2 0.5	3450 1.3 0.6	3070 1.5 0.6	2740 1.6 0.6	2450 1.7 0.7	2210 1.8 0.7	1990 1.9 0.7	1800 2.0 0.7
16RB36	20	5.40						9940 0.6 0.3	5800 0.7 0.3	7340 0.8 0.4	6390 0.9 0.4	5600 1.0 0.4	4940 1.1 0.5	4380 1.2 0.5	3900 1.3 0.5	3490 1.4 0.6	3130 1.5 0.6	2820 1.6 0.6	2550 1.7 0.6	2310 1.8 0.6
16RB40	22	6.00								9120 0.7 0.3	7940 0.8 0.4	6970 0.9 0.4	6160 1.0 0.4	5470 1.1 0.5	4880 1.2 0.5	4370 1.3 0.5	3930 1.4 0.5	3550 1.5 0.6	3210 1.6 0.6	2910 1.7 0.6

L-BEAM & T-BEAM (in plf)

3.10 L-Beam Load Tables

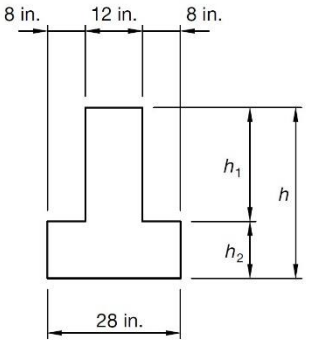


f'_c = 5000 psi
f_{pu} = 270,000 psi
½ in. diameter,
low-relaxation strand

Normalweight concrete

Section Properties								
Designation	h in.	h_1/h_2 in.	A in.^{2}	I in.^{4}	y_b in.	S_b in.^{3}	S_t in.^{3}	wt lb/ft
20LB20	20	12/8	304	10,160	8.74	1163	902	317
20LB24	24	12/12	384	17,568	10.5	1673	1301	400
20LB28	28	16/12	432	27,883	12.22	2282	1767	450
20LB32	32	20/12	480	41,600	14.00	2971	2311	500
20LB36	36	24/12	528	59,119	15.82	3737	2930	550
20LB40	40	24/16	608	81,282	17.47	4653	3608	633
20LB44	44	28/16	656	108,107	19.27	5610	4372	683
20LB48	48	32/16	704	140,133	21.09	6645	5208	733
20LB52	52	36/16	752	177,752	22.94	7749	6117	783
20LB56	56	40/16	800	221,355	24.8	8926	7095	833
20LB60	60	44/16	848	271,332	26.68	10,170	8143	883

3.11 Inverted T Beam Load Tables



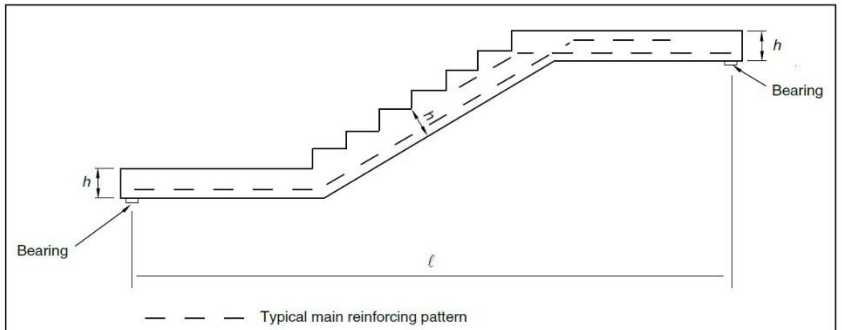
f'_c = 5000 psi
f_{pu} = 270,000 psi
½ in. diameter,
low-relaxation strand

Normalweight concrete

Section Properties								
Designation	h in.	h_1/h_2 in.	A in.^{2}	I in.^{4}	y_b in.	S_b in.^{3}	S_t in.^{3}	wt lb/ft
28IT20	20	12/8	368	11,688	7.91	1478	967	383
28IT24	24	12/12	480	20,275	9.60	2112	1408	500
28IT28	28	16/12	528	32,076	11.09	2892	1897	550
28IT32	32	20/12	576	47,872	12.67	3778	2477	600
28IT36	36	24/12	624	68,101	14.31	4759	3140	650
28IT40	40	24/16	736	93,503	15.83	5907	3869	767
28IT44	44	28/16	784	124,437	17.43	7139	4683	817
28IT48	48	32/16	832	161,424	19.08	8460	5582	867
28IT52	52	36/16	880	204,884	20.76	9869	6558	917
28IT56	56	40/16	928	255,229	22.48	11,354	7614	967
28IT60	60	44/16	976	312,866	24.23	12,912	8747	1017

PRECAST STAIR (100 psf load assumed, thickness based on span)

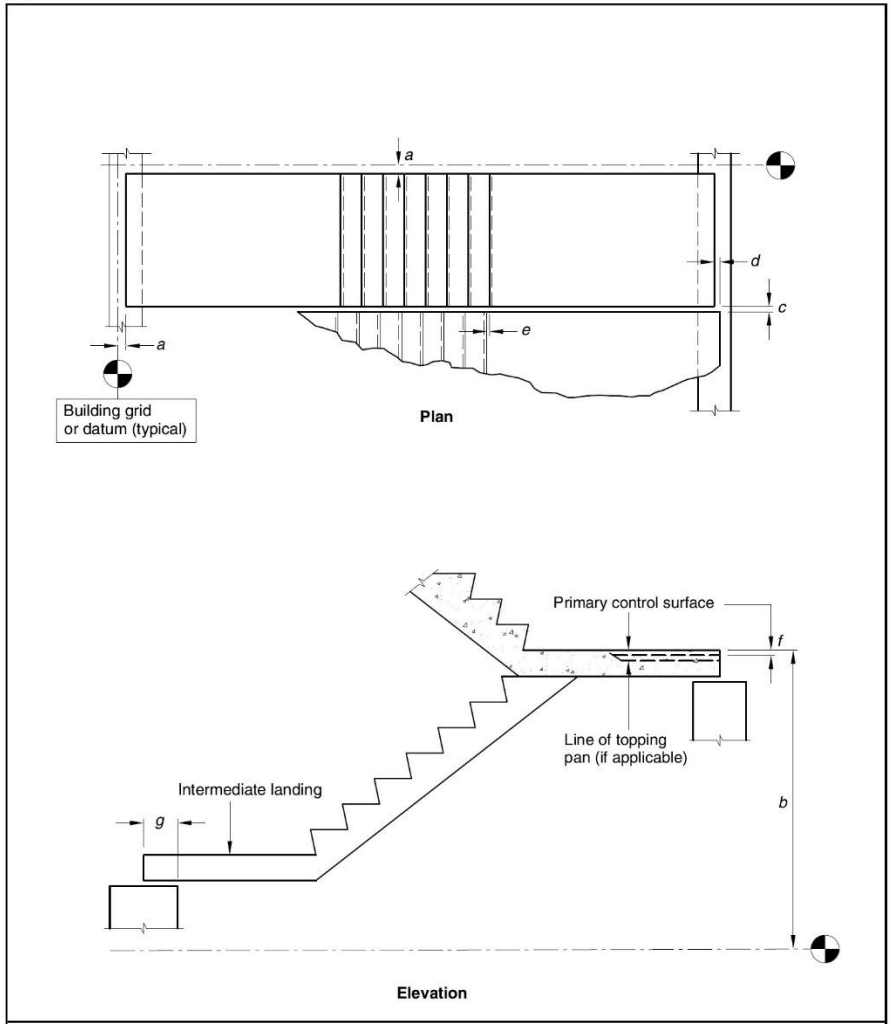
Design Aid 3.12.10 Recommended Maximum Spans for Precast Concrete Stairs



Thickness h, in.	Span ℓ, ft
6	14
8	19
10	23
12	26

Notes:
1. Stair reinforcing must be designed for all factored loads.
2. Normalweight concrete.
3. Recommended maximum span ℓ is based on dead weight of stair plus 100 lb/ft^2 live load.
4. Minimum concrete 28-day compressive strength is 5000 psi.
5. When deflections are required to be checked in accordance with ACI 318-05 Section 9.5, cracked section properties are required to be used. Top steel may be required to control deflections.
6. Longer spans can be obtained with higher concrete strength or special reinforcing.

Fig. 13.3.9 Stair unit erection tolerance



CONCRETE BUILDING project

…what to do by this Thursday?

CONCRETE BUILDING project

…what to do by this Thursday?

…review the Concrete Building parameters for your prototype model.

…review your prototype model and begin to understand the logic of the structure

…diagram (to scale) a few iterations of possible prototype transformations

…prepare for a walking field-trip to the LTU CIMR Lab & Southfield Town Center

… Concrete Building due on November 16 (11:59pm upload to Canvas)

CONCRETE BUILDING project

…contemporary precast reference model?

POPULUS: Denver, Colorado JEANNE GANG

POPULUS: Denver, Colorado JEANNE GANG

…vertical Lite-Wall?

2021 PREFABRICATION (ETH experimental precast project)